

LEHMER'S TOTIENT PROBLEM WITH FEWER THAN SIXTEEN PRIME FACTORS

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ABSTRACT. A composite number n whose totient divides $n - 1$ was known to have at least fourteen prime factors. We prove that it has at least sixteen, at least 16001 if $(n - 1)/\varphi(n) \geq 3$, and more than 10^8 if $3 \mid n$, and that $n < 2^{2^{k-7}}$ if n has k prime factors. For the companion equation $\varphi(n) \mid n + 1$ we show that the nine known solutions are the only ones with at most eight prime factors, and that a further solution prime to 3 has at least sixteen. Each statement is reduced by hand to a finite computation, which independent programs carry out. Lehmer's conjecture remains open, and we state the assertion that would settle it.

1. INTRODUCTION

1.1. Two questions of Lehmer. Every prime p satisfies $\varphi(p) = p - 1$, and so $\varphi(p)$ divides $p - 1$. In 1932 Lehmer [19] asked whether any composite number n has the same property,

$$\varphi(n) \mid n - 1. \quad (1.1)$$

Nobody has found one, and nobody has shown that none exists. Lehmer proved that a composite solution of (1.1) must be odd, squarefree and divisible by at least seven distinct primes. The question is Problem B37 in Guy's collection [16]. Lehmer's paper also lists solutions of the companion equation¹

$$\varphi(n) \mid n + 1. \quad (1.2)$$

The two equations differ only in a sign. Equation (1.1) has no known composite solution, whereas (1.2) has the solutions

$$1, 2, 3, 15, 255, 65535, 83623935, 4294967295, 6992962672132095, \quad (1.3)$$

and no others are known. Except for 2, each of them satisfies $n + 1 = 2\varphi(n)$. Five of them are products of the first Fermat primes², namely $3, 15 = 3 \cdot 5, 255 = 3 \cdot 5 \cdot 17, 65535 = 3 \cdot 5 \cdot 17 \cdot 257$ and $4294967295 = 3 \cdot 5 \cdot 17 \cdot 257 \cdot 65537$. Apart from 1 and 2, the other two are $83623935 = 3 \cdot 5 \cdot 17 \cdot 353 \cdot 929$ and $6992962672132095 = 83623935 \cdot 83623937$. These numbers form sequence A203966 of the OEIS [29].

Much of the literature on (1.1) concerns the number $\omega(n)$ of distinct prime factors of a composite solution. Pomerance [32] proved that a composite solution with k prime factors satisfies $n < k^{2^k}$, so that for each k the question is a finite one, and Burek and Žmija [7] sharpened this to $n \leq 2^{2^k} - 2^{2^{k-1}}$, using a lemma that goes back to Heath-Brown's work on

2020 *Mathematics Subject Classification*. Primary 11A25; Secondary 11A41, 11A51, 11D68, 11N36, 11Y11, 11Y16, 11Y70.

Key words and phrases. Lehmer's totient problem, Euler's totient function, large sieve, Fermat primes.

¹The comments to sequence A050474 of the OEIS [29] record this, and a misprint in Lehmer's summary where $3 \cdot 5 \cdot 353 \cdot 929$ stands for $83623935 = 3 \cdot 5 \cdot 17 \cdot 353 \cdot 929$.

²The Fermat numbers $F_i = 2^{2^i} + 1$ are prime for $0 \leq i \leq 4$, and $F_0 F_1 \cdots F_{r-1} = 2^{2^r} - 1$. For $1 \leq r \leq 5$ the number $n = 2^{2^r} - 1$ is therefore a product of primes F_i , and $\varphi(n) = 2^{1+2+\cdots+2^{r-1}} = 2^{2^r-1} = (n+1)/2$. The chain breaks at $r = 6$ because $F_5 = 641 \cdot 6700417$, as Euler found.

odd perfect numbers [18] and was sharpened by Cook [10] and Nielsen [27].³ In binary, their bound allows n one digit for each subset of the set of its prime factors. Cohen and Hagis [8] settled all $k \leq 13$ and obtained $\omega(n) \geq 14$. When $3 \mid n$ much more is true: Hagis [17] proved $\omega(n) \geq 298848$, and Burcsi, Czirbusz and Farkas [6] raised this to $\omega(n) \geq 40000000$. In the same paper they report that a computation of Renze [34], made available as a Mathematica notebook, gives $\omega(n) \geq 15$ in general, and Pinch [31] outlined a proof of the same bound in a poster presented at ANTS VII. Neither account has been refereed. Since $p - 1 \mid n - 1$ for every prime p dividing a solution, a composite solution of (1.1) would be a Carmichael number;⁴ Banks, Nevans and Pomerance [3] discuss this and the connection with Giuga's conjecture.⁵

More is known when the quotient $M = (n - 1)/\varphi(n)$ exceeds 2. Hagis [17] proved $\omega(n) \geq 1991$ for $M = 3$, and Grytczuk and Wójtowicz [15] gave lower bounds for $\omega(n)$ that grow exponentially with $M \geq 4$ (see [38] for both). The prime factors of a solution form what Golomb [14] called a G -sequence: none of them divides another minus one. Meijer [22], who points out the connection with (1.1), proved with the Brun–Selberg sieve that the product of $p/(p - 1)$ over the elements of a G -sequence up to x is at most a constant times $\log \log x$. Yamada [38] made this explicit with the large sieve and deduced that $M < 16.03235 \log \log \omega(n)$ for every composite solution of $\varphi(n) \mid n \pm 1$ with $\omega(n) \geq 4$, so that $\omega(n)$ is at least doubly exponential in M .

Norman [28] claimed that $M \geq 3$ for every composite solution of (1.1), and hence $\omega(n) \geq 1991$. The proof rests on [28, Lemma 2.1], which asserts that $\psi(G)/|G|^2 > \varphi(m)/(2m)$ for every noncyclic nilpotent group G of order $4m$ with m odd and squarefree, where $\psi(G)$ is the sum of the orders of the elements of G . The lemma is false. For $G = C_2 \times C_2 \times C_m$ with m prime to 3 we have $\psi(G) = 7 \prod_{p \mid m} (p^2 - p + 1)$, since ψ is multiplicative over factors of coprime order, and the ratio of $\psi(G)/|G|^2$ to $\varphi(m)/(2m)$ is $\frac{7}{8} \prod_{p \mid m} (1 + \frac{1}{p(p-1)})$. The product over all primes $p \geq 5$ is $\frac{4}{7} \zeta(2) \zeta(3) / \zeta(6) = 1.1106 \dots < \frac{8}{7}$, so the ratio is less than 1; for $m = 5$ the two sides are $147/400$ and $2/5$. The proof of the lemma uses $\psi(V)/16 \geq \frac{1}{2}$ for the Sylow 2-subgroup V , whereas $\psi(C_2 \times C_2) = 7$. The claim $M \geq 3$ is therefore unproved; if it holds, Theorem 1.1 below gives $\omega(n) \geq 16001$ for every composite solution.

Less is known about (1.2). The comments to sequences A050474 and A203966 of the OEIS [29] report that the M.Sc. thesis of Wong [37] shows that a further solution of $n + 1 = 2\varphi(n)$ would need at least eight prime factors, that there are no further solutions of $n + 1 = 2\varphi(n)$ below 10^{31} and none of (1.2) below $1.6 \cdot 10^{30}$ (Aleksyev), and that $(n + 1)/\varphi(n) = 3$ would force at least 33 prime factors. None of this has appeared in a refereed paper, and we know of no refereed bound on $\omega(n)$ for (1.2).

1.2. Results. We write $\omega(n)$ for the number of distinct prime factors of n . Our first theorem goes one prime further than the bound announced by Renze and by Pinch, and its proof includes a complete proof of that bound.

Theorem 1.1. *Let n be composite with $\varphi(n) \mid n - 1$. Then $\omega(n) \geq 16$. If $(n - 1)/\varphi(n) \geq 3$, then $\omega(n) \geq 16001$, and if $3 \mid n$, then $\omega(n) > 10^8$.*

³Heath-Brown proved that an odd perfect number with k distinct prime factors is less than 4^{4^k} , and Nielsen lowered this to 2^{4^k} . The lemma of Cook and Nielsen concerns integers $x_1 \leq \dots \leq x_m$ for which $\prod (1 - 1/x_i)$ lies just below a given fraction; we state and sharpen it in Section 6.1.

⁴By Korselt's criterion a composite n satisfies $a^{n-1} \equiv 1 \pmod{n}$ for every a prime to n exactly when n is squarefree and $p - 1 \mid n - 1$ for every prime $p \mid n$. A solution of (1.1) is squarefree by Lemma 2.1, and $p - 1 \mid \varphi(n) \mid n - 1$.

⁵Giuga conjectured in 1950 that an integer $n > 1$ with $1^{n-1} + 2^{n-1} + \dots + (n - 1)^{n-1} \equiv -1 \pmod{n}$ is prime. He showed that a composite n satisfies this congruence exactly when $p \mid n/p - 1$ and $p - 1 \mid n/p - 1$ for every prime $p \mid n$; the second condition makes n a Carmichael number.

Apart from Norman's claim, we know of no earlier claim beyond fifteen prime factors in general. The second assertion raises the bound 1991 of Hags for the quotient 3 to 16001, and for $3 \nmid n$ it also improves the bounds of Grytczuk and Wójtowicz for the quotients from 4 to 7. For $3 \mid n$ the third assertion raises the bound of Bursi, Czirbusz and Farkas from 40000000 to more than 10^8 . Our second theorem bounds the size of a solution in terms of the number of its prime factors.

Theorem 1.2. *Let n be composite with $\varphi(n) \mid n - 1$, let $k = \omega(n)$, and put $M = (n - 1)/\varphi(n)$.*

- (i) $n < 2^{2^{k-7}}$.
- (ii) If $M \geq 3$, then $n < 2^{2^{k-15981}}$.
- (iii) If $3 \mid n$, then $n < 2^{2^{k-10^8}}$.

Part (i) divides the number of binary digits allowed by the bound of Burek and Żmija by $2^7 = 128$. Together with Theorem 1.1 it shows that a solution with sixteen prime factors, the first case not yet excluded, lies below $2^{512} < 1.35 \cdot 10^{154}$.

The other results concern (1.2).

Theorem 1.3. *The solutions of $\varphi(n) \mid n + 1$ with $\omega(n) \leq 8$ are exactly the nine numbers (1.3).*

Theorem 1.4. *Every solution of $\varphi(n) \mid n + 1$ with $n > 3$ and $3 \nmid n$ has $\omega(n) \geq 16$.*

Theorem 1.5. *Let $n > 3$ satisfy $\varphi(n) \mid n + 1$ with $(n + 1)/\varphi(n) \geq 3$. Then $\omega(n) \geq 16001$, and $\omega(n) > 10^8$ if $3 \mid n$.*

The next theorem explains the list (1.3). Say that n is a *Fermat-type* solution if $n + 1 = 2\varphi(n)$; by convention $n = 1$ is one.

Theorem 1.6. *Let n_0 be a Fermat-type solution.*

- (i) *For a prime $q \nmid n_0$, the number n_0q is a Fermat-type solution if and only if $q = n_0 + 2$.*
- (ii) *For primes $p < q$ not dividing n_0 , the number n_0pq is a Fermat-type solution if and only if*

$$(p - n_0 - 1)(q - n_0 - 1) = n_0^2 + n_0 + 1.$$

- (iii) *The Fermat-type solutions in (1.3), which are all of them except 2, are exactly the numbers obtained from 1 by repeated use of (i) and (ii): every such extension of one of them is again in (1.3).*

Theorem 1.3 contains the bound reported from Wong's thesis and goes one prime factor further: a solution of (1.2) outside (1.3) has at least nine prime factors, whatever the quotient $(n + 1)/\varphi(n)$. Theorem 1.5 raises the 33 prime factors of the observation on the quotient 3 recorded in the OEIS to 16001. Part (iii) of Theorem 1.6 places 83623935 in the same family as the products of Fermat primes: it is the two-prime extension of 255 coming from the factorisation $255^2 + 255 + 1 = 65281 = 97 \cdot 673$, just as 4294967295 comes from the trivial factorisation $65281 = 1 \cdot 65281$ (Figure 15).

The last result of this section is not new. It follows from the bound of Meijer [22], and Yamada [38] proved it with $c = 1/16.03235$ when $\omega(n) \geq 4$. We give a short proof by the large sieve in Section 2.2, because the estimate behind it explains the size of the bounds in Theorems 1.1 and 1.5.

Theorem 1.7 (Meijer, Yamada). *There is an absolute and effective constant $c > 0$ with the following property. Let n be composite with $\varphi(n) \mid n - 1$, or let $n > 3$ with $\varphi(n) \mid n + 1$, and let M be the quotient $(n - 1)/\varphi(n)$ or $(n + 1)/\varphi(n)$. Then*

$$\omega(n) \geq \exp(\exp(cM)) - 2.$$

The first primes alone give only $\omega(n) \geq \exp(c'M)$, since the product of $p/(p-1)$ over the first K primes grows like $\log K$. Theorem 1.7 uses the fact that no prime factor of a solution divides another prime factor minus one, and it is proved by the large sieve (Section 2.2). It explains why the bounds 16001 and 10^8 in Theorems 1.1 and 1.5 lie so far above the bounds 33 and 1540 that the first primes give.

1.3. Theory and computation. Every search in this paper rests on a theorem, proved by hand, that reduces the statement to finitely many cases and shows that the search covers all of them. The results divide as follows.

Proved entirely by hand. Theorem 1.7 and Meijer's bound (Theorem 2.7), by the large sieve; the sharp product lemma (Theorem 6.2) and the size bound $n < 2^{2^{k-2}}$ (Proposition 6.11); the reductions of Section 2, up to the evaluation of the explicit finite products in Lemmas 2.2 and 2.3; parts (i) and (ii) of Theorem 1.6; and the structural results of Section 9 (Propositions 9.1, 9.4, 9.6 and 9.7, Lemmas 9.2, 9.8, 9.9, 9.11 and 9.12, Proposition 9.10(i), and the equivalence of Remark 9.16).

Proved by hand up to a finite computation.

- $\omega(n) \geq 16$ in Theorem 1.1, and Theorem 1.4: the interval bounds, the divisibility condition for the last two primes and the completeness of the search (Propositions 3.2 and 3.5, Theorem 3.6, Corollary 3.3), and the reduction of the last three primes to divisors in a residue class (Proposition 4.1, Theorem 4.4, Proposition 4.5), leave a search over at most fifteen primes, which ends in 33865004 two-prime problems for fifteen (Section 5).
- The bounds 16001 and 10^8 in Theorems 1.1 and 1.5: Corollary 2.6 reduces them to the branch-and-bound of Proposition 2.5.
- Theorem 1.2: Corollary 6.10 reduces part (i) to the tree $\mathcal{T}_7$ of 33678 nodes (Theorem 6.15), and parts (ii) and (iii) to the evaluation of two explicit rational numbers.
- Theorem 1.3: the same reductions leave 228260561 two-prime problems for eight primes (Section 7.4).
- Theorem 1.6(iii): the factorisation of eight integers.
- Theorems 9.3 and 9.5 and Propositions 9.10(ii), (iii), 9.13 and 9.14: searches of the same kind, and the check of one explicit tuple.

Every search was carried out by at least two programs written independently, and every count reported here is the same for all of them (Section 8 and Appendix A). A search that finds nothing proves something only if it would have found what it looks for, and the programs were therefore also run on equations that have solutions.

1.4. Scope. Theorems 1.1–1.7 bound the number of prime factors and the size of a new solution and describe the known ones; they do not decide whether (1.1) has a composite solution or whether (1.2) has solutions beyond (1.3). Section 9 estimates the cost of the next cases. For (1.1) with sixteen prime factors the number of two-prime problems of the kind solved here grows from $3.4 \cdot 10^7$ to about $9 \cdot 10^{13}$, and for (1.2) with nine prime factors and $3 \mid n$ the Fermat primes alone leave some 10^{17} of them. In both cases most of the cost comes from a few prefixes whose product $\prod p/(p-1)$ differs from 2 by less than 10^{-7} . Section 9 also shows that the product equation, the interval bounds and the congruence conditions used by the search are satisfied by tuples of integers that are not all prime (Proposition 9.1), so a proof for all k has to use primality in an essential way. These tuples all have two entries divisible by 3. Tuples with no entry divisible by 3 do not occur for $k \leq 13$, nor for $k \leq 15$ when all entries but the last three are prime (Theorem 9.3), and no congruence condition other than that of Lemma 2.2 can exclude them (Proposition 9.4). They do occur, for every $k \geq 25$ with the sign +1 and every $k \geq 26$ with the sign −1 (Theorem 9.5), so for Lehmer's equation, and for the companion equation with $3 \nmid n$,

a proof for all k has to use the primality of entries other than 3. Section 9.7 proves two facts about the structure of the problem: the last three primes, unlike the last two, cannot be read off from the divisors of a single integer (Proposition 9.7), and an entry that passes Fermat's test for the exponent $n + \varepsilon$ is squarefree and satisfies Korselt's condition (Lemma 9.8), which the tuples of Theorem 9.5 violate. Section 9.8 shows that for Lehmer's equation with $M = 2$ the first $k - 1$ prime factors determine the last through a single divisibility, and that for $k \leq 15$ no solution exists even when the largest entry need only be an integer prime to the others (Proposition 9.10); for the companion equation such a solution exists with six entries. Section 9.9 shows that a small prime can divide the modulus C of this divisibility, the last defect, only when the solution has many prime factors: $5 \mid C$ needs at least 201 of them, and a solution with sixteen prime factors has no prime factor of C below 29 (Corollary 9.15). Remark 9.16 states the assertion, equivalent to Lehmer's conjecture, that this paper does not prove.

1.5. Method. The bound $\omega(n) \geq 16$ of Theorem 1.1 and Theorems 1.3 and 1.4 rest on three reductions proved in Sections 2 to 4. The first (Section 2) shows that one may assume $n + \varepsilon = 2\varphi(n)$, with $\varepsilon = -1$ for Lehmer's equation and $\varepsilon = +1$ for the companion, and, except in Theorem 1.3, that $3 \nmid n$. The second (Section 3) shows that once the first j prime factors are fixed, the next one lies in an interval whose ends are explicit functions of them (Proposition 3.2), that for each k only finitely many sequences of primes satisfy these conditions (Corollary 3.3), and that the last two prime factors are determined by a divisibility condition (Proposition 3.5). Each value of k is therefore a finite problem, and the traversal of an explicit finite tree finds every solution (Theorem 3.6). The bounds for the two signs of ε differ only by terms of size $1/\varphi(n)$; for $p_1 \geq 5$ they give identical trees for every $k \leq 14$, and for $k = 15$ the same nodes down to depth 13 (Figure 8). The third reduction (Section 4) treats the last three primes together: once p_{k-2} is fixed, the last two correspond to a divisor of an explicit integer N in a known residue class modulo an integer c (Proposition 4.1). Such divisors are the lattice points of a coset near a hyperbola (Lemmas 4.2 and 4.3). Since the two complementary divisors lie in the same class, their sum lies in a single class modulo c , which reduces the problem to one square test for each value of the sum (Section 4.4), and when $c^3 \geq N + 2c^2$ there are at most four such divisors (Proposition 4.5). With these reductions the case $k \leq 14$ becomes a tree of at most 31962 nodes and the case $k = 15$ a set of 33865004 problems of this kind, which we solved by computer. One of the routes never factors an integer, so that for every $k \leq 15$ the conclusion depends on no primality proof.

The companion equation with eight prime factors and $3 \mid n$ behaves differently. After the Fermat primes 3, 5, 17, 257, 65537 the product $\prod p/(p - 1)$ falls short of 2 by only 2^{-31} , the interval for the sixth prime has length 2^{33} , and each of the $1.3 \cdot 10^8$ admissible choices of that prime leaves a two-prime problem whose modulus is far below the cube root of the integer to be split. We treat these problems by the sum route of Section 4.4, combined with trial division for the small divisors and sieved by congruences that primes satisfy, and we settle the 1967265 cases in which this would be slow by factoring the integer completely, with every prime factor proved prime (Section 7.4). Theorem 1.6 rests on a short identity together with the factorisation of eight integers.

The bounds for a quotient of at least 3 and for $3 \mid n$, in Theorems 1.1 and 1.5, come from independence. The prime factors of a solution form a set in which no prime divides another minus one, and over such a set the product of $p/(p - 1)$ grows only like the double logarithm of its size (Theorem 2.7). A branch-and-bound over these sets (Proposition 2.5) shows that such a set must be very large before the product reaches 3, as it must when the quotient is at least 3 and $3 \nmid n$, or reaches $\frac{8}{3}$ with all its primes congruent to 2 modulo 3, as it must for the prime factors other than 3 when $3 \mid n$.

Theorem 1.2 is proved in Section 6. Its first ingredient is a sharp form of the lemma of Cook and Nielsen (Theorem 6.2), in which the bound for the product of the remaining prime factors depends on how far the product $\prod p/(p-1)$ over the prime factors already chosen falls short of M . This lemma alone gives $n < 2^{2^{k-2}}$ (Proposition 6.11), and more generally $n < 2^{2^{k-s}}$ for every solution whose first prime factors are small in a precise sense. In every other solution the prime factors grow rapidly, and for $s = 7$ an exhaustive search over their possible initial segments, which visits 33678 of them, shows that there are none. Parts (ii) and (iii) combine the lemma of Cook and Nielsen with Proposition 2.5.

The paper is organised as follows. Section 2 proves the reductions, the bounds for large quotients and for $3 \mid n$, Theorem 1.5, and, by the large sieve, Theorem 1.7. Section 3 proves the interval bounds and the completeness of the search, and Section 4 the reduction of the last three primes to divisors in a residue class. Section 5 completes the proof of Theorem 1.1, Section 6 proves Theorem 1.2, and Section 7 proves Theorems 1.3, 1.4 and 1.6. Section 8 summarises the checks on the computations, whose details are in Appendix A, and Section 9 treats the next cases, the pseudo-solutions and the structure of the equation, and ends with the assertion that remains open (Remark 9.16).

Notation. Throughout, $n = p_1 p_2 \cdots p_k$ with primes $p_1 < \cdots < p_k$, and $\varepsilon \in \{-1, +1\}$ selects the equation $\varphi(n) \mid n + \varepsilon$; we put $M = (n + \varepsilon)/\varphi(n)$. For $0 \leq j \leq k$ let

$$A_j = \prod_{i \leq j} p_i, \quad B_j = \prod_{i \leq j} (p_i - 1), \quad P_j = \frac{A_j}{B_j} = \prod_{i \leq j} \frac{p_i}{p_i - 1},$$

with $A_0 = B_0 = P_0 = 1$ and $p_0 = 1$. We write $m = k - j$ for the number of primes still to be chosen at depth j , and $r_1 < r_2 < \cdots$ for the primes from 5 on.

2. REDUCTIONS

Lemma 2.1. *Let $n > 1$ satisfy $\varphi(n) \mid n + \varepsilon$ and put $M = (n + \varepsilon)/\varphi(n)$. Suppose n is composite if $\varepsilon = -1$, and $n > 3$ if $\varepsilon = +1$.*

- (i) *n is odd, squarefree and composite, and $M \geq 2$.*
- (ii) *For primes p, q dividing n , possibly equal, $p \nmid q - 1$ and $p \nmid M$.*
- (iii) $\prod_{p \mid n} \frac{p}{p-1} = M - \frac{\varepsilon}{\varphi(n)}.$

Proof. If $n > 2$ is even then $\varphi(n)$ is even and $n + \varepsilon$ is odd. If $p^2 \mid n$ then p divides $\varphi(n)$, hence $n + \varepsilon$, which is impossible since $p \mid n$. For $\varepsilon = +1$ a prime solution satisfies $n - 1 \mid 2$, so $n \leq 3$; thus n is composite in both cases. For odd squarefree composite n we have $\varphi(n) < n - 1$, so $M \geq 2$. If $p, q \mid n$ then $q - 1 \mid \varphi(n) \mid n + \varepsilon$ while $p \mid n$, so $p \nmid q - 1$; similarly $p \nmid M\varphi(n) = n + \varepsilon$. Finally $n/\varphi(n) = (n + \varepsilon)/\varphi(n) - \varepsilon/\varphi(n)$. $\square$

The next lemma shows how the prime 3 enters, and that it does so differently for the two equations.

Lemma 2.2. *In the situation of Lemma 2.1, suppose $3 \mid n$. Then every other prime factor of n is congruent to 2 modulo 3, and*

$$M \equiv \begin{cases} 1 & (\text{mod } 3), \quad \varepsilon = -1, \\ 2 & (\text{mod } 3), \quad \varepsilon = +1. \end{cases}$$

Consequently $M \geq 4$ if $\varepsilon = -1$, and $M = 2$ or $M \geq 5$ if $\varepsilon = +1$. If $M \geq 4$ then $\omega(n) \geq 1540$.

Proof. For a prime $p \mid n$ with $p \neq 3$, Lemma 2.1(ii) gives $3 \nmid p-1$, so $p \equiv 2$ and $p-1 \equiv 1 \pmod{3}$. Hence $\varphi(n) = 2 \prod_{p \neq 3} (p-1) \equiv 2 \pmod{3}$, while $n + \varepsilon \equiv \varepsilon \pmod{3}$. The relation $n + \varepsilon = M\varphi(n)$ becomes $2M \equiv \varepsilon$, which is the stated congruence. Suppose $M \geq 4$. Then $n/\varphi(n) > 4$: for $\varepsilon = -1$ because $n/\varphi(n) = M + 1/\varphi(n)$ by Lemma 2.1(iii), and for $\varepsilon = +1$ because then $M \geq 5$ and $n/\varphi(n) = M - 1/\varphi(n) \geq 5 - \frac{1}{2}$. On the other hand, if $q_1 < q_2 < \dots$ are the primes congruent to 2 modulo 3 from 5 on, then

$$\frac{n}{\varphi(n)} = \frac{3}{2} \prod_{p \mid n, p \neq 3} \frac{p}{p-1} \leq \frac{3}{2} \prod_{i=1}^{k-1} \frac{q_i}{q_i-1},$$

since $p/(p-1)$ decreases in p . An exact computation in rational arithmetic shows that the right-hand side is $3.99985\dots$ for $k = 1539$ and first exceeds 4 at $k = 1540$, where the last prime in the product is $q_{1539} = 27947$. $\square$

For $\varepsilon = -1$ this is a weak form of the bounds of Hagis and of Burcsi, Czirbusz and Farkas mentioned in the introduction, and Corollary 2.6 below goes much further, but the lemma needs no search and suffices wherever we use it. For $\varepsilon = +1$ it shows why 3 cannot be removed in the same way: $M = 2$ is compatible with $3 \mid n$, and each Fermat prime 5, 17, 257, 65537 is congruent to 2 modulo 3, as are 353 and 929.

Lemma 2.3. *In the situation of Lemma 2.1, suppose $3 \nmid n$ and $\omega(n) \leq 32$. Then $M = 2$.*

Proof. All prime factors are at least 5. By Lemma 2.1(iii) and monotonicity,

$$M = \frac{n}{\varphi(n)} + \frac{\varepsilon}{\varphi(n)} \leq \prod_{i \leq k} \frac{r_i}{r_i-1} + \frac{1}{\prod_{i \leq k} (r_i-1)},$$

where $k = \omega(n)$, since $n/\varphi(n) = \prod_{p \mid n} p/(p-1)$ and $\varphi(n) = \prod_{p \mid n} (p-1)$, and the i -th smallest prime factor of n is at least r_i . An exact computation shows that the right-hand side is less than 3 for every $k \leq 32$. (For $k = 32$ the product is $2.99315\dots$ and the second term is below 10^{-53} ; at $k = 33$ the product exceeds 3.) Since M is an integer and $M \geq 2$ by Lemma 2.1(i), $M = 2$. $\square$

Corollary 2.4. *Let $n > 3$ satisfy $\varphi(n) \mid n+1$ with $\omega(n) \leq 7$. Then $n+1 = 2\varphi(n)$.*

Proof. Let $k = \omega(n) \geq 2$. The prime factors of n , in increasing order, are at least 3, $r_1, \dots, r_{k-1}$, and Lemma 2.1(iii) gives, as in the proof of Lemma 2.3,

$$M \leq \frac{3}{2} \prod_{i < k} \frac{r_i}{r_i-1} + \frac{1}{2 \prod_{i < k} (r_i-1)}.$$

The maximum of the right-hand side over $2 \leq k \leq 7$ is $2.92357\dots < 3$, attained at $k = 7$, and so $M = 2$. $\square$

2.1. Independent sets of primes. Call a finite set of primes *independent* if none of its elements divides another minus one. Golomb [14] and Meijer [22] call an increasing sequence of odd primes with this property a G -sequence. By Lemma 2.1(ii) the prime factors of a solution form an independent set, whereas the proofs above used only that the i -th smallest of them is at least the i -th prime of a given list. Independence is a much stronger condition. Each prime q of the set excludes the primes congruent to 1 modulo q , which make up the proportion $1/(q-1)$ of all primes by the prime number theorem for arithmetic progressions, and so the product of $p/(p-1)$ over an independent set grows far more slowly with its size than the product over the first primes. Choosing at each step the least prime from 5 on that the primes already chosen allow gives the independent set

5, 7, 13, 17, 19, 23, 37, 59, 67, 73, 83, $\dots$

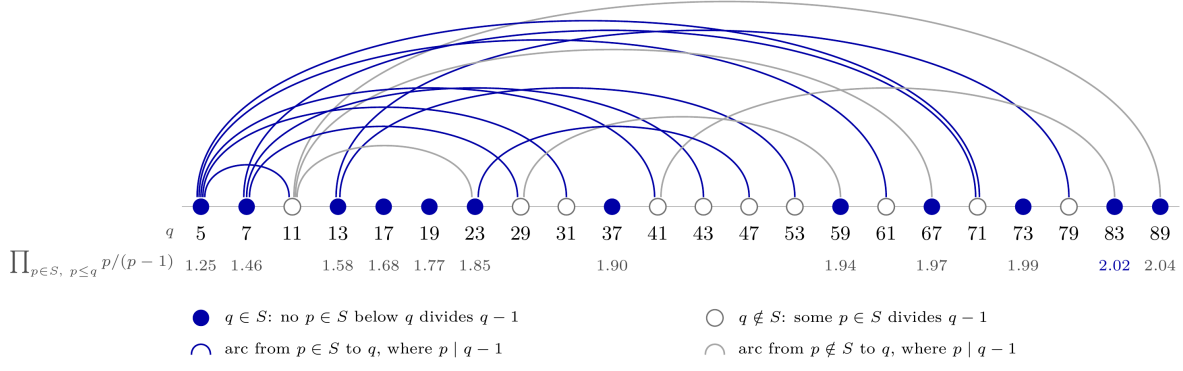


FIGURE 1. The primes from 5 to 89, with an arc from p to q whenever p divides $q - 1$; an independent set contains no two primes joined by an arc. The set S of filled primes is chosen one prime at a time, each the least prime joined to none of those chosen before it. Below each prime of S is the product of $p/(p - 1)$ over the primes of S up to it, which first exceeds 2 at 83, the eleventh.

(Figure 1). Its product passes 2 at the eleventh prime, where the first primes from 5 on need only seven, and it passes 3 only at the 100470-th prime, 5160959, where they need 33 (Figure 2).

- Proposition 2.5.** (i) *No independent set of at most 16000 primes $p \geq 5$ has $\prod p/(p - 1) \geq 2.999999$.*
(ii) *No independent set of at most 10^8 primes $p \equiv 2 \pmod{3}$ with $p \geq 5$ has $\prod p/(p - 1) \geq \frac{8}{3}(1 - 10^{-6})$.*
(iii) *No independent set of at most 10^8 primes $p \equiv 2 \pmod{3}$ with $p \geq 5$ has $\prod p/(p - 1) \geq \frac{10}{3}(1 - 10^{-6})$.*

Proof. Let Q be the set of primes allowed, K the size and V the target. For an independent set $S \subseteq Q$ whose largest element is p (with $p = 3$ if S is empty), call the primes $q \in Q$ with $q > p$ and $q \not\equiv 1$ modulo every element of S the *candidates* for S . If $|S| < K$, let $B(S)$ be the product of $q/(q - 1)$ over S and over the $K - |S|$ smallest candidates for S . An independent set T of at most K primes of Q whose $|S|$ smallest elements form S has its other elements among the candidates for S , and there are at most $K - |S|$ of them; as $q/(q - 1)$ decreases in q , the product over T is at most $B(S)$.

The search starts from the empty set and treats each set S that it creates as follows. If the product over S is at least V , it stops and reports S . If $|S| = K$ or $B(S) < V$, it discards S . Otherwise it creates the sets $S \cup \{q\}$ for the candidates q with

$$\prod_{p \in S} \frac{p}{p - 1} \cdot \frac{q}{q - 1} \cdot \prod_{q'} \frac{q'}{q' - 1} \geq V, \quad (2.1)$$

where q' runs over the $K - |S| - 1$ candidates for S that follow q . The left-hand side of (2.1) is at least $B(S \cup \{q\})$, because every candidate for $S \cup \{q\}$ is a candidate for S , and it decreases as q increases, so the sets created from S come from its first few candidates. Let T be as above with product at least V . Every initial segment S of T has $B(S) \geq V$, and if $S \cup \{q\}$ is also an initial segment, then q satisfies (2.1); so the search creates the initial segments of T one after another until it stops.

In each of the three cases the search ends without reporting a set. It creates 90320 sets in case (i), the largest of them with 46 elements, and 9 sets in case (ii) and 2 in case (iii). Two programs written independently, one of them in PARI/GP [30], carry out the three searches and

create the same sets. For (ii) and (iii), which involve primes up to $2 \cdot 10^{10}$, they run through the primes once for each set created. The products are handled through their logarithms, and every bound is a sum of a few terms $\log(q/(q-1))$ and a difference of two running sums of such terms. Every running sum here stays below 4; it has fewer than $2 \cdot 10^6$ terms in case (i), where the primes below $3 \cdot 10^7$ suffice, and fewer than $1.1 \cdot 10^8$ in cases (ii) and (iii). In double precision each term is computed with an absolute error below $3 \cdot 10^{-16}$ and each addition with one below 2^{-52} , so a bound carries an error below 10^{-8} in case (i) and below $2 \cdot 10^{-7}$ in cases (ii) and (iii). The PARI/GP programs work with 57 significant digits in case (i) and 38 in cases (ii) and (iii), and their errors are below 10^{-45} and 10^{-25} . A set is discarded, or a candidate passed over, only when exact rational arithmetic confirms it or when the logarithm of its bound lies below $\log V$ by more than a margin that exceeds these errors: 10^{-7} in case (i) and 10^{-6} in cases (ii) and (iii) in double precision, and 10^{-40} and 10^{-20} in PARI/GP. Rounding can therefore add sets to the search, but never remove one. $\square$

The search for (i) grows quickly with K : it creates 2567 sets for $K = 10^4$ and 19711 for $K = 14000$. The least size of an independent set of primes from 5 on with product at least 3 therefore lies between 16001 and 100470, the length of the set chosen one prime at a time above. For the primes $p \equiv 2 \pmod{3}$ the products grow much more slowly: the independent set chosen from them in the same way has 2307386 elements below $2 \cdot 10^8$, and its product is only $2.0703 \dots$, far below $\frac{8}{3}$.

Corollary 2.6. *Let n be as in Lemma 2.1, with $M = (n + \varepsilon)/\varphi(n)$.*

- (i) *If $M \geq 3$ and $3 \nmid n$, then $\omega(n) \geq 16001$.*
- (ii) *If $3 \mid n$ and $\varepsilon = -1$, then $\omega(n) \geq 10^8 + 2$.*
- (iii) *If $3 \mid n$, $\varepsilon = +1$ and $M \geq 3$, then $\omega(n) \geq 10^8 + 2$.*

Proof. Let $k = \omega(n)$. By Lemma 2.1 the prime factors of n form an independent set, and $n/\varphi(n) = M - \varepsilon/\varphi(n)$.

(i) The prime factors are at least 5, and $n/\varphi(n) \geq 3 - 1/\varphi(n) > 2$. The product of $p/(p-1)$ over the six smallest primes from 5 on is $1.9490 \dots$, so $k \geq 7$ and $\varphi(n) \geq 4 \cdot 6 \cdot 10 \cdot 12 \cdot 16 \cdot 18 \cdot 22 > 10^6$. Hence $n/\varphi(n) > 2.999999$, and Proposition 2.5(i) gives $k > 16000$.

(ii) By Lemma 2.2, $M \geq 4$, and the $k-1$ prime factors other than 3 are congruent to 2 modulo 3. The product of $p/(p-1)$ over them is $\frac{2}{3} n/\varphi(n) = \frac{2}{3}(M + 1/\varphi(n)) > \frac{8}{3}$, so $k-1 > 10^8$ by Proposition 2.5(ii).

(iii) By Lemma 2.2, $M \geq 5$, and the prime factors other than 3 are as in (ii). The product over them is $\frac{2}{3}(M - 1/\varphi(n)) \geq \frac{10}{3} - \frac{2}{3\varphi(n)} \geq 3$, while over the five smallest primes $p \equiv 2 \pmod{3}$ from 5 on it is $1.5818 \dots$. So there are at least six of them, $\varphi(n) \geq 2 \cdot 4 \cdot 10 \cdot 16 \cdot 22 \cdot 28 \cdot 40 > 10^6$, and the product exceeds $\frac{10}{3}(1 - 10^{-6})$. Proposition 2.5(iii) gives $k-1 > 10^8$. $\square$

For $\varepsilon = -1$, part (ii) improves the bound $\omega(n) \geq 40000000$ of Burcsi, Czirbusz and Farkas [6], who completed sequences of primes $p \equiv 2 \pmod{3}$ in a similar way by the least primes allowed. Part (i) replaces the bound 33 that the products over the first primes give, and for $M = 3$ the bound 1991 of Hagis [17]; part (iii) replaces the bound 1540 that the products give.

Proof of Theorem 1.5 and of the last two assertions of Theorem 1.1. Let n satisfy (1.1) or (1.2), with n composite in the first case and $n > 3$ in the second. If $3 \nmid n$ and $M \geq 3$, then $\omega(n) \geq 16001$ by Corollary 2.6(i). If $3 \mid n$, then $\omega(n) > 10^8$ by Corollary 2.6(ii) for (1.1), where $M \geq 4$ holds automatically, and by Corollary 2.6(iii) for (1.2) when $M \geq 3$. $\square$

By Corollary 2.6, a solution with at most 16000 prime factors has $M = 2$, and if it solves (1.1) it is prime to 3. The searches below concern the case $n + \varepsilon = 2\varphi(n)$.

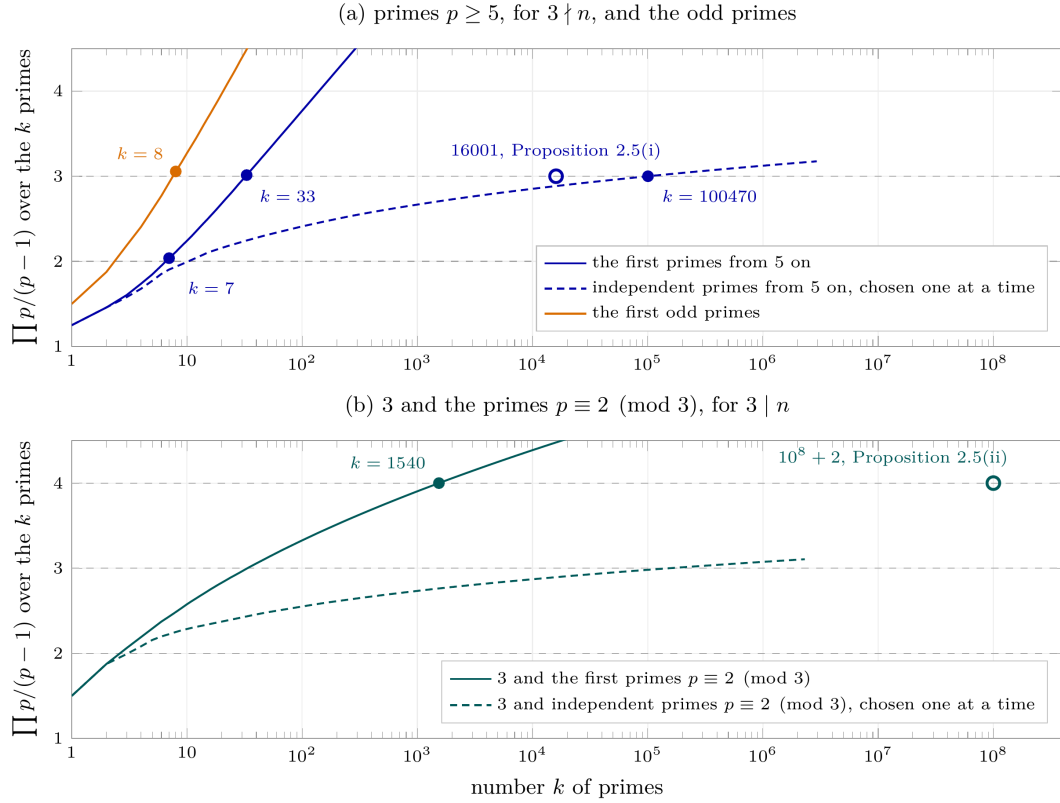


FIGURE 2. The product of $p/(p-1)$ over the first k primes of a list (solid) and over the independent set chosen from the same list one prime at a time (dashed), for the primes below $2 \cdot 10^8$, with k on a logarithmic scale. Where a list has both curves, the largest product over an independent set of k of its primes lies between them. (a) For the primes from 5 on the product passes 2 at $k = 7$, so a solution of (3.1) prime to 3 has at least seven prime factors (Section 5), and 3 at $k = 33$ (Lemma 2.3); for the odd primes it passes 3 at $k = 8$, where the argument of Corollary 2.4 stops. The independent set passes 3 only at $k = 100470$, and no independent set of primes from 5 on reaches 3 before $k = 16001$ (open circle). (b) For 3 and the primes $p \equiv 2 \pmod{3}$ the product passes 4 at $k = 1540$ (Lemma 2.2), and no independent set formed by 3 and such primes reaches 4 before $k = 10^8 + 2$ (open circle).

2.2. The growth of independent sets. Proposition 2.5 needs a search because the product of $p/(p-1)$ over an independent set grows very slowly with the size of the set. Over the first K primes the product is asymptotic to $e^\gamma \log K$ by Mertens' theorem, so that the products alone give only $\omega(n) \geq \exp(c'M)$ for some $c' > 0$. Over an independent set of K primes the product is at most a constant multiple of $\log \log K$, and this forces $\omega(n)$ to be at least doubly exponential in M (Theorem 1.7).

For a set S of primes and $x \geq 2$ put

$$P_S(x) = \prod_{p \in S, p \leq x} \frac{p}{p-1}, \quad \delta_S(x) = \prod_{p \in S, p \leq x} \left(1 - \frac{1}{p-1}\right).$$

An independent set with at least two elements does not contain 2, since 2 divides $q - 1$ for every odd prime q . For a set S of odd primes every factor of $\delta_S(x)$ is at least $\frac{1}{2}$, and since $\frac{p}{p-1}(1 - \frac{1}{p-1}) = 1 - \frac{1}{(p-1)^2}$,

$$\delta_S(x) P_S(x) \leq 1. \quad (2.2)$$

Theorem 2.7 (Meijer [22]). *There is an absolute constant c_0 such that every independent set S of $K \geq 3$ primes satisfies*

$$\prod_{p \in S} \frac{p}{p-1} \leq c_0 \log \log K.$$

Meijer proved the bound $P_S(x) \leq c \log \log x$ with the Brun–Selberg sieve, and Yamada [38] gave explicit constants with the large sieve. We include a short proof by the large sieve, with constants left unspecified. Each element p of S excludes from S the primes congruent to 1 modulo p , so that a prime above x can belong to S only if it avoids one residue class modulo every element of S up to x ; the large sieve bounds the number of such primes by $\delta_S(x)$ times the number of all primes in the range.

Lemma 2.8. *Let S be an independent set of odd primes and $N \geq 2^{88}$. Then*

$$\#\{q \in S : N^{1/44} < q \leq N\} \leq \frac{132 N \delta_S(N^{1/44})}{\log N}.$$

Proof. Put $Q = N^{1/2}$ and $z = N^{1/44} = Q^{1/22}$, so that $z \geq 4$. For a prime $p \leq z$ let Ω_p consist of the residue class 0 modulo p and, if $p \in S$, also of the class 1; for $z < p \leq Q$ let Ω_p be empty. Let $\omega(p) < p$ be the number of classes in Ω_p . A prime $q \in S$ with $z < q \leq N$ lies in none of these classes: it is not divisible by a prime $p \leq z < q$, and for $p \in S$ it is not congruent to 1 modulo p , because S is independent. The large sieve [24, 25] states that a set of integers in $[1, N]$ that avoids the classes of Ω_p for every prime $p \leq Q$ has at most $(N + Q^2)/L$ elements, where

$$L = \sum_{q \leq Q} \mu^2(q) G(q), \quad G(q) = \prod_{p|q} g(p), \quad g(p) = \frac{\omega(p)}{p - \omega(p)}.$$

Here $g(p) = 1/(p-1)$ for $p \leq z$ with $p \notin S$, $g(p) = 2/(p-2)$ for $p \leq z$ with $p \in S$, and $g(p) = 0$ for $p > z$; in every case $g(p) \leq 6/p$.

Let Π be the product of the primes up to z and put $\sigma = 1/\log z$, so that $Q^\sigma = e^{22}$. Then L is the sum of $G(q)$ over the divisors $q \leq Q$ of Π . A divisor $q > Q$ has $(q/Q)^\sigma > 1$, and so

$$\sum_{q|\Pi, q > Q} G(q) \leq Q^{-\sigma} \sum_{q|\Pi} G(q) q^\sigma = e^{-22} \prod_{p \leq z} (1 + g(p)p^\sigma).$$

For $p \leq z$ we have $0 < \sigma \log p \leq 1$, hence $p^\sigma - 1 \leq (e-1)\sigma \log p$ by convexity, and $1 + g(p)p^\sigma \leq (1 + g(p)) \exp(g(p)(p^\sigma - 1))$. With $g(p) \leq 6/p$ and the elementary bound⁶ $\sum_{p \leq z} (\log p)/p < \log z + \log 4$ this gives

$$\prod_{p \leq z} (1 + g(p)p^\sigma) \leq \exp\left(6(e-1)\left(1 + \frac{\log 4}{\log z}\right)\right) \prod_{p \leq z} (1 + g(p)) < e^{20.7} \prod_{p \leq z} (1 + g(p)),$$

because $z \geq 4$ and $12(e-1) < 20.7$. Since $\prod_{p \leq z} (1 + g(p)) = \sum_{q|\Pi} G(q)$ and $e^{20.7-22} < \frac{1}{3}$, the divisors $q > Q$ contribute less than a third of this sum, and $L > \frac{2}{3} \prod_{p \leq z} (1 + g(p))$. Finally

⁶Let $m = \lfloor z \rfloor$. Legendre's formula gives $\log m! \geq \sum_{p \leq m} \lfloor m/p \rfloor \log p > m \sum_{p \leq m} (\log p)/p - \sum_{p \leq m} \log p$, while $\log m! \leq m \log m$ and $\sum_{p \leq m} \log p < m \log 4$, because the product of the primes up to m is less than 4^m . Dividing by m gives the bound.

$1 + g(p) = p/(p-1)$ for $p \notin S$ and $1 + g(p) = \frac{p}{p-1} \left(1 - \frac{1}{p-1}\right)^{-1}$ for $p \in S$, so that

$$\prod_{p \leq z} (1 + g(p)) = \frac{1}{\delta_S(z)} \prod_{p \leq z} \left(1 - \frac{1}{p}\right)^{-1} > \frac{\log z}{\delta_S(z)},$$

since the product $\prod_{p \leq z} (1 - 1/p)^{-1}$ is at least $\sum_{n \leq z} 1/n > \log z$. With $\log z = (\log N)/44$ we obtain $L > \log N/(66 \delta_S(z))$, and the large sieve, with $N + Q^2 = 2N$, gives the lemma. $\square$

Lemma 2.9. *Let S be an independent set of odd primes and $y \geq 2^{88}$. Then*

$$\sum_{q \in S, y < q \leq y^2} \frac{1}{q} \leq 269 \delta_S(y^{1/44}).$$

Proof. Let $a = \lfloor \log_2 y \rfloor \geq 88$ and $b = \lceil 2 \log_2 y \rceil$, so that $b \leq 2a + 2$. The interval $(y, y^2]$ is covered by the intervals $(2^{i-1}, 2^i]$ with $a < i \leq b$. For each such i , Lemma 2.8 with $N = 2^i$ applies to the elements of S in $(2^{i-1}, 2^i]$, because $2^{i/44} \leq 2^{i-1}$, and $\delta_S(2^{i/44}) \leq \delta_S(y^{1/44})$, because $2^i > y$ and δ_S is decreasing. Each of these elements exceeds 2^{i-1} , so their reciprocals add up to at most

$$\frac{132 \cdot 2^i \delta_S(y^{1/44})}{2^{i-1} i \log 2} = \frac{264 \delta_S(y^{1/44})}{i \log 2}.$$

Since $\sum_{a < i \leq b} 1/i \leq \log(b/a) \leq \log(2 + 2/88) < 0.705$ and $264 \cdot 0.705 / \log 2 < 269$, the lemma follows. $\square$

Proof of Theorem 2.7. By Mertens' theorem $\prod_{p \leq x} (1 - 1/p)^{-1} = e^\gamma \log x + O(1)$, so there is an absolute constant c_1 with $\prod_{u < p \leq u^{44}} p/(p-1) \leq c_1$ for all $u \geq 2$. Let S be an independent set of odd primes. Then $P_S(y) \leq c_1 P_S(y^{1/44})$ for $y \geq 2^{44}$, and Lemma 2.9 and (2.2) give, for $y \geq 2^{88}$,

$$\sum_{q \in S, y < q \leq y^2} \frac{1}{q} \leq \frac{269}{P_S(y^{1/44})} \leq \frac{269 c_1}{P_S(y)}. \quad (2.3)$$

Put $y_0 = 2^{88}$, $y_{j+1} = y_j^2$ and $a_j = P_S(y_j)$. Since $-\log(1 - 1/q) \leq 2/q$, (2.3) gives $a_{j+1} \leq a_j e^{c_2/a_j}$ with $c_2 = 538 c_1$, and also $a_{j+1} \leq c_1 a_j$. If $a_j \geq c_2$, then $a_{j+1} \leq a_j(1 + 2c_2/a_j) = a_j + 2c_2$, because $e^t \leq 1 + 2t$ for $0 \leq t \leq 1$; if $a_j < c_2$, then $a_{j+1} < c_1 c_2$. By induction $a_j \leq c_3 + 2c_2 j$ for all $j \geq 0$, where $c_3 = \max\{c_1 c_2, \prod_{p \leq 2^{88}} p/(p-1)\} \geq a_0$. For $x > y_0$ let j be the least index with $y_j \geq x$. Then $2^{j-1} \log y_0 < \log x$, and $P_S(x) \leq a_j \leq c_3 + 2c_2(1 + \log_2(\log x / \log y_0))$. Together with $P_S(x) \leq a_0 \leq c_3$ for $x \leq y_0$, this gives $P_S(x) \leq c_4 \log \log x$ for all $x \geq 3$, with an absolute constant c_4 .

Now let S have $K \geq 3$ elements, so that its elements are odd. At most K of them exceed K^2 , and each of those contributes a factor at most $1 + 1/K^2$, so that together they contribute less than $e^{1/K} < 2$. Hence

$$\prod_{p \in S} \frac{p}{p-1} < 2 P_S(K^2) \leq 2 c_4 \log \log K^2 = 2 c_4 (\log 2 + \log \log K) \leq 20 c_4 \log \log K,$$

since $\log 2 < 9 \log \log 3$. $\square$

Proof of Theorem 1.7. Let n be as in Lemma 2.1 and $k = \omega(n) \geq 2$. The prime factors of n form an independent set, and $n/\varphi(n)$ is the product of $p/(p-1)$ over them. If $k \geq 3$, Theorem 2.7 and Lemma 2.1(iii) give $M \leq c_0 \log \log k + 1/\varphi(n) < (c_0 + 3) \log \log(k+2)$, since $\log \log 5 > \frac{1}{3}$. If $k = 2$, then $n/\varphi(n) \leq \frac{3}{2} \cdot \frac{5}{4}$ and $\varphi(n) \geq 8$, so that $M \leq 2 < 7 \log \log 4$. In both cases $\log \log(k+2) \geq cM$ with $c = 1/\max\{c_0 + 3, 7\}$, which is the assertion. $\square$

The constants in Theorems 2.7 and 1.7 are effective but large, and the numerical bounds of Corollary 2.6 come from Proposition 2.5, not from the sieve. What the sieve explains is the shape of those bounds. Heuristically, the elements of S up to x leave a proportion $\delta_S(x)$ of the primes near x available, and a set that takes every available prime gains $\log P_S$ at the rate $\delta_S(x) d \log \log x$, since $\sum_{q \leq x} 1/q$ grows like $\log \log x$. The derivative of P_S with respect to $\log \log x$ is then $\delta_S(x) P_S(x)$, which is at most 1 by (2.2), so that $P_S(x)$ grows at most like $\log \log x$, the rate of Theorem 2.7. The independent sets chosen one prime at a time in Section 2.1 grow at this rate (Table 1). Between $x = 10^3$ and $x = 10^8$ the product of $p/(p-1)$ over the elements up to x differs from a linear function of $\log \log x$ by less than 0.003, with slope 0.82 for the primes from 5 on and 0.41 for the primes $p \equiv 2 \pmod{3}$, which are half of all primes. The first set needs 100470 primes to reach the product 3, where the first primes from 5 on need 33, and the second has not reached $\frac{8}{3}$ below $2 \cdot 10^8$ (Section 2.1). This is why the bounds of Corollary 2.6 exceed 33 and 1540 by so much.

TABLE 1. The independent sets chosen one prime at a time from the primes from 5 on and from the primes $p \equiv 2 \pmod{3}$ from 5 on: the number of elements up to x and the product of $p/(p-1)$ over them.

x	$\log \log x$	primes from 5 on		$p \equiv 2 \pmod{3}$	
		elements	product	elements	product
10^3	1.933	62	2.3430	45	1.6489
10^4	2.220	407	2.5792	293	1.7626
10^5	2.443	2 908	2.7593	2 185	1.8564
10^6	2.626	22 602	2.9078	17 396	1.9328
10^7	2.780	183 730	3.0344	142 723	1.9979
10^8	2.913	1 538 281	3.1447	1 207 989	2.0546

3. INTERVALS FOR THE PRIME FACTORS

Fix $k \geq 2$, a sign ε , and a squarefree odd $n = p_1 \cdots p_k$ with

$$n + \varepsilon = 2\varphi(n). \quad (3.1)$$

By Lemma 2.1(iii),

$$\prod_{i=1}^k \frac{p_i}{p_i - 1} = 2 - \frac{\varepsilon}{\varphi(n)}. \quad (3.2)$$

For $\varepsilon = -1$ the product lies just above 2, and for $\varepsilon = +1$ just below it. Everything in this section is a consequence of (3.2), and the two signs are treated together.

Proposition 3.1. $P_j < 2$ for every $j < k$.

Proof. For $\varepsilon = +1$ this is clear, since $P_j \leq P_k = 2 - 1/\varphi(n)$. For $\varepsilon = -1$, suppose $P_j \geq 2$ with $j < k$. Since $p_{j+1} - 1 < n - 1$ we have $p_{j+1}/(p_{j+1} - 1) > 1 + 1/(n - 1)$, and the remaining factors exceed 1, so $P_k > 2(1 + 1/(n - 1)) = 2 + 1/\varphi(n)$ by (3.1), contradicting (3.2). $\square$

At depth j put $T = 2/P_j$, which exceeds 1 by the proposition, and

$$C_j = 2B_j - A_j, \quad T - 1 = \frac{C_j}{A_j}.$$

Thus C_j is a positive integer, and C_j/A_j measures how far the primes chosen so far fall short of the target 2.

Proposition 3.2. *Let $0 \leq j \leq k-2$, $m = k-j$, $W = (p_j + 1)^m$ and $x = p_{j+1}/(p_{j+1} - 1)$. Then*

$$\begin{aligned} x^m &> T \quad \text{and} \quad x < T + \frac{1}{A_j W} \quad \text{if } \varepsilon = -1, \\ x^m &\geq T - \frac{1}{A_j W} \quad \text{and} \quad x < T \quad \text{if } \varepsilon = +1. \end{aligned} \tag{3.3}$$

In particular p_{j+1} lies in an interval $[L_j, U_j]$ whose endpoints are determined by $p_1, \dots, p_j$, k and ε .

Proof. Let $\rho = \prod_{i>j} p_i/(p_i - 1)$. By (3.2),

$$\rho = \frac{2 - \varepsilon/\varphi(n)}{P_j} = T - \frac{\varepsilon}{P_j \varphi(n)}.$$

There are $m \geq 2$ factors in ρ ; each exceeds 1 and none exceeds x , because $p/(p-1)$ decreases in p . Hence $x < \rho \leq x^m$. The remaining primes are odd and larger than p_j (larger than 2 when $j = 0$), so $p_i - 1 \geq p_j + 1$ for $i > j$, and $\varphi(n) = B_j \prod_{i>j} (p_i - 1) \geq B_j W$. Therefore

$$0 < \frac{1}{P_j \varphi(n)} \leq \frac{1}{P_j B_j W} = \frac{1}{A_j W},$$

and the four inequalities follow from $x < \rho \leq x^m$. Finally, for $\varepsilon = +1$ the quantity $T - 1/(A_j W)$ exceeds 1, since $T - 1 = C_j/A_j \geq 1/A_j$. $\square$

Apart from the correction terms, (3.3) says $x < T \leq x^m$, that is $1/(T-1) < p_{j+1} - 1 \leq 1/(T^{1/m} - 1)$. Since $T^{1/m} - 1$ is close to $(T-1)/m$, the interval has length about $(m-1)/(T-1) = (m-1)A_j/C_j$ (Figure 3). The upper end obeys a bound that makes the search finite.

Corollary 3.3. *Let $0 \leq j \leq k-2$, let $p_1 < \dots < p_j$ be odd primes with $P_j < 2$, and let $p_{j+1} > p_j$ be a prime that satisfies the inequality for x^m in (3.3). Then $p_{j+1} < 1 + 3(k-j)B_j/C_j < 4kA_j$. If $n = p_1 \cdots p_k$ satisfies (3.1), then moreover $p_k \leq 2B_{k-1} + 1$, and*

$$A_j < (4k)^{2^j} \quad (0 \leq j \leq k), \quad n < (4k)^{2^k}.$$

Proof. Let $m = k-j$ and $x = p_{j+1}/(p_{j+1} - 1)$, so that $p_{j+1} - 1 = 1/(x-1)$. By (3.3), $x^m \geq T'$, where $T' = T$ if $\varepsilon = -1$ and $T' = T - 1/(A_j W)$ if $\varepsilon = +1$. As $T - 1 = C_j/A_j$ with $C_j \geq 1$, and $W \geq 4$, in both cases $T' - 1 \geq (C_j - \frac{1}{4})/A_j \geq 3C_j/(4A_j) > 0$ and $T' \leq T$. For $t > 1$ we have $t^{1/m} - 1 = (t-1)/\sum_{i<m} t^{i/m} \geq (t-1)/(mt)$, and so

$$p_{j+1} - 1 = \frac{1}{x-1} \leq \frac{1}{T^{1/m} - 1} \leq \frac{mT'}{T' - 1} \leq \frac{2mB_j}{A_j} \cdot \frac{4A_j}{3C_j} < \frac{3mB_j}{C_j}.$$

Since $C_j \geq 1$ and $B_j \leq A_j - 1$ for $j \geq 1$, while $A_0 = B_0 = C_0 = 1$, this gives $p_{j+1} < 4kA_j$. For a solution, the equation $A_{k-1}p_k + \varepsilon = 2B_{k-1}(p_k - 1)$ gives $p_k C_{k-1} = 2B_{k-1} + \varepsilon$, and $C_{k-1} \geq 1$ by Proposition 3.1, so that $p_k \leq 2B_{k-1} + 1 < 4kA_{k-1}$. By Propositions 3.1 and 3.2 the first part applies to p_{j+1} for every $j \leq k-2$. Hence $A_{j+1} < 4kA_j^2$ for $0 \leq j < k$, and induction gives $A_j \leq (4k)^{2^j-1}$. $\square$

Every node of the search below satisfies the hypotheses of Corollary 3.3, so the tree has finitely many nodes and the search terminates. The bound $n < (4k)^{2^k}$ has the shape of the bound $n < k^{2^k}$ of Pomerance [32]; Theorem 1.2 replaces it by $n < 2^{2^{k-7}}$.

The inequalities (3.3) are tested in integers: for instance $x^m > T$ reads $p^m A_j > 2B_j(p-1)^m$. A floating-point estimate is used only to find a starting point, which is then corrected by exact comparisons.

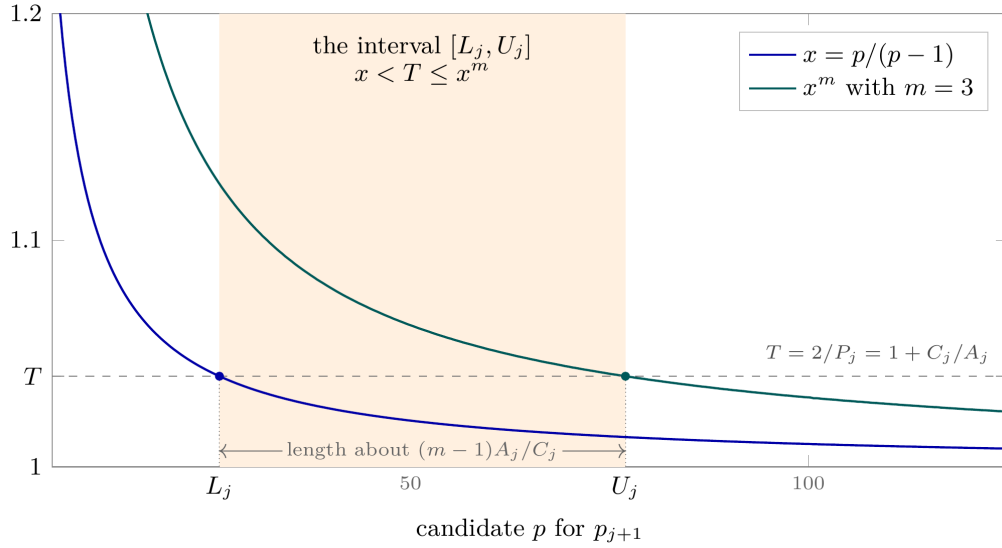


FIGURE 3. The interval of Proposition 3.2 for $T = 1.04$ and $m = 3$, with the correction terms $1/(A_j W)$ left out. The candidate p must satisfy $x < T$, because the other $m - 1$ factors of ρ exceed 1, and $x^m \geq T$, because none of them exceeds x . Here the two conditions give $26 < p < 77$.

Lemma 3.4 (congruence prune). *No prime factor of n divides $p - 1$ for another prime factor p of n .*

This is Lemma 2.1(ii). A candidate p for p_{j+1} is discarded if $p \mid p_i - 1$ or $p_i \mid p - 1$ for some $i \leq j$. The primes of the interval $[L_j, U_j]$ that are not discarded are called *admissible*.

The last two primes are determined together by a divisibility condition.

Proposition 3.5. *Let $A = A_{k-2}$, $B = B_{k-2}$, $C = 2B - A$, $p = p_{k-1}$ and $q = p_k$. Then*

$$t := Cp - 2B > 0, \quad t \mid Ap + \varepsilon, \quad q = 1 + \frac{Ap + \varepsilon}{t}, \quad (3.4)$$

and equivalently

$$(Cp - 2B)(Cq - 2B) = 2AB + \varepsilon C. \quad (3.5)$$

Proof. Equation (3.1) reads $Apq + \varepsilon = 2B(p - 1)(q - 1)$. Collecting the terms in q gives $q(Cp - 2B) = 2B(p - 1) + \varepsilon$. The right-hand side is positive, so $t > 0$, and $q - 1 = (2Bp - 2B + \varepsilon - Cp + 2B)/t = (Ap + \varepsilon)/t$. Multiplying $Cpq - 2B(p + q) + 2B - \varepsilon = 0$ by C and completing the product gives $(Cp - 2B)(Cq - 2B) = 4B^2 - 2BC + \varepsilon C = 2AB + \varepsilon C$. $\square$

Once $p_1, \dots, p_{k-2}$ are fixed, the prime p_{k-1} runs over the interval of Proposition 3.2 with $m = 2$, and for each candidate p the last prime is determined by (3.4). Alternatively, by (3.5), t is a divisor of the integer

$$D = 2AB + \varepsilon C$$

lying in a known residue class modulo C , so the candidates can be read off from the divisors of D when D can be factored. The first route costs time proportional to the length of the interval, the second the cost of factoring D . Figure 4 summarises the procedure.

The search is a depth-first traversal of the tree whose nodes at depth $j \leq k - 2$ are the increasing sequences $(p_1, \dots, p_j)$ of primes with $p_1 \geq p_{\min}$, where each p_{i+1} lies in the interval determined by $(p_1, \dots, p_i)$ and passes the congruence prune, and where $P_i < 2$ for all $i \leq j$. At

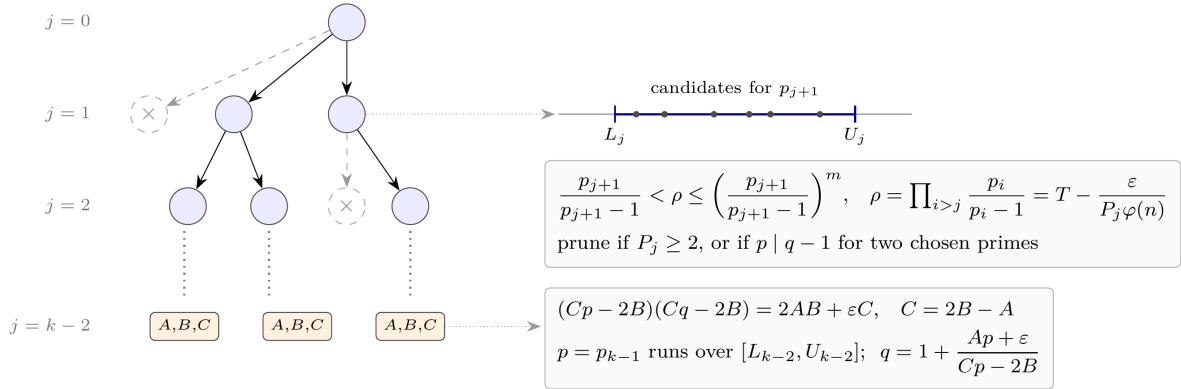


FIGURE 4. The search for $n = p_1 \cdots p_k$ with $n + \varepsilon = 2\varphi(n)$. A node at depth j is an increasing sequence of primes $p_1 < \cdots < p_j$. Its children are the primes in the interval $[L_j, U_j]$ of Proposition 3.2 that pass the congruence prune; dashed branches are cut. At depth $k-2$ the pair (p_{k-1}, p_k) is a lattice point on the hyperbola (3.5), found by running p over $[L_{k-2}, U_{k-2}]$ or by factoring $2AB + \varepsilon C$.

depth $k-2$ each candidate p_{k-1} is tested with (3.4): one checks that $t > 0$, that $t \mid Ap + \varepsilon$, that $q = 1 + (Ap + \varepsilon)/t$ is a prime larger than p , that p and q pass the congruence prune, and finally that (3.1) holds exactly.

Theorem 3.6 (completeness). *If $n = p_1 \cdots p_k$ with primes $p_{\min} \leq p_1 < \cdots < p_k$ satisfies (3.1), then the traversal visits $(p_1, \dots, p_{k-2})$, examines the candidate p_{k-1} , and records n .*

Proof. By induction on j the node $(p_1, \dots, p_j)$ is visited for every $j \leq k-2$: the prime p_{j+1} lies in the interval of Proposition 3.2, it passes the prune by Lemma 3.4, and $P_{j+1} < 2$ by Proposition 3.1. At depth $k-2$ the candidate $p = p_{k-1}$ satisfies (3.4) with $q = p_k$. A primality test never declares a prime to be composite, so q is accepted, and the final check of (3.1) succeeds. $\square$

The search is rigorous for two reasons. First, all arithmetic is exact: A_j, B_j, C_j , the interval endpoints and the quantities in (3.4) are integers or ratios of integers compared by cross multiplication. Second, the probabilistic primality tests used by the programs⁷ can err only by declaring a composite number prime, never by declaring a prime composite. Such an error could add a node or accept a composite q , but it can never remove an actual solution. A search that finds nothing is therefore a proof that there is nothing, and the completeness of the enumeration route depends on no primality proof. The exact check of (3.1) would not catch such an error, since it uses $\prod(p_i - 1)$ in place of $\varphi(n)$. This matters only for solutions that are reported, and every solution reported in this paper, including those of Appendix B, has all its prime factors below 10^{14} , where the tests are proofs. The divisor route depends on a complete factorisation of D , which is certified whenever $D < \psi_{13}$, since then every prime factor of D lies below that bound.

⁷Below the number $\psi_{13} = 3317044064679887385961981$ the programs use a strong probable-prime test to the thirteen bases 2, 3, $\dots$, 41, and above it a Baillie–PSW test. Sorenson and Webster [35] showed that every composite number below ψ_{13} fails the first test for at least one of these bases, so there it is a proof. The Baillie–PSW test [2, 33] combines a strong probable-prime test to base 2 with a strong Lucas test; no composite number is known to pass it.

4. THE LAST THREE PRIMES

For $k \leq 14$ the search of Section 3 is cheap because the interval for p_{k-1} is never longer than about $5 \cdot 10^7$. For $k = 15$ this is no longer true. After the prime p_{13} has been chosen the interval for p_{14} can be longer than 10^{14} , and there are tens of millions of choices of p_{13} . Neither route of Section 3 copes with this: running p_{14} over its interval is far too slow, and factoring tens of millions of integers of about fifty digits is a computation of a different order. In this section we describe two further ways of finding the divisors of $2AB + \varepsilon C$ that can occur in Proposition 3.5. Both use the residue class in which these divisors lie: the first lists lattice points near a hyperbola, and the second scans the possible values of the sum of two complementary divisors.

4.1. Reduction to divisors in a residue class. Fix a node $(p_1, \dots, p_{k-3})$ at depth $k-3$ and a candidate $s = p_{k-2}$ in its interval that passes the congruence prune. Put

$$A' = A_{k-3}s, \quad B' = B_{k-3}(s-1), \quad c = 2B' - A', \quad N = 2A'B' + \varepsilon c.$$

If $c \leq 0$ then $P_{k-2} = A'/B' \geq 2$, and by Proposition 3.1 no solution has $p_1, \dots, p_{k-3}, s$ as its first $k-2$ primes; such s are discarded. So let $c \geq 1$. By Proposition 3.5, applied with $A = A'$ and $B = B'$, the remaining primes $p < q$ of a solution satisfy $(cp - 2B')(cq - 2B') = N$ with both factors positive. Let r be the least positive residue of $-2B'$ modulo c .

By (3.2) the last three primes satisfy

$$\frac{s}{s-1} \cdot \frac{p}{p-1} \cdot \frac{q}{q-1} = T - \frac{\varepsilon}{P_{k-3}\varphi(n)}, \quad T = \frac{2}{P_{k-3}},$$

so the point (s, p, q) lies next to the surface on which this product equals T . Fixing s cuts the surface in a curve, and the pair (p, q) lies exactly on the hyperbola $(cp - 2B')(cq - 2B') = N$ beside that curve (Figure 5).

Proposition 4.1. *The integer $t = cp - 2B'$ is a divisor of N with*

$$t \equiv r \pmod{c}, \quad 1 \leq t \leq \sqrt{N}.$$

Moreover $\gcd(r, c) = 1$ and $N \equiv r^2 \pmod{c}$. Conversely, every divisor t of N with these properties gives integers $p = (t + 2B')/c$ and $q = (N/t + 2B')/c$ satisfying $A'pq + \varepsilon = 2B'(p-1)(q-1)$.

Proof. The congruence holds because $t \equiv -2B' \pmod{c}$, and $t \geq 1$ because both factors are positive, and $t \leq \sqrt{N}$ because $p < q$ makes t the smaller of the two factors. Since $c = 2B' - A'$ we have $\gcd(2B', c) = \gcd(2B', A')$, which is 1 because A' is odd and, by the congruence prune, no prime factor of A' divides B' ; hence $\gcd(r, c) = 1$. Further $N \equiv 2A'B' \equiv (2B')^2 \equiv r^2 \pmod{c}$, using $A' \equiv 2B'$. Conversely, let t be such a divisor. Then $N/t \equiv r^2 r^{-1} = r \pmod{c}$, so both t and N/t are congruent to $-2B'$ and p and q are integers. Expanding $(cp - 2B')(cq - 2B') = N$ and using $4B'^2 - 2A'B' = 2B'c$ gives $c(cpq - 2B'(p+q) + 2B' - \varepsilon) = 0$, which, as $c \geq 1$ and $c = 2B' - A'$, is the equation $A'pq + \varepsilon = 2B'(p-1)(q-1)$. $\square$

So the two-prime problem at depth $k-2$ becomes the following question: given N , a modulus c and a class r prime to c , list the divisors of N up to $\sqrt{N}$ that are congruent to r modulo c . Lenstra [20] proved that there are at most 11 such divisors when $c > N^{1/3}$,⁸ and gave a polynomial-time algorithm to find them; Coppersmith, Howgrave-Graham and Nagaraj [11] extended the algorithmic side down to $c > N^{1/4+\delta}$. We use a more elementary method suited to

⁸In the form used here: if $0 \leq r < c < N$, $\gcd(r, c) = 1$ and $c > N^{1/3}$, then at most 11 positive divisors of N are congruent to r modulo c . The bound counts all divisors in the class, not only those up to $\sqrt{N}$.

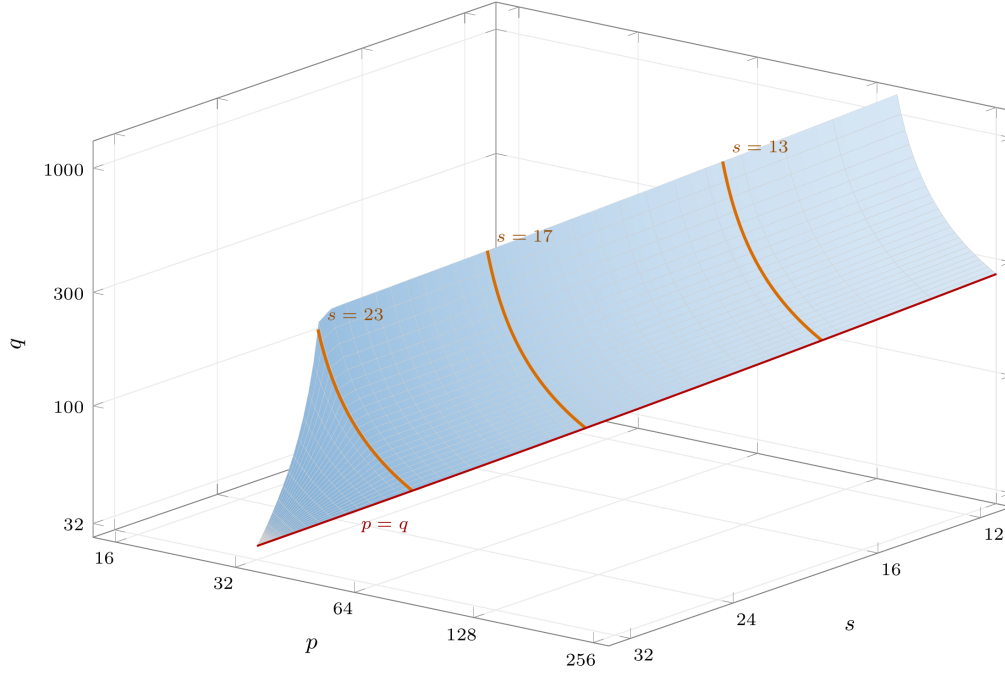


FIGURE 5. The surface $\frac{s}{s-1} \cdot \frac{p}{p-1} \cdot \frac{q}{q-1} = T$ with $s < p < q$, for $T = 1.1$ on logarithmic axes. Each slice with fixed s , three of which are marked, falls steeply from large values of q to the edge $p = q$. The slices shrink as s grows, and the surface ends at the point $s = p = q$ where $(s/(s-1))^3 = T$.

our range, and in Section 4.4 a still simpler one based on the fact that t and N/t lie in the same class.

4.2. Lattice points near a hyperbola. Let r' be the residue of Nr^{-1} modulo c , so that $r' = r$ in the situation of Proposition 4.1, and put

$$w = \frac{N - rr'}{c}, \quad \alpha \equiv wr^{-1}, \quad \beta \equiv r'r^{-1} \pmod{c}.$$

A divisor $t \equiv r$ is written $t = r + cu$ with $u \geq 0$, and then $N/t = r' + cv$ for an integer v . Expanding $(r + cu)(r' + cv) = N$ gives

$$cuv + r'u + rv = w, \tag{4.1}$$

so the pairs (u, v) are the lattice points on a hyperbola. Reducing (4.1) modulo c shows that each of them lies in the coset

$$v \equiv \alpha - \beta u \pmod{c}. \tag{4.2}$$

The coset has one point in every c consecutive values of v above a given u , so by itself it excludes little; combined with the shape of the hyperbola it leaves only a few points to test.

Lemma 4.2. *Let $0 \leq u_0 \leq u_1$ and put*

$$v^+ = \left\lfloor \frac{\lfloor N/(r + cu_0) \rfloor - r'}{c} \right\rfloor, \quad v^- = \left\lceil \frac{\lceil N/(r + cu_1) \rceil - r'}{c} \right\rceil.$$

If $t = r + cu$ divides N and $u_0 \leq u \leq u_1$, then $N/t = r' + cv$ with $v^- \leq v \leq v^+$ and v satisfying (4.2). Consequently, with $H = v^+ - v^- + 1$, the integer u lies in the set

$$\{u \in [u_0, u_1] : (\alpha - \beta u - v^-) \bmod c < H\}, \quad (4.3)$$

and there is no such t at all if $H \leq 0$.

Proof. Since $t \equiv r \pmod{c}$ and $\gcd(r, c) = 1$, we have $N/t \equiv Nr^{-1} \equiv r' \pmod{c}$, so $v = (N/t - r')/c$ is an integer. The cofactor N/t is an integer between $N/(r + cu_1)$ and $N/(r + cu_0)$, hence between the ceiling of the first and the floor of the second, and this gives $v^- \leq v \leq v^+$; in particular $H \geq 1$. The congruence (4.2) was shown above. Put $x = (\alpha - \beta u - v^-) \bmod c$, so that $0 \leq x < c$ and $v - v^- \equiv x \pmod{c}$. As $v - v^- \geq 0$, it is at least the least non-negative member x of its class, and so $x \leq v - v^- \leq H - 1$. $\square$

If $H \geq c$ the set (4.3) is the whole of $[u_0, u_1]$, and every u in the box is tested. If $H < c$ it consists of the u for which an arithmetic progression modulo c falls into a window of length H , and its elements can be listed in increasing order without looking at the others, at a cost of $O(\log c)$ arithmetic operations each, by the following elementary recursion.

Lemma 4.3. *Let $0 \leq a < g$ and $0 < L \leq R < g$. The least $y \geq 0$ with $L \leq ay \bmod g \leq R$ is found as follows. If $a = 0$ there is no such y . If $a \geq 1$ and $a[L/a] \leq R$, it is $\lceil L/a \rceil$. Otherwise it is $\lceil (L + gz)/a \rceil$, where z is the least integer ≥ 0 with*

$$(-R) \bmod a \leq (g \bmod a)z \bmod a \leq (-L) \bmod a,$$

and if no such z exists there is no y . The last problem is of the same kind, with (g, a, L, R) replaced by $(a, g \bmod a, (-R) \bmod a, (-L) \bmod a)$.

Proof. If $a = 0$ then $ay \bmod g = 0 < L$ for every y . In the second case $ay \leq R < g$ for $y = \lceil L/a \rceil$, so $ay \bmod g = ay \in [L, R]$, while $0 \leq ay < L$ for smaller $y \geq 0$. Otherwise no multiple of a lies in $[L, R]$. Since any a consecutive integers contain a multiple of a , this gives $R - L \leq a - 2$; moreover $a \nmid R$, and no multiple of a lies in $[-R, -L]$, so the residues $(-R) \bmod a, \dots, (-L) \bmod a$ of $-R, \dots, -L$ increase without wrapping around and $0 < (-R) \bmod a \leq (-L) \bmod a < a$. A solution y satisfies $ay = gz + x$ with $z = \lfloor ay/g \rfloor$ and $x = ay \bmod g \in [L, R]$, and $z \geq 1$ because $z = 0$ would make $x = ay$ a multiple of a in $[L, R]$. Conversely, for a given $z \geq 0$ a solution with this z exists exactly when $[L + gz, R + gz]$ contains a multiple of a ; that multiple is unique, because $R - L < a$, and the solution is $y = \lceil (L + gz)/a \rceil$. The interval contains a multiple of a exactly when $gz \equiv -x \pmod{a}$ for some $x \in [L, R]$, that is, when $gz \bmod a = (g \bmod a)z \bmod a$ lies in the stated range. The intervals $[L + gz, R + gz]$ are disjoint and increase with z , since $R - L < g$, so the least z gives the least y . $\square$

The recursion replaces (g, a) by $(a, g \bmod a)$, as in Euclid's algorithm, so it stops after $O(\log g)$ steps; Figure 6 follows one step. It is applied with $g = c$ and $a = -\beta \bmod c$ as follows. Starting from $u = u_0$, let $b = (\alpha - \beta u - v^-) \bmod c$. If $b < H$, then u is in (4.3). Otherwise the next element of (4.3), if it does not exceed u_1 , is $u + y$, where y is the least integer ≥ 0 with $c - b \leq ay \bmod c \leq c - b + H - 1$; this window lies inside $[1, c - 1]$, so Lemma 4.3 applies. Every u found in this way is tested by one exact division of N by $r + cu$, and the search resumes at $u + 1$.

4.3. Choosing the boxes. The range $0 \leq u \leq U = \lfloor (\sqrt{N} - r)/c \rfloor$ is cut into consecutive intervals $[u_0, u_1]$, each giving a box $[u_0, u_1] \times [v^-, v^+]$ in Lemma 4.2 (Figure 7). Correctness does not depend on how this is done; the choice only affects the running time. Since $v \approx K/u$ with $K = N/c^2$, a box with $u_1 - u_0 = \lambda u_0$ has area close to $K\lambda^2/(1 + \lambda)$, and as the coset has one point per area c we choose λ with $\lambda^2/(1 + \lambda) = c^3/N$, so that each box holds about one

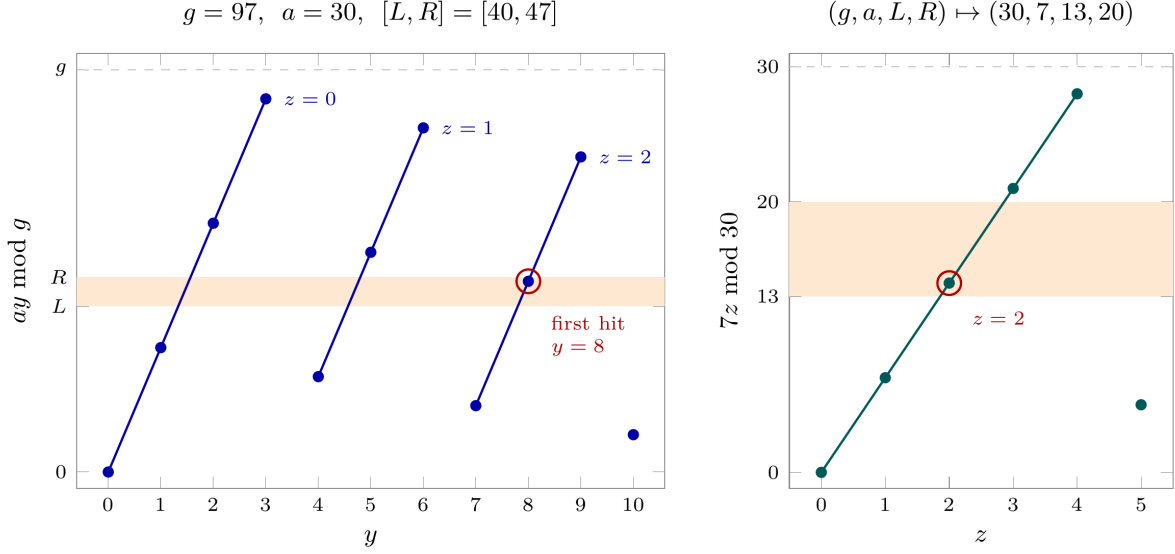


FIGURE 6. One step of Lemma 4.3 with $g = 97$, $a = 30$ and $[L, R] = [40, 47]$. Left: the residues $ay \bmod g$ rise in runs, one for each value of $z = \lfloor ay/g \rfloor$, and the first run misses the window because $a\lceil L/a \rceil = 60 > R$. Right: the reduced problem $(30, 7, 13, 20)$ asks for the least z with $7z \bmod 30$ in $[13, 20]$. It is $z = 2$, and the lemma returns $y = \lceil (40 + 2 \cdot 97)/30 \rceil = 8$, for which $ay \bmod g = 46$.

point of (4.2). There are two regimes. If $c^3 > N$, then λ exceeds 1.6 and grows like c^3/N , so that $O(\log N / \log(2 + c^3/N))$ boxes cover the whole range: this is Lenstra's range, and the work is a handful of Euclidean recursions. If $c^3 < N$, the boxes shrink to width about $u_0(c^3/N)^{1/2}$, the values of u below $(N/c^3)^{1/2}$ are tested one at a time, and the number of boxes grows like $(N/c^3)^{1/2} \log N$. When this estimate exceeds a fixed threshold we factor N completely instead and read off its divisors in the class.

Theorem 4.4. *Let $(p_1, \dots, p_{k-3})$ be a node at depth $k-3$ of the search of Section 3. Every $n = p_1 \cdots p_k$ with $n + \varepsilon = 2\varphi(n)$ extending this node is found by running $s = p_{k-2}$ over the admissible primes of its interval and, for each s , listing the divisors of Proposition 4.1 by Lemmas 4.2 and 4.3 or by a complete factorisation of N .*

Proof. The prime p_{k-2} lies in the interval and passes the prune, as in the proof of Theorem 3.6. For $s = p_{k-2}$, Proposition 4.1 shows that $t = cp_{k-1} - 2B'$ is one of the divisors listed: Lemma 4.2 shows that $u = (t - r)/c$ lies in the set (4.3) of the box that contains it, and Lemma 4.3 lists every element of that set, or, if N is factored, t is read off from its factorisation. The converse part of Proposition 4.1 then recovers p_{k-1} and p_k . $\square$

Every divisor that is found yields integers $p \leq q$ with $A_{k-3}spq + \varepsilon = 2B_{k-3}(s-1)(p-1)(q-1)$, by the converse part of Proposition 4.1, and $n = p_1 \cdots p_{k-3}spq$ is a solution exactly when $s < p < q$ and p and q are prime. As before, the correctness of a negative answer depends on no primality proof except in the factoring route, where every prime factor of N is proved prime (Appendix A.10).

4.4. The sum of the two factors. The classes of Proposition 4.1 are special: t and N/t lie in the same class modulo c , because $N \equiv r^2$. This allows a third treatment which uses neither

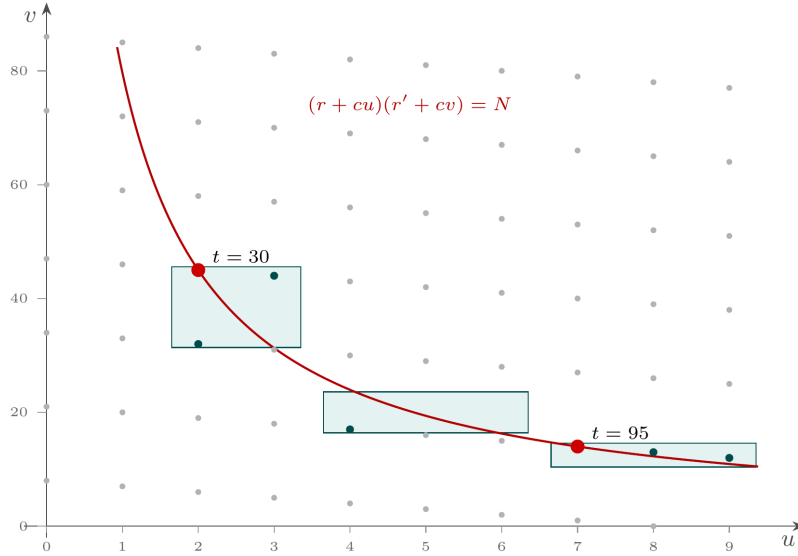


FIGURE 7. The divisors of $N = 17670$ in the class 4 modulo 13 up to $\sqrt{N}$, found as in Lemma 4.2. The grey points form the coset (4.2), here $v \equiv 8 - u \pmod{13}$; the curve is the hyperbola (4.1). The boxes cover the curve for $2 \leq u \leq 9$ (the values $u = 0, 1$ are tested directly), and only the points inside a box are examined. Two of them lie on the curve and give the divisors $t = 30$ and $t = 95$.

boxes nor factoring. Write $t = r + cu$ and $N/t = r + cv$ with $0 \leq u \leq v$. Since $r' = r$, the integer w of Section 4.2 is $w = (N - r^2)/c$, and (4.1) becomes

$$cuv + r(u + v) = w, \quad (4.4)$$

so the sum $S = u + v$ lies in the single class $S \equiv wr^{-1} \pmod{c}$, and u and v are the roots of $X^2 - SX + (w - rS)/c$. Conversely, if $S \equiv wr^{-1} \pmod{c}$, then $(w - rS)/c$ is an integer, and if $S^2 - 4(w - rS)/c = e^2$ with an integer $e \geq 0$ of the parity of S , then $u = (S - e)/2$ and $v = (S + e)/2$ are integers satisfying (4.4), so that $(r + cu)(r + cv) = N$. Fix $u_d \geq 1$ and put $t_d = r + cu_d$; if $t_d > \sqrt{N}$ there is no divisor with $u \geq u_d$ to look for. Since $x + N/x$ decreases on $(0, \sqrt{N}]$, every divisor $t \leq \sqrt{N}$ with $u \geq u_d$ has $2r + cS = t + N/t \leq t_d + N/t_d$, while $t + N/t \geq 2\sqrt{N}$. So S runs over at most $N/(c^3 u_d) + 1$ values of one residue class, and each value needs a single test whether $S^2 - 4(w - rS)/c$ is a perfect square. Every divisor with $u \geq u_d$ is found in this way, and by the converse every square found gives a factorisation $N = (r + cu)(r + cv)$. The divisors with $u < u_d$ are found by trial division. With u_d close to $(N/c^3)^{1/2}$ the work is about $2(N/c^3)^{1/2} + 2$ operations. This is comparable with the number of boxes above, but unlike the work in the boxes it does not depend on where the divisors lie.

The same argument bounds the number of divisors.

Proposition 4.5. *In the situation of Proposition 4.1, suppose that $c^3 \geq N + 2c^2$. Then at most one divisor $t \leq \sqrt{N}$ of N with $t \equiv r \pmod{c}$ is different from r , and N has at most four positive divisors congruent to r modulo c .*

Proof. By Proposition 4.1, every divisor t of N in the class has $N/t \equiv r^2 r^{-1} = r \pmod{c}$. Let $t_1 < t_2 \leq \sqrt{N}$ be divisors of N in the class, both different from r . The hypothesis gives $c^3 > 2c^2$, so $c > 2$ and $0 < r < c$. Hence $t_i = r + cu_i$ with $u_i \geq 1$, and $N/t_i = r + cv_i$ with $v_i \geq u_i$. By (4.4) the sums $u_i + v_i$ lie in one class modulo c , so the numbers $t_i + N/t_i = 2r + c(u_i + v_i)$ differ

by a multiple of c^2 . The difference is positive, because $x + N/x$ decreases on $(0, \sqrt{N}]$, and both numbers lie between $2\sqrt{N}$ and $r+c+N/(r+c) < 2c+N/c$. Hence $c^2 < 2c+N/c - 2\sqrt{N} < 2c+N/c$, that is, $c^3 < N + 2c^2$, contrary to the hypothesis. Finally, every divisor of N above $\sqrt{N}$ in the class is N/t for a divisor $t < \sqrt{N}$ in the class, and so there are at most four divisors in the class: r , N/r , t and N/t . $\square$

For classes of this special kind, and at the price of the term $2c^2$, Proposition 4.5 sharpens Lenstra's bound of 11 to four.

We ran this route, too, over every pair of a node at depth $k - 3$ and an admissible s . It takes its primes s from a sieve of its own and uses no factorisation, so its negative answers depend on no primality test at all.

5. LEHMER'S EQUATION: PROOF OF THEOREM 1.1

The last two assertions of Theorem 1.1 were proved in Section 2.1. For the first, let n be composite with $\varphi(n) \mid n - 1$ and suppose $\omega(n) \leq 15$. By Lemma 2.2, $3 \nmid n$; by Lemma 2.1(i), $n = p_1 \cdots p_k$ with $5 \leq p_1 < \cdots < p_k$ and $k \leq 15$; and by Lemma 2.3, $n - 1 = 2\varphi(n)$. If $k \leq 6$ then $\prod p_i/(p_i - 1) \leq \prod_{i \leq 6} r_i/(r_i - 1) = 1.9490\dots$, which contradicts (3.2). It therefore suffices to run the search with $\varepsilon = -1$ and $p_{\min} = 5$ for $7 \leq k \leq 15$, and to find nothing. For $k \leq 14$ we use the search of Section 3, and for $k = 15$ the method of Section 4.

5.1. Up to fourteen primes. Table 2 records the outcome. The column *terminal* counts the nodes at depth $k - 2$ whose interval for p_{k-1} is non-empty, and the last column gives the largest width $U_{k-2} - L_{k-2}$ of such an interval. No solution was found for any $k \leq 14$, so by Theorem 3.6 there is none. The case $k \leq 13$ recovers the theorem of Cohen and Hagis, and the case $k = 14$ is the bound $\omega(n) \geq 15$ announced by Renze and by Pinch.

TABLE 2. The search for $n - 1 = 2\varphi(n)$ with $5 \leq p_1$. Both implementations visit the same number of nodes for every k .

k	nodes	terminal	largest width for p_{k-1}
7	8	0	—
8	18	1	2
9	29	1	3
10	42	1	2
11	73	7	49
12	187	61	298
13	1 092	725	25 854
14	31 962	29 549	54 962 375

For $k = 14$ the numbers of nodes at depths $0, 1, \dots, 12$ are

$$1, 6, 16, 27, 29, 23, 22, 33, 58, 127, 335, 1654, 29631,$$

as in the row $k = 14$ of Figure 8, and 82 of the 29631 nodes at depth 12 have an empty interval for p_{13} . The largest prime occurring at depth 12 is 77563. Most of the work lies in a few terminal nodes. Exactly fifteen have an interval for p_{13} of width more than three million; they are listed in Table 3. Each of these intervals was enumerated in full, and independently the integer $D = 2AB - C$ was factored completely for each of the fifteen; the factorisations are part of the archived data.

TABLE 3. The fifteen terminal nodes of the case $k = 14$ with an interval for p_{13} of width more than $3 \cdot 10^6$, with the number of digits of $D = 2AB - C$.

$p_1, \dots, p_{12}$	width	digits of D
5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, 25867	54 962 375	41
5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, 25889	19 589 597	41
5, 7, 13, 17, 19, 23, 37, 59, 67, 73, 317, 5483	14 242 697	40
5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, 25903	13 903 362	41
5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, 25913	11 517 582	41
5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, 25919	10 443 048	41
5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, 25933	8 577 304	41
5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, 25939	7 967 703	41
5, 7, 13, 17, 19, 23, 37, 59, 83, 97, 107, 3343	6 121 542	39
5, 7, 13, 17, 19, 23, 37, 59, 67, 97, 199, 577	6 086 511	38
5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, 25969	5 880 810	41
5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, 26003	4 537 275	41
5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, 26017	4 147 802	41
5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, 26029	3 863 803	41
5, 7, 13, 17, 19, 23, 37, 59, 83, 109, 173, 199	3 320 179	37

Eleven of the fifteen share the prefix 5, 7, 13, 17, 19, 23, 37, 59, 67, 89, 173, whose product $\prod p/(p-1)$ is already within $8 \cdot 10^{-5}$ of 2; the twelfth prime then closes most of the remaining gap. For the node ending in 25867 one has $T-1 = C/A = 1.8 \cdot 10^{-8}$, and for the node ending in 5483 one has $T-1 = 7.0 \cdot 10^{-8}$. These near misses are the reason the width of the last interval grows so quickly with k : by Proposition 3.2 with $m = 2$ the interval for p_{k-1} runs from about A/C to about $2A/C$, and a small C means a long interval. For the node ending in 25867 this gives exactly the interval [54962376, 109924751]. Among the fifteen integers D one, for the node ending in 25913, is itself a prime of 41 digits⁹.

5.2. Fifteen primes. For $k = 15$ the traversal of Section 3 is stopped at depth 12. The numbers of nodes at depths 0, 1, $\dots$, 12 are

$$1, 6, 18, 30, 34, 37, 41, 56, 103, 218, 616, 3227, 55125,$$

and 54985 of the nodes at depth 12 have a non-empty interval for p_{13} . Both implementations of the search (Appendix A.1) produce the same 54985 prefixes with the same intervals, and so do the bounds for $\varepsilon = +1$. The intervals have total length $1.005 \cdot 10^9$ and contain 33865004 primes s that pass the congruence prune. For each of these pairs of a prefix and a prime $s = p_{13}$ we solved the two-prime problem of Proposition 4.1, in which N has between 40 and 58 digits, three times: by the two implementations of the lattice method (Section 4.2 and Appendix A.2) and by the route of Section 4.4.

Figure 9 shows how the pairs are distributed according to the ratio c^3/N , which by Section 4 decides how hard the two-prime problem is. For 33786783 of them, all but 0.23 per cent, the modulus satisfies $c^3 > N$, so that Lenstra's bound applies and a few boxes suffice. The remaining 78221 pairs come from values of s just above the lower end of long intervals, where $c = C_{12}s - 2B_{12}$ is small. The most extreme is the prefix ending in 25867 of Table 3, whose

⁹For this node (3.5) has only the trivial factorisations $D = 1 \cdot D$ and $D = D \cdot 1$, and neither gives a pair of primes in the required range. The enumeration of the $1.15 \cdot 10^7$ candidates for p_{13} reaches the same conclusion without knowing that D is prime.

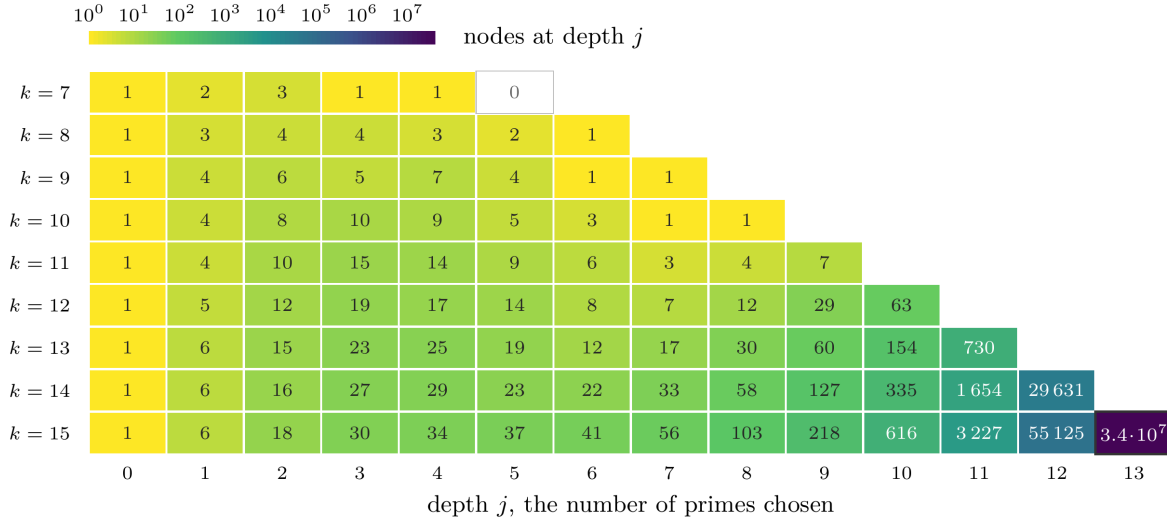


FIGURE 8. Number of nodes at each depth j of the search for $k = 7, \dots, 15$ primes, with the colour on a logarithmic scale. The counts are the same for $n - 1 = 2\varphi(n)$ and $n + 1 = 2\varphi(n)$ with $p_1 \geq 5$. The tree stays small until the last two levels. The last cell of each row counts the nodes at depth $k - 2$: for $k \leq 14$ these are the nodes where p_{k-1} is enumerated, and for $k = 15$ (framed) they are the 33865004 choices of p_{13} , each completed by the method of Section 4.

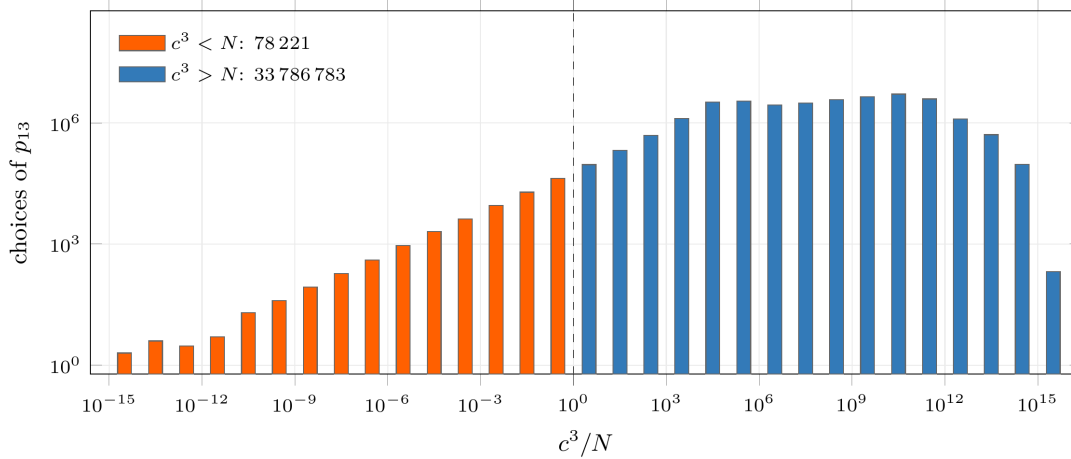


FIGURE 9. The 33865004 two-prime problems of the case $k = 15$, by the ratio c^3/N on a logarithmic scale (bins of one decade). To the right of the dashed line is Lenstra's range, where N has at most 11 divisors in the class and a few boxes find them. The left tail comes from the lower ends of the longest intervals for p_{13} .

first admissible s gives $c^3/N \approx 2 \cdot 10^{-15}$. For these pairs the route of Section 4.4 needs about $2(N/c^3)^{1/2}$ operations, and Appendix A.3 records how the work was divided.

For none of the 33865004 pairs does N have a divisor in the class of Proposition 4.1 that yields integers $s < p < q$, and the three computations agree on this. So there is not even a

completion by composite integers, and in particular no solution with $k = 15$. Together with the case $k \leq 14$ this proves Theorem 1.1.

6. THE SIZE OF A SOLUTION

This section proves Theorem 1.2. Of what precedes it uses only Section 2, and unlike Sections 3 to 5 it does not assume $M = 2$. Throughout, $n = p_1 \cdots p_k$ is a composite solution of (1.1) with $p_1 < \cdots < p_k$, so that $\varepsilon = -1$ and $M = (n-1)/\varphi(n)$, and A_j, B_j and P_j are as in the Notation of Section 1; thus $B_j = \varphi(A_j)$. By Lemma 2.1, n is odd and squarefree, $k \geq 2$, $M \geq 2$, and no prime factor of n divides $p-1$ for a prime factor p of n . By Lemma 2.2, if $3 \mid n$ then $p_1 = 3$ and $M \equiv 1 \pmod{3}$, so that $M \geq 4$.

Lemma 6.1. (i) $P_1 < P_2 < \cdots < P_{k-1} < M < P_k = M + 1/\varphi(n)$.
(ii) $p_k = (MB_{k-1} - 1)/(MB_{k-1} - A_{k-1})$.

Proof. The equality $P_k = M + 1/\varphi(n)$ is Lemma 2.1(iii), and P_j increases with j . Writing $n-1 = M\varphi(n)$ as $A_{k-1}p_k - 1 = MB_{k-1}(p_k - 1)$ gives $p_k(MB_{k-1} - A_{k-1}) = MB_{k-1} - 1 > 0$. Hence $MB_{k-1} > A_{k-1}$, that is $P_{k-1} < M$, and (ii) follows. $\square$

6.1. A sharp product lemma. For integers $m \geq 1$ and real $y \geq 1$ put

$$E_m(y) = y^{2^m} - y^{2^{m-1}}.$$

The function E_m is increasing on $y \geq 1$, satisfies $E_m(y) < y^{2^m}$, and

$$E_m(y^{2^u}) = E_{m+u}(y) \quad (m \geq 1, u \geq 0). \quad (6.1)$$

Theorem 6.2. Let $a < b$ be positive integers, $d = b - a$, $m \geq 2$, and let $2 \leq x_1 \leq x_2 \leq \cdots \leq x_m$ be integers with

$$\prod_{i=1}^m \left(1 - \frac{1}{x_i}\right) \leq \frac{a}{b} < \prod_{i=1}^{m-1} \left(1 - \frac{1}{x_i}\right). \quad (6.2)$$

Then

$$a x_1 x_2 \cdots x_m \leq E_{m-1}\left(\frac{b(a+1)}{d}\right).$$

If d divides $b+1$, equality holds for a suitable choice of the x_i .

For $d = 1$ the bound is $E_{m-1}((a+1)^2) = E_m(a+1)$, which is the lemma of Cook and Nielsen (Lemma 6.5 below) behind the bound of Burek and Žmija. For $d \geq 2$ it is smaller, and it improves the bound $E_{m-1}(b^2/d)$ of Stone [36, Corollary 1], since $b(a+1) < b^2$. Applied to the prime factors of n that follow the first j , Theorem 6.2 is strong whenever P_j is still well below M .

Throughout this subsection $a < b$ are positive integers, $d = b - a$, and $x_1 \leq \cdots \leq x_m$ are integers at least 2 satisfying (6.2). We write

$$X_u = x_1 \cdots x_u, \quad Y_u = (x_1 - 1) \cdots (x_u - 1),$$

with $X_0 = Y_0 = 1$, and use the same notation $Z_u = z_1 \cdots z_u$ for other sequences.

Lemma 6.3. For $0 \leq u \leq m-1$ the tail $x_{u+1}, \dots, x_m$ satisfies (6.2) with (a, b) replaced by (aX_u, bY_u) , and $aX_u < bY_u$.

Proof. Divide (6.2) by $\prod_{i \leq u} (1 - 1/x_i) = Y_u/X_u$. Since $u \leq m-1$, we have $Y_u/X_u \geq \prod_{i \leq m-1} (1 - 1/x_i) > a/b$. $\square$

The next lemma is a majorisation inequality; only the sequence x needs to be ordered.

Lemma 6.4. *Let $x_1 \leq \dots \leq x_m$ and $z_1, \dots, z_m$ be real numbers greater than 1 with $X_u \geq Z_u$ for $1 \leq u \leq m$. Then*

$$\prod_{i=1}^m \left(1 - \frac{1}{x_i}\right) \geq \prod_{i=1}^m \left(1 - \frac{1}{z_i}\right),$$

with strict inequality if $X_m > Z_m$.

Proof. The function $f(t) = \log(1 - e^{-t})$ on $t > 0$ has derivative $f'(t) = 1/(e^t - 1)$, which is positive and decreasing, so f is concave. Put $\tau_i = \log x_i$ and $\sigma_i = \log z_i$, so that $f(\tau_i) = \log(1 - 1/x_i)$ and $f(\sigma_i) = \log(1 - 1/z_i)$. By concavity

$$f(\sigma_i) \leq f(\tau_i) + c_i(\sigma_i - \tau_i), \quad c_i = f'(\tau_i) = \frac{1}{x_i - 1},$$

and $c_1 \geq c_2 \geq \dots \geq c_m > 0$ because the x_i are nondecreasing. The partial sums $\Delta_u = \sum_{i \leq u} (\sigma_i - \tau_i) = \log(Z_u/X_u)$ are at most 0 by hypothesis, and summation by parts gives

$$\sum_{i=1}^m c_i(\sigma_i - \tau_i) = \sum_{u=1}^{m-1} (c_u - c_{u+1})\Delta_u + c_m\Delta_m \leq c_m\Delta_m \leq 0,$$

with $c_m\Delta_m < 0$ when $X_m > Z_m$. Hence $\sum f(\sigma_i) \leq \sum f(\tau_i)$, strictly in that case, and the lemma follows on exponentiating. $\square$

Lemma 6.5 (Cook, Nielsen). *Under (6.2), with $m \geq 1$, we have $aX_m \leq E_m(a+1)$.*

This is the lemma of Cook [10] and Nielsen [27] mentioned in the introduction. We include its proof, on which the proof of Theorem 6.2 builds.

Proof. We use induction on m . Define

$$z_u = (a+1)^{2^{u-1}} + 1 \quad (1 \leq u \leq m-1), \quad z_m = (a+1)^{2^{m-1}}.$$

Induction on u gives $aZ_u = (a+1)^{2^u} - 1$ for $0 \leq u \leq m-1$, and then $aZ_m = E_m(a+1)$. Writing $c_u = (a+1)^{2^{u-1}}$, so that $c_{u+1} = c_u^2$ and $1 - 1/z_u = c_u(c_u - 1)/(c_{u+1} - 1)$ for $u < m$, the product telescopes:

$$\prod_{u=1}^m \left(1 - \frac{1}{z_u}\right) = \frac{(a+1)^{2^{m-1}-1} a}{c_m - 1} \cdot \frac{c_m - 1}{c_m} = \frac{a}{a+1}. \quad (6.3)$$

These identities use only that a is a positive real number, and we shall apply them once more with a value of a that need not be an integer.

Suppose first that $X_u \leq Z_u$ for some $1 \leq u \leq m-1$, so that $aX_u + 1 \leq (a+1)^{2^u}$. By Lemma 6.3 and the induction hypothesis applied to the $m-u$ terms $x_{u+1}, \dots, x_m$ with numerator aX_u , and by (6.1),

$$aX_m = aX_u \prod_{i>u} x_i \leq E_{m-u}(aX_u + 1) \leq E_{m-u}((a+1)^{2^u}) = E_m(a+1).$$

Otherwise $X_u > Z_u$ for every $u \leq m-1$ (for $m=1$ this is vacuous). If also $X_m > Z_m$, Lemma 6.4 and (6.3) give $\prod(1 - 1/x_i) > a/(a+1) \geq a/b$, contrary to (6.2). Hence $X_m \leq Z_m$ and $aX_m \leq E_m(a+1)$. $\square$

The proof compares $x_1, \dots, x_m$ with an extremal sequence $z_1, \dots, z_m$ through the partial products X_u and Z_u , and splits according to whether $X_u \leq Z_u$ for some $u \leq m-1$ (Figure 10).

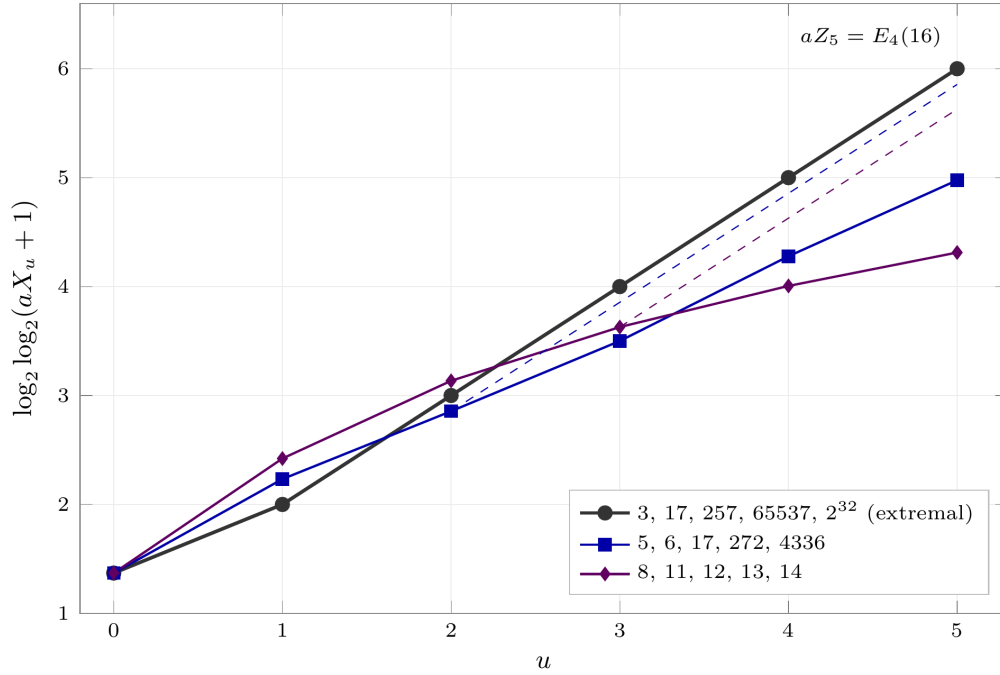


FIGURE 10. The proof of Theorem 6.2 for $a/b = 5/8$ and $m = 5$, on the scale $\log_2 \log_2(aX_u + 1)$. The extremal sequence $3, 17, 257, 65537, 2^{32}$ rises by exactly 1 per step from $u = 1$ on and ends at $aZ_5 = E_4(16)$. The two other sequences satisfy (6.2), start above it, and fall below it at $u = 2$ and $u = 3$. From that point Lemmas 6.3 and 6.5 give $aX_5 < (aX_u + 1)^{2^{5-u}}$, the dashed line of slope 1, which ends below $E_4(16)$. A sequence that stayed above the extremal one up to $u = 4$ would end below it at $u = 5$, by Lemma 6.4.

Proof of Theorem 6.2. Put $G = b(a+1)/d$, so that $G-1 = a(b+1)/d$, and define real numbers

$$z_1 = \frac{b+1}{d}, \quad z_u = G^{2^{u-2}} + 1 \quad (2 \leq u \leq m-1), \quad z_m = G^{2^{m-2}}.$$

Then $aZ_1 = G-1$, induction gives $aZ_u = G^{2^{u-1}} - 1$ for $1 \leq u \leq m-1$, and $aZ_m = (G^{2^{m-2}} - 1)G^{2^{m-2}} = E_{m-1}(G)$. The numbers $z_2, \dots, z_m$ are the sequence of the proof of Lemma 6.5 with $G-1$ in place of a and $m-1$ terms, so by (6.3)

$$\prod_{u=1}^m \left(1 - \frac{1}{z_u}\right) = \frac{a+1}{b+1} \cdot \frac{G-1}{G} = \frac{a+1}{b+1} \cdot \frac{a(b+1)}{b(a+1)} = \frac{a}{b}.$$

Suppose first that $X_u \leq Z_u$ for some $1 \leq u \leq m-1$, so that $aX_u + 1 \leq G^{2^{u-1}}$. Lemmas 6.3 and 6.5, applied to the $m-u \geq 1$ terms $x_{u+1}, \dots, x_m$, and (6.1) give

$$aX_m \leq E_{m-u}(aX_u + 1) \leq E_{m-u}(G^{2^{u-1}}) = E_{m-1}(G).$$

Otherwise $X_u > Z_u$ for all $u \leq m-1$. If also $X_m > Z_m$, then Lemma 6.4, which applies because every z_u exceeds 1 (for z_1 because $b+1 > d$), gives $\prod(1 - 1/x_i) > \prod(1 - 1/z_i) = a/b$, contrary to (6.2). Hence $X_m \leq Z_m$, that is $aX_m \leq E_{m-1}(G)$.

If $d \mid b+1$, then z_1 is an integer greater than 1, so $z_1 \geq 2$, and $G-1 = az_1$, $G \geq 3$ and all z_u are integers. They are nondecreasing: $z_2 \geq G \geq z_1$, since $G \geq z_1$ amounts to $ab \geq 1$; the middle terms increase; and for $m \geq 3$ we have $z_m = (z_{m-1} - 1)^2 \geq z_{m-1}$ because $z_{m-1} - 1 \geq G \geq 3$.

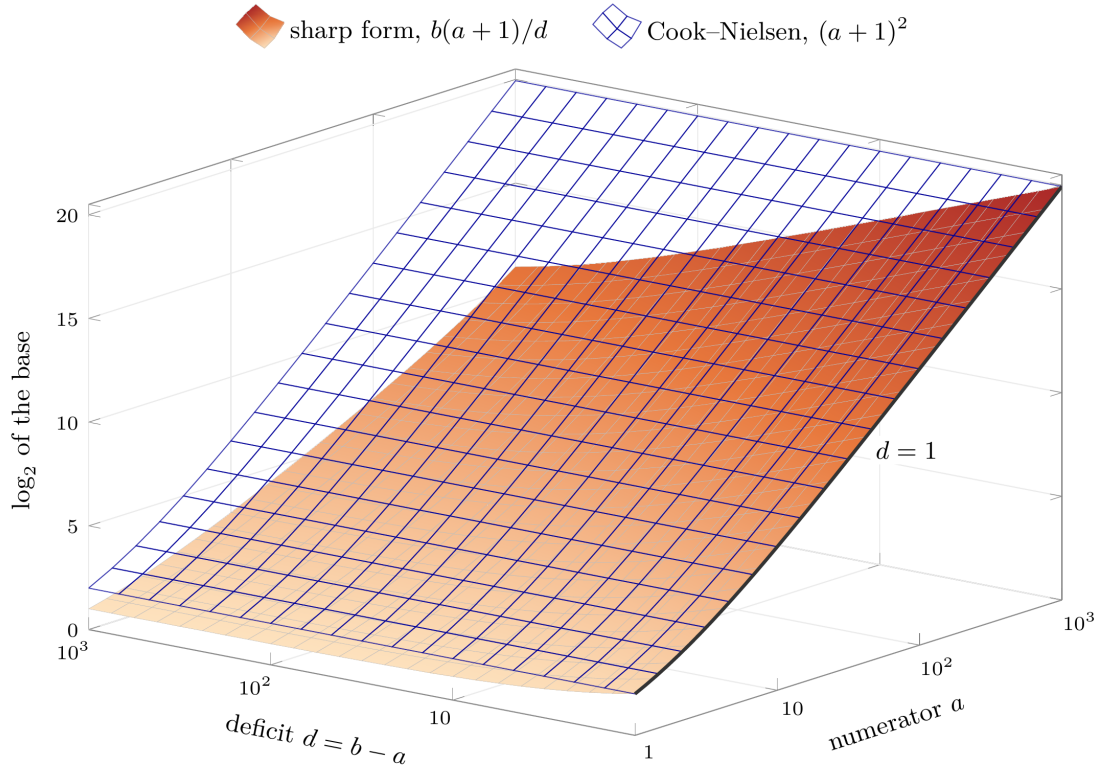


FIGURE 11. The base of the bound for aX_m written as $E_{m-1}(\cdot)$, on a $\log_2$ scale, over the numerator a and the deficit $d = b - a$. The wire mesh is the base $(a + 1)^2$ of the lemma of Cook and Nielsen, the shaded sheet below it the base $b(a + 1)/d$ of Theorem 6.2. They meet along $d = 1$, where the two bounds coincide, and separate as d grows.

The product of $1 - 1/z_u$ is a/b over all m terms and larger over the first $m - 1$, so the z_u satisfy (6.2) and attain $aZ_m = E_{m-1}(G)$. $\square$

Remark 6.6. Since $G = (a + 1)/(1 - a/b)$, the bound of Theorem 6.2 depends on b only through the distance of a/b from 1. Its base is smaller than the base $(a + 1)^2$ of Lemma 6.5 by the factor $(a + 1)d/b$, which is at least 1 because $(a + 1)d - b = a(d - 1) \geq 0$; Figure 11 shows the two bases. Stone's bound $E_{m-1}(b^2/d)$ exceeds ours by the factor $b/(a + 1)$ in the base. In the application below $a = A_j$ and $b = MB_j$, and then the base $G = (A_j + 1)/(1 - P_j/M)$ decreases as M increases, so the smallest admissible value of M yields an upper bound. The base $b^2/d = M^2 B_j^2 / (MB_j - A_j)$ of Stone has no such property: it decreases for $M < 2P_j$ and increases for $M > 2P_j$.

Remark 6.7. The extremal sequences of Theorem 6.2 are explicit. For $a/b = 5/8$ we have $d = 3$, $z_1 = 3$ and $G = 16$, and with $m = 4$ the sequence is 3, 17, 257, 65536:

$$\frac{2}{3} \cdot \frac{16}{17} \cdot \frac{256}{257} \cdot \frac{65535}{65536} = \frac{5}{8}, \quad 5 \cdot 3 \cdot 17 \cdot 257 \cdot 65536 = 2^{32} - 2^{16} = E_3(16).$$

The first three terms are Fermat primes. For $a = 1$ the sequence of the proof of Lemma 6.5 consists of the Fermat numbers $F_0, \dots, F_{m-2}$ followed by $2^{2^{m-1}}$, and the products $F_0 F_1 \cdots F_{r-1} = 2^{2^r} - 1$ with $1 \leq r \leq 5$ are the solutions 3, 15, 255, 65535 and 4294967295 of (1.2) in (1.3). Such

products have $n/\varphi(n)$ just below 2, and so do the deepest nodes of the search of Section 6.3 (Section 6.6).

6.2. Heavy solutions. For $0 \leq j \leq k-1$ put

$$V_j = V_j(M) = \frac{MB_j(A_j+1)}{MB_j - A_j} = \frac{A_j+1}{1 - P_j/M}. \quad (6.4)$$

By Lemma 6.1(i) the denominator is positive, and $V_j(M)$ decreases as M increases.

Proposition 6.8. For $0 \leq j \leq k-1$,

$$n < (A_j+1)^{2^{k-j}} \quad \text{and} \quad n < V_j^{2^{k-j-1}}.$$

Proof. Take $m = k-j$, $x_i = p_{j+i}$, $a = A_j$ and $b = MB_j$. Then

$$\prod_{i=j+1}^k \left(1 - \frac{1}{p_i}\right) = \frac{P_j}{P_k} < \frac{P_j}{M} = \frac{a}{b} < \frac{P_j}{P_{k-1}} = \prod_{i=j+1}^{k-1} \left(1 - \frac{1}{p_i}\right)$$

by Lemma 6.1(i), so (6.2) holds, $a < b$, and $ax_1 \cdots x_m = n$. Lemma 6.5 gives $n \leq E_{k-j}(A_j+1)$, which implies the first bound. For $j \leq k-2$, Theorem 6.2 gives $n \leq E_{k-j-1}(V_j)$, because $b(a+1)/d = V_j$. For $j = k-1$, Lemma 6.1(ii) gives $n = A_{k-1}p_k < A_{k-1}MB_{k-1}/(MB_{k-1} - A_{k-1}) < V_{k-1}$. $\square$

Definition 6.9. Let $s \geq 0$ be an integer, and put $\theta_i = 2^{2^{i-s}}$ for $i \geq s$ and $\theta_i = 1$ for $i < s$. A composite solution n of (1.1) is *s-heavy* if for every j with $0 \leq j \leq k-1$

- (H1) $A_j \geq \theta_j$, and
- (H2) $V_j \geq \theta_{j+1}$.

Corollary 6.10. A composite solution of (1.1) with k prime factors that is not s -heavy satisfies $n < 2^{2^{k-s}}$.

Proof. If (H1) fails at j , then $j \geq s$ and $A_j+1 \leq 2^{2^{j-s}}$, and Proposition 6.8 gives $n < (A_j+1)^{2^{k-j}} \leq 2^{2^{k-s}}$. If (H2) fails at j , then $j+1 \geq s$ because $V_j > 1$, and $n < V_j^{2^{k-j-1}} < 2^{2^{k-s}}$. $\square$

For $s = 2$ condition (H1) alone contradicts Lemma 6.1(i), and the argument can be carried out by hand.

Proposition 6.11. No composite solution of (1.1) is 2-heavy. Consequently every composite solution with k prime factors satisfies $n < 2^{2^{k-2}}$.

Proof. Let n be 2-heavy, and let $1 \leq i \leq k-1$. Condition (H1) at $j = i$ and $A_i \leq p_i^i$ give $p_i \geq \theta_i^{1/i} = 2^{2^{i-2}/i}$. Hence $p_7 \geq 24$, $p_8 \geq 256$, $p_9 > 19000$ and $p_i > 4^i$ for $i \geq 10$, so that

$$\sum_{8 \leq i \leq k-1} \frac{1}{p_i - 1} < \frac{1}{256} + \frac{1}{19000} + \sum_{i \geq 10} \frac{1}{4^i - 1} < 0.004.$$

Put $f(x) = x/(x-1)$, which decreases for $x > 1$ and satisfies $f(x) \leq e^{1/(x-1)}$. Then $P_k = \prod_{i \leq k} f(p_i)$ and $f(p_k) < f(p_{k-1})$. Let $b_1, \dots, b_7$ be lower bounds for $p_1, \dots, p_7$ and $\beta_j = \prod_{i \leq j} f(b_i)$. If $k \leq 8$, then $P_k < \beta_{k-1}f(b_{k-1})$. If $k \geq 9$, the prime factors $p_8, \dots, p_k$ contribute less than $e^{0.004}f(p_{k-1}) < e^{0.008}$, and $P_k < \beta_7 e^{0.008}$. We distinguish four cases.

- (a) If $3 \mid n$, then $p_1 = 3$ and $M \geq 4$ by Lemma 2.2, and the next prime factors are at least 5, 7, 11, 13, 17 and, as $p_7 \geq 24$, 29.
- (b) If $p_1 = 5$ and $p_2 = 7$, then no other prime factor is congruent to 1 modulo 5 or 7 by Lemma 2.1(ii), which excludes 11, 29 and 31.

- (c) If $p_1 = 5$ and $p_2 \neq 7$, then $p_2 \geq 13$, and 11 and 31 are excluded because they are congruent to 1 modulo 5.
- (d) If $p_1 \geq 7$, the prime factors are at least the primes from 7 on, and $p_7 \geq 29$.

In cases (b) to (d), $3 \nmid n$ and $M \geq 2$. Table 4 lists the bounds $b_1, \dots, b_7$ in each case and the larger of $\max_{j \leq 7} \beta_j f(b_j)$ and $\beta_7 e^{0.008}$. In every case it is less than M , so that $P_k < M$, contrary to Lemma 6.1(i). Hence no composite solution is 2-heavy, and Corollary 6.10 with $s = 2$ gives the bound. $\square$

TABLE 4. The four cases in the proof of Proposition 6.11: lower bounds $b_1, \dots, b_7$ for the seven smallest prime factors and the resulting upper bound for P_k , rounded up.

case	$b_1, \dots, b_7$	bound for P_k	$M \geq$
(a) $p_1 = 3$	3, 5, 7, 11, 13, 17, 29	2.972	4
(b) $p_1 = 5, p_2 = 7$	5, 7, 13, 17, 19, 23, 37	1.957	2
(c) $p_1 = 5, p_2 \neq 7$	5, 13, 17, 19, 23, 29, 37	1.738	2
(d) $p_1 \geq 7$	7, 11, 13, 17, 19, 23, 29	1.749	2

The proof uses only condition (H1), whose failure gives the bound through the lemma of Cook and Nielsen, and the fact that no prime factor divides another minus one. For $s = 1$ the same argument needs only the cases $3 \mid n$ and $3 \nmid n$, with the primes from 5 on as lower bounds and $p_6 \geq 41$, $p_7 \geq 569$, $p_8 \geq 65537$ from (H1); it shows that no composite solution is 1-heavy, which gives $n < 2^{2^{k-1}}$, half the number of binary digits allowed by the bound of Burek and Žmija. For larger s the estimates of this proof are too weak, and Theorem 1.2(i) follows once we know that no solution is 7-heavy, which Section 6.3 establishes with exact bounds at every node and with condition (H2) and Theorem 6.2 as well.

We first record how fast the prime factors of a heavy solution grow. Fix s , an index $j \geq 0$ and the product A_j of j odd primes. For $i > j$ let λ_i be the least positive integer λ with $A_j \lambda^{i-j} \geq \theta_i$, and put $y_i = (\theta_i/A_j)^{1/(i-j)}$. Then $\lambda_i = \max\{1, \lceil y_i \rceil\}$, so $\lambda_i \geq y_i$, and $\lambda_i < y_i + 1$ whenever $\lambda_i > 1$.

Lemma 6.12. *Let $i > j$.*

- (i) *If $i \geq j + 2$ and $y_i > 1$, then $y_{i+1} \geq y_i^{4/3}$.*
- (ii) *If n is an s -heavy solution, A_j is the product of its j smallest prime factors and $i \leq k - 1$, then $p_i \geq \lambda_i$.*

Proof. (i) If $y_i > 1$, then $\theta_i > A_j \geq 1$, so $i \geq s$ and $\theta_{i+1} = \theta_i^2$. Hence

$$(i + 1 - j) \log y_{i+1} = 2 \log \theta_i - \log A_j \geq 2(\log \theta_i - \log A_j) = 2(i - j) \log y_i,$$

so $\log y_{i+1} \geq \frac{2(i-j)}{i+1-j} \log y_i \geq \frac{4}{3} \log y_i$, since $i - j \geq 2$ and $\log y_i > 0$.

- (ii) By (H1) at i , $\theta_i \leq A_i \leq A_j p_i^{i-j}$, so $p_i \geq \lambda_i$. $\square$

The search needs an upper bound for P_k/P_j that uses only $p_1, \dots, p_j$ and a lower bound for p_{j+1} . For an integer $q \geq 3$ and $i > j$ let $\pi_i(q)$ be the $(i - j)$ -th smallest prime that is at least q , if this prime is less than $2 \cdot 10^7$, and $\pi_i(q) = q + 2(i - j - 1)$ otherwise. Put $b_i = \max\{\pi_i(q), \lambda_i\}$, let i_0 be the least index $i \geq j + 3$ with $\lambda_i > 10^{30}$, and define

$$\Gamma_j(q) = \max \left\{ \frac{q}{q-1}, \max_{j+1 \leq i < i_0} \frac{b_i + 2}{b_i + 1} \prod_{l=j+1}^i \frac{b_l}{b_l - 1}, \left(1 + \frac{16}{\lambda_{i_0} - 2}\right) \prod_{l=j+1}^{i_0-1} \frac{b_l}{b_l - 1} \right\}. \quad (6.5)$$

The quantity $\Gamma_j(q)$ depends only on j , A_j , q and s , and it is a finite computation because λ_i grows doubly exponentially in i .

Lemma 6.13. *Let n be an s -heavy solution, $0 \leq j \leq k-1$, and let q be an integer with $3 \leq q \leq p_{j+1}$. Then $P_k \leq P_j \Gamma_j(q)$.*

Proof. The quotient P_k/P_j is the product of $p_i/(p_i-1)$ over $j < i \leq k$, and $x/(x-1)$ decreases in x . For $j < i \leq k-1$ we have $p_i \geq \pi_i(q)$, since $p_{j+1} < \dots < p_i$ are odd primes at least q , and $p_i \geq \lambda_i$ by Lemma 6.12(ii); so $p_i \geq b_i$. For the last prime, $p_k \geq q$ if $k = j+1$ and $p_k \geq p_{k-1} + 2 \geq b_{k-1} + 2$ otherwise. This bounds P_k/P_j by the first term of (6.5) when $k = j+1$, and by the second term, with $i = k-1$, when $j+2 \leq k \leq i_0$.

Suppose $k > i_0$. The primes $p_{j+1}, \dots, p_{i_0-1}$ contribute at most $\prod_{l=j+1}^{i_0-1} b_l/(b_l-1)$, as before. Put $Y = y_{i_0}$. Since $\lambda_{i_0} > 10^{30}$, we have $Y > \lambda_{i_0} - 1 \geq 10^{30}$, and Lemma 6.12 gives $p_i \geq \lambda_i \geq y_i \geq Y^{(4/3)^{i-i_0}}$ for $i_0 \leq i \leq k-1$. As $p_k > p_{k-1}$, the factor of p_k is at most that of p_{k-1} , so

$$\prod_{i=i_0}^k \frac{p_i}{p_i-1} \leq \prod_{i=i_0}^{k-1} \left(1 + \frac{1}{p_i-1}\right)^2 \leq \exp\left(2 \sum_{u \geq 0} \frac{1}{Y^{(4/3)^u} - 1}\right).$$

Because $(4/3)^u \geq 1 + u/3$ and $Y \geq 10^{30}$, the terms with $u \geq 1$ add up to less than $1/(Y-1)$, so the exponent is less than $4/(Y-1) \leq 1$. With $e^x \leq 1 + 2x$ for $0 \leq x \leq 1$ the product is at most $1 + 8/(Y-1) < 1 + 8/(\lambda_{i_0} - 2)$, which is smaller than the factor $1 + 16/(\lambda_{i_0} - 2)$ of the third term of (6.5). $\square$

6.3. The search. Fix s . Let $S = (p_1, \dots, p_j)$, $j \geq 0$, be an increasing sequence of odd primes in which no p_i divides any $p_l - 1$, and write A_j , B_j and P_j as before. Let $\mu(S)$ be the least integer $M > P_j$ with $M \geq 2$ and, if $p_1 = 3$, $M \equiv 1 \pmod{3}$, and let $\mu'(S)$ be the next such integer. We say that a solution n *extends* S if $j \leq k-1$ and S consists of the j smallest prime factors of n . By Lemmas 2.1, 2.2 and 6.1(i), the quotient M of such an n satisfies $M \geq \mu(S)$. For $M > P_j$ we use $V_j(M)$ as in (6.4), computed from S .

The tree $\mathcal{T}_s$ is built from the empty sequence by the following rules, applied to each sequence S of length j that is generated; Figure 12 summarises them.

- (R1) *Ratio test.* Let q_1 be the least prime greater than p_j ($q_1 = 3$ if $j = 0$). If $P_j \Gamma_j(q_1) \leq \mu(S)$, the sequence S is discarded.
- (R2) *Deficit test.* If $V_j(\mu(S)) < \theta_{j+1}$, the sequence S is discarded.

A sequence that survives both tests is a *node* of $\mathcal{T}_s$. At a node:

- (R3) *Last prime.* If $j \geq 1$, then for each integer M with $\mu(S) \leq M < P_j(p_j+2)/(p_j+1)$, and $M \equiv 1 \pmod{3}$ if $p_1 = 3$, test whether $(MB_j - 1)/(MB_j - A_j)$ is a prime greater than p_j that is not 1 modulo any p_i .
- (R4) *Children.* Put $\mu = \mu(S)$ and $q_c = \lfloor \mu B_j / (\mu B_j - A_j) \rfloor$. Generate the sequences $(p_1, \dots, p_j, q)$ for the primes $q > p_j$ with $q \not\equiv 1 \pmod{p_i}$ for $i \leq j$ and with $A_j q \geq \theta_{j+1}$, taking q in increasing order and omitting q in the following cases.
 - (i) If $P_j \Gamma_j(q) \leq \mu$, omit q and every larger prime.
 - (ii) If $q \leq q_c$ and $P_j \Gamma_j(q) \leq \mu'(S)$, omit every prime in $[q, q_c]$.
 - (iii) If $q > q_c$ and $V_{j+1}(\mu)$, computed for the sequence $(p_1, \dots, p_j, q)$, is less than θ_{j+2} , omit q .

A prime q satisfies $q \leq q_c$ exactly when $P_j q / (q-1) \geq \mu$, so in (ii) a solution extending $(p_1, \dots, p_j, q)$ has $P_{j+1} \geq \mu$ and hence $M \geq \mu'(S)$. For $q > q_c$ we have $P_{j+1} < \mu$, so $V_{j+1}(\mu)$ is defined, and after multiplication by its positive denominator the inequality in (iii) becomes a

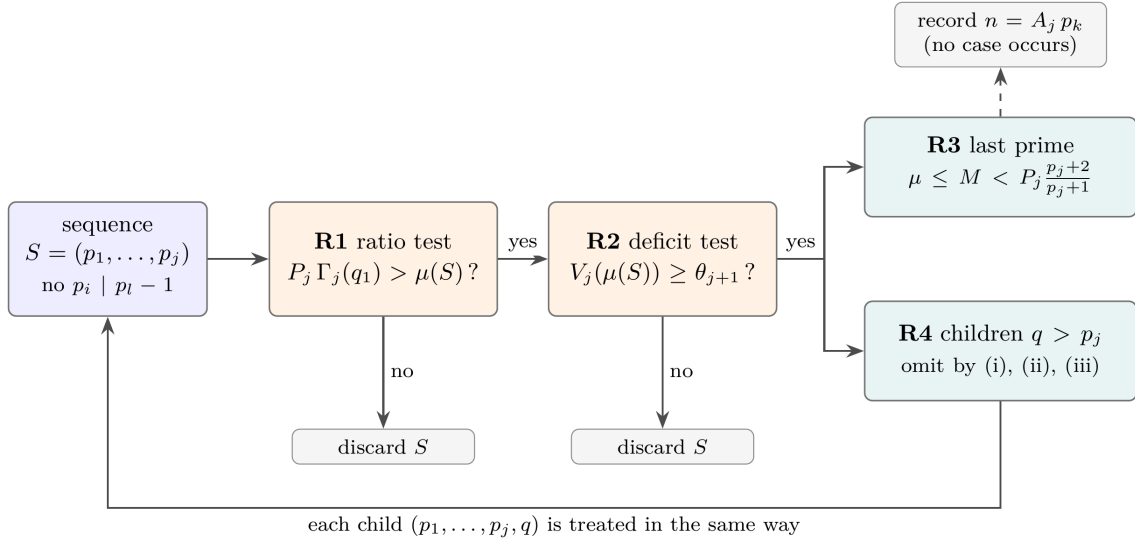


FIGURE 12. The treatment of a sequence $S = (p_1, \dots, p_j)$ in the tree $\mathcal{T}_s$. A sequence that fails the ratio test or the deficit test cannot be extended by an s -heavy solution. A node is tested for a last prime factor and passes to its children every prime that the three omission rules allow.

quadratic inequality in q with positive leading coefficient;¹⁰ the primes omitted by (iii) therefore form an interval, which is computed once per node.

Proposition 6.14. *Let n be an s -heavy solution. Then $(p_1, \dots, p_{k-1})$ is a node of $\mathcal{T}_s$, and rule (R3) at that node finds p_k .*

Proof. Write $S_j = (p_1, \dots, p_j)$. We show by induction on j that S_j is generated and is a node, for $0 \leq j \leq k-1$. The empty sequence S_0 is generated by definition. Suppose that S_j is generated. As n extends S_j , we have $M \geq \mu(S_j)$. Lemma 6.13 with $q = q_1 \leq p_{j+1}$ gives $P_j \Gamma_j(q_1) \geq P_k > M \geq \mu(S_j)$, so (R1) does not discard S_j , and $V_j(\mu(S_j)) \geq V_j(M) \geq \theta_{j+1}$ by (H2), so (R2) does not either. Thus S_j is a node.

Now let $j \leq k-2$. The prime p_{j+1} is a candidate of (R4) at S_j , by Lemma 2.1(ii) and (H1) at $j+1$. Lemma 6.13 gives $P_j \Gamma_j(q) \geq P_k > M \geq \mu(S_j)$ for every integer q with $3 \leq q \leq p_{j+1}$, so rule (i) applies at no prime $q \leq p_{j+1}$. If $p_{j+1} \leq q_c$, then $P_{j+1} \geq \mu(S_j)$, and $M > P_{j+1}$ forces $M \geq \mu'(S_j)$; hence $P_j \Gamma_j(q) > \mu'(S_j)$ for $q \leq p_{j+1}$, rule (ii) applies at no prime $q \leq p_{j+1}$, and an application at a larger prime does not reach p_{j+1} . If $p_{j+1} > q_c$, rule (ii) omits only primes up to q_c , and $V_{j+1}(\mu(S_j)) \geq V_{j+1}(M) \geq \theta_{j+2}$ by (H2) at $j+1$, so rule (iii) does not omit p_{j+1} . Hence S_{j+1} is generated.

At the node S_{k-1} we have $\mu(S_{k-1}) \leq M < P_k$ and $P_k = P_{k-1}p_k/(p_k-1) \leq P_{k-1}(p_{k-1}+2)/(p_{k-1}+1)$, so M is one of the values tested in (R3), and p_k is the quotient given by Lemma 6.1(ii). $\square$

6.4. The computation. All quantities in the rules are rational numbers, and the search evaluates them exactly. The primes $\pi_i(q)$ below $2 \cdot 10^7$ are taken from a sieve. Primes above that are reached by stepping to the next probable prime; such a step never skips a prime, so a composite number may occasionally be treated as a prime, which can only enlarge the tree.

¹⁰With $D = \mu B_j - A_j$ and $\theta = \theta_{j+2}$ one has $V_{j+1}(\mu) = \mu B_j(q-1)(A_j q + 1)/(Dq - \mu B_j)$, and $V_{j+1}(\mu) < \theta$ reads $\mu A_j B_j q^2 + (\mu B_j - \mu A_j B_j - \theta D)q + (\theta - 1)\mu B_j < 0$.

It is equally harmless to replace q_1 in (R1) by the smaller number $p_j + 2$, as one of the two programs of Appendix A.9 does beyond the sieve, because the proof of Proposition 6.14 uses only Lemma 6.13, which holds for every integer q with $3 \leq q \leq p_{j+1}$. In rule (R3) the quotient was never an integer, so no primality proof is needed.

TABLE 5. The tree $\mathcal{T}_7$ by depth j : the number of nodes and the number of values of M tested by rule (R3).

j	0	1	2	3	4	5	6	7	8	9	10	11	12	13
nodes	1	1	1	3	6	7	7	13	31	79	266	1501	29073	2689
tests	0	0	0	0	0	0	0	0	0	0	16	215	11894	2689

Theorem 6.15. *The tree $\mathcal{T}_7$ has 33678 nodes and depth 13. Rule (R3) tests 14814 values of M , and for none of them is the quotient $(MB_j - 1)/(MB_j - A_j)$ an integer. Consequently, by Proposition 6.14, no composite solution of (1.1) is 7-heavy.*

Table 5 lists the nodes by depth. Every node has $\mu(S) = 2$, so every node has $p_1 \neq 3$ and $P_j < 2$; in particular the sequence (3), for which $\mu = 4$, is discarded at once by the ratio test. The first branch of the tree is 5, 7, 13, 17, 19, 23, 37, 59, 67, 73, the beginning of the independent set of Section 2.1 chosen one prime at a time, and the deepest nodes have thirteen entries.

Proof of Theorem 1.2(i). By Theorem 6.15 no composite solution of (1.1) is 7-heavy, and Corollary 6.10 with $s = 7$ gives $n < 2^{2^{k-7}}$. $\square$

For $s = 1, \dots, 7$ the trees $\mathcal{T}_s$ have 1, 3, 7, 12, 24, 118 and 33678 nodes. The deficit test does most of the work: without rules (R2) and (iii), the tree $\mathcal{T}_6$ already has 4469 nodes and depth 12.

6.5. Larger quotients. When $M \geq 3$, condition (H1) for a suitable s is incompatible with Lemma 6.1(i) and the bounds of Section 2.1, and no further search is needed. Let $\ell_1 < \ell_2 < \dots$ be integers with $p_i \geq \ell_i$ for $i \leq k$, let $J \geq 1$, and let V^* be a number with $P_{\min\{k, J\}} < V^*$. If (H1) holds for all $j \leq k - 1$, then $A_i \leq p_i^i$ gives $p_i \geq \rho_i$ for $i \leq k - 1$, where ρ_i is the least positive integer x with $x^i \geq \theta_i$. With $w_i = \max\{\ell_i, \rho_i\}$ we obtain, as in the proof of Lemma 6.13,

$$P_k < R_s = V^* \max\left\{1, \max_{J \leq i < i_0} \frac{w'_i + 2}{w'_i + 1} \prod_{l=J+1}^i \frac{w_l}{w_l - 1}, \left(1 + \frac{16}{\rho_{i_0} - 2}\right) \prod_{l=J+1}^{i_0-1} \frac{w_l}{w_l - 1}\right\}, \quad (6.6)$$

where $w'_i = \max\{w_i, \ell_{i+1} - 2\}$ accounts for $p_{i+1} \geq \max\{\ell_{i+1}, w_i + 2\}$, and $i_0 \geq \max\{J+1, s+4\}$ is an index with $\rho_{i_0} > 10^{30}$. The three terms cover $k \leq J$, then $J < k \leq i_0$ with $k = i+1$, and $k > i_0$. The tail estimate is that of Lemma 6.13, applied to $y_i = \theta_i^{1/i}$, for which $p_i \geq \rho_i \geq y_i > \rho_i - 1$ when $\rho_i > 1$, and $\log y_{i+1} = \frac{2i}{i+1} \log y_i \geq \frac{4}{3} \log y_i$ when $i \geq \max\{s, 2\}$. We evaluate R_s in exact rational arithmetic, with integer lower bounds in place of the ρ_i , which can only enlarge it.

Proof of Theorem 1.2(ii) and (iii). Suppose $3 \mid n$. Then $M \geq 4$, every prime factor other than 3 is congruent to 2 modulo 3, by Lemma 2.2, and $k \geq 10^8 + 2$ by Corollary 2.6(ii). Take $J = 10^8 + 1$, $\ell_i = 2i + 1$, which is at most the i -th odd prime, and $V^* = 4(1 - 10^{-6})$. The J smallest prime factors are 3 and an independent set of 10^8 primes $p \equiv 2 \pmod{3}$, so $P_J < \frac{3}{2} \cdot \frac{8}{3}(1 - 10^{-6}) = V^*$ by Proposition 2.5(ii). The computation gives $R_{10^8} = 3.9999966\dots < 4$. If (H1) held for all $j \leq k - 1$ with $s = 10^8$, then (6.6) would give $P_k < 4 \leq M$, contrary to Lemma 6.1(i). So n is not 10^8 -heavy, and Corollary 6.10 gives (iii).

Suppose $M \geq 3$ and $3 \nmid n$. Then $k \geq 16001$ by Corollary 2.6(i). Take $J = 16000$, $\ell_i = r_i$, the primes from 5 on, and $V^* = 2.999999$; the J smallest prime factors form an independent set, so

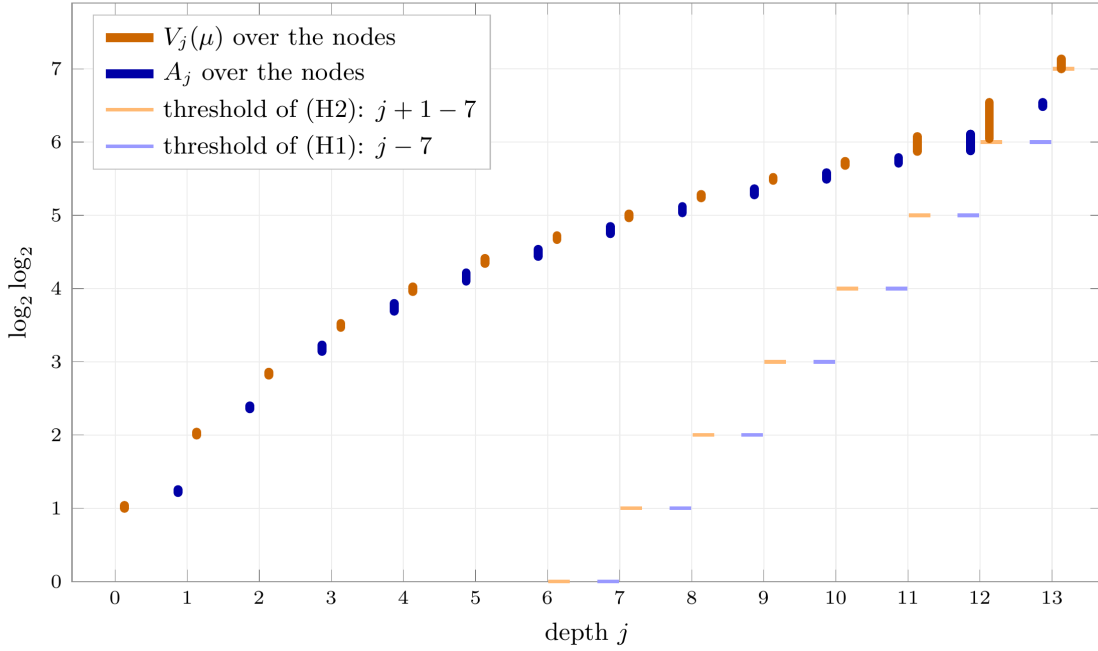


FIGURE 13. The nodes of $\mathcal{T}_7$ by depth j . The bars span the range of $\log_2 \log_2 V_j(\mu)$ (right of each depth) and of $\log_2 \log_2 A_j$ (left of each depth) over the nodes of depth j , and the short marks below them are the thresholds $j + 1 - 7$ of (H2) and $j - 7$ of (H1). The deficit condition (H2) is the one that binds: at depth 13 the smallest value of $\log_2 \log_2 V_{13}$ is $7.0000075\dots$

$P_J < V^*$ by Proposition 2.5(i). Then $R_{15981} = 2.99999900041\dots < 3$, and the same argument gives $n < 2^{2^{k-15981}}$. If $M \geq 3$ and $3 \mid n$, part (iii) gives a stronger bound. $\square$

The exponent 15981 is the largest that this argument gives with $J = 16000$, since $R_{15982} = 3.000016\dots$. Everything is decided by the prime factors around the 16000-th. For $s = 15981$, (H1) at $j = 16000$ gives $p_{16000} \geq \rho_{16000} > 7 \cdot 10^9$, so that the remaining prime factors raise $P_{16000} < 2.999999$ by a factor less than $1 + 10^{-9}$. For $s = 15982$ the corresponding bound $\rho_{16000} < 10^5$ lies below $r_{16000} = 176089$, and the next prime factor alone may raise the product by the factor $1 + 1/176122$.

6.6. Where the search is hard. The nodes that survive longest are prefixes with P_j just below 2, the same kind of prefix that makes the next cases of Section 9 expensive. Every node at depth 13 has $2 - P_{13} < 10^{-10}$: there A_{13} has between 27 and 29 digits, while $C_{13} = 2B_{13} - A_{13}$ has between 14 and 18. In this respect the deep nodes behave like the products of Fermat primes of Remark 6.7, for which $2\varphi(n) - n = 1$. For such a prefix $1 - P_j/2$ is tiny, so V_j is large and the deficit test cannot discard it. Its children are few for the same reason: a prime $q \leq q_c = \lfloor 1/(1 - P_j/2) \rfloor$, which exceeds $3 \cdot 10^{10}$ at depth 13, would raise P_{j+1} to 2 or more and force $M \geq 3$, and rule (ii) removes these primes. Figure 13 shows how close to the threshold of (H2) the deepest nodes come.

The tree grows quickly with s (Figure 14), and for $s = 7$ more than nine tenths of its nodes lie at depths 12 and 13. Each further unit of s halves the number of binary digits allowed by Theorem 1.2(i). The method proves $n < 2^{2^{k-s}}$ for every s for which the tree $\mathcal{T}_s$ can be searched, and the cost of the search is dominated by the deep prefixes with P_j close to 2. We have not

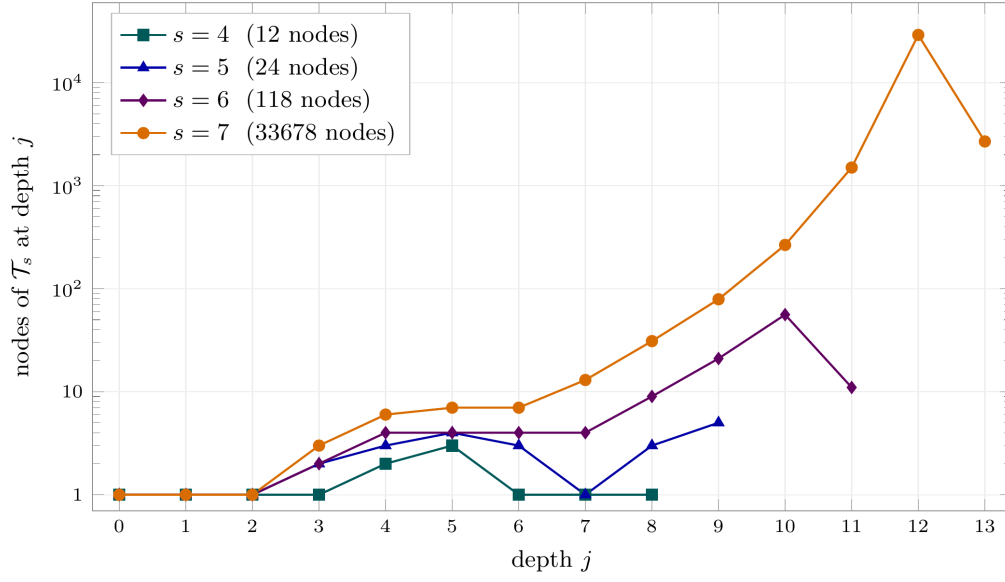


FIGURE 14. The number of nodes of $\mathcal{T}_s$ at each depth, for $s = 4, 5, 6, 7$, on a logarithmic scale.

completed the search for $s = 8$: at the node 5, 7, 13, 17, 19, 23, 37, 59, 67, 73, 317, 5483 of depth 12, where $2/P_{12} - 1 = 7.0 \cdot 10^{-8}$, rule (R4) already generates more than 10^6 sequences.

6.7. Comparison with other bounds. Stone [36, (52), (53)] treated tuples of primes with $\prod(1 - 1/p_i) \leq \frac{1}{2} < \prod_{i < k}(1 - 1/p_i)$, which include the prime factors of solutions with $M = 2$. For $p_1 = 5$ he obtained $\prod p_i \leq E_k(\beta)$ with $\beta \approx 1.66127$, and with $\beta \approx 1.50188$ when also $p_2 = 7$. Theorem 1.2(i) gives the base $2^{1/128} = 1.00543 \dots$ for solutions of (1.1). Table 6 compares, along the first branch of $\mathcal{T}_7$, the exponents e for which the three bases give $n < 2^{2^{k-e}}$. The deficit gains about three quarters of a unit of e over Lemma 6.5, and Theorem 6.2 adds a further gain over Stone's form that is largest for short prefixes.

TABLE 6. Along the branch 5, 7, 13, ... of $\mathcal{T}_7$ with $M = 2$: the exponent e such that the bound for n obtained at depth j is $2^{2^{k-e}}$, for the base $(A_j + 1)^2$ of Lemma 6.5, the base b^2/d of Stone and the base V_j of Theorem 6.2.

j	p_j	P_j	Cook–Nielsen	Stone	Theorem 6.2
1	5	1.25000	−0.370	−0.142	0.000
2	7	1.45833	−0.370	0.099	0.181
3	13	1.57986	−0.143	0.486	0.530
4	17	1.67860	0.309	1.017	1.041
5	19	1.77186	0.899	1.644	1.657
6	23	1.85240	1.561	2.324	2.330
7	37	1.90385	2.251	3.030	3.033
8	59	1.93668	2.965	3.759	3.761
9	67	1.96602	3.720	4.516	4.517
10	73	1.99333	4.507	5.265	5.265

7. THE COMPANION EQUATION

7.1. Up to seven prime factors. Let n satisfy (1.2) with $\omega(n) \leq 7$. If $n \leq 3$ there is nothing to prove. Otherwise n is odd, squarefree and composite by Lemma 2.1, and $n + 1 = 2\varphi(n)$ by Corollary 2.4. Since 3 may divide n , the search now starts from $p_{\min} = 3$, and we run it with $\varepsilon = +1$ for $2 \leq k \leq 7$. The outcome is in Table 7.

TABLE 7. The search for $n + 1 = 2\varphi(n)$ with $3 \leq p_1$ and $k \leq 7$.

k	nodes	terminal	largest width for p_{k-1}	solutions found
2	1	1	1	15
3	3	2	2	255
4	5	2	15	65535
5	11	4	256	83623935, 4294967295
6	60	44	65 536	6992962672132095
7	6 914	6 826	2^{32}	none

For $k \leq 6$ the search returns exactly the known solutions, each at its own depth, and this is also a test of the programs. For $k = 7$ it returns nothing. Here the enumeration route is too slow, because one terminal node, the prefix 3, 5, 17, 257, 65537, has an interval for p_6 of width 2^{32} ; for this prefix $C = 1$ and $D = 2AB + C = 2^{64} - 2^{32} + 1$, which is prime¹¹. We therefore use the divisor route throughout, which is rigorous in this range: the largest integer D factored in the case $k \leq 7$ has 21 digits, so every factorisation is certified. For $k \leq 6$ the enumeration route was also run and agrees node for node. This proves Theorem 1.3 for solutions with at most seven prime factors; the case of eight is treated in Section 7.4.

7.2. Solutions prime to three. Let $n > 3$ satisfy (1.2) with $3 \nmid n$ and $\omega(n) \leq 15$. Then $n = p_1 \cdots p_k$ with $5 \leq p_1$, and $n + 1 = 2\varphi(n)$ by Lemma 2.3. For $k \leq 6$ the product $\prod p_i/(p_i - 1)$ is at most 1.9490..., whereas (3.2) requires it to be $2 - 1/\varphi(n)$. For $k \geq 2$ this is at least $2 - \frac{1}{24} > 1.95$, as $\varphi(n) \geq 4 \cdot 6$, and for $k = 1$ it is at least $\frac{7}{4}$, whereas $p_1/(p_1 - 1) \leq \frac{5}{4}$. For $7 \leq k \leq 14$ we ran both implementations with $\varepsilon = +1$ and $p_{\min} = 5$. Neither found a solution, and the node counts, the numbers of terminal nodes and the largest widths coincide with those of Table 2 for every k . For $k = 15$ the nodes at depth 12 and their intervals are again those of Section 5.2, and the method of Section 4, run with $\varepsilon = +1$ by both implementations and by the route of Section 4.4 over the same 33865004 pairs, finds no completion in integers. This proves Theorem 1.4.

The coincidence with Table 2 has a simple cause. The two sets of bounds in (3.3) differ only by the terms $1/(A_j W)$, which in this range are far smaller than the gaps between consecutive admissible values of $p/(p - 1)$ near the endpoints, and the computation confirms that no endpoint moves. The tests (3.4) at the last level do differ, but by then the tree is fixed. For $3 \nmid n$ and $k \leq 15$ the two equations therefore lead to the same search.

7.3. Extensions of Fermat-type solutions.

Proof of Theorem 1.6. Let $2\varphi(n_0) = n_0 + 1$ and let $Q > 1$ be squarefree and coprime to n_0 . Then $n = n_0 Q$ satisfies $2\varphi(n) = n + 1$ if and only if $2\varphi(n_0)\varphi(Q) = n_0 Q + 1$, that is, $(n_0 + 1)\varphi(Q) = n_0 Q + 1$, or

$$n_0(Q - \varphi(Q)) = \varphi(Q) - 1. \quad (7.1)$$

¹¹Indeed $A = 2^{32} - 1$, $B = 2^{31}$ and $C = 2B - A = 1$, so $D = 2^{32}(2^{32} - 1) + 1 = \Phi_6(2^{32})$, the sixth cyclotomic polynomial at 2^{32} . Its primality is well known and is confirmed by a deterministic test, since $D < 3.3 \cdot 10^{24}$.

- (i) For $Q = q$ prime, (7.1) reads $n_0 = q - 2$.
(ii) For $Q = pq$ with primes $p < q$ we have $Q - \varphi(Q) = p + q - 1$ and $\varphi(Q) - 1 = pq - p - q$, so (7.1) becomes $pq - (n_0 + 1)(p + q) + n_0 = 0$, and adding $(n_0 + 1)^2 - n_0$ to both sides gives

$$(p - n_0 - 1)(q - n_0 - 1) = n_0^2 + n_0 + 1.$$

The two factors on the left have the same sign, and they are positive: if both were negative we would have $2 \leq p < q \leq n_0$, and the left-hand side would be at most $(n_0 - 1)^2 < n_0^2 + n_0 + 1$. Conversely, a factorisation $n_0^2 + n_0 + 1 = de$ with $d < e$ gives $p = n_0 + 1 + d$ and $q = n_0 + 1 + e$, and if both are prime then $n_0 pq$ is a Fermat-type solution; the conditions $p, q \nmid n_0$ hold because $p, q > n_0$. The divisor $d = 1$ gives $p = n_0 + 2$ and $q = (n_0 + 1)^2 + 1 = n_0(n_0 + 2) + 2$, which is the composition of two one-prime extensions.

(iii) Factoring $n_0^2 + n_0 + 1$ and testing $n_0 + 2$ for each of the eight numbers gives Table 8. The one-prime extensions are $1 \rightarrow 3 \rightarrow 15 \rightarrow 255 \rightarrow 65535 \rightarrow 4294967295$ and $83623935 \rightarrow 6992962672132095$. The two-prime extensions are those listed in the table, and the only one that is not a composition of one-prime extensions is $255 \rightarrow 83623935$. Every extension of each of the eight numbers is therefore again one of the eight, and each of them is reached from 1, so the eight numbers are exactly the closure of $\{1\}$. The rows for 65535, 83623935, 4294967295 and 6992962672132095 show where the chains stop. Every factor in Table 8 is below 2^{64} , where the primality tests we use are proofs, and a test that declares a candidate p or q composite is never wrong. $\square$

TABLE 8. One- and two-prime extensions of the Fermat-type solutions.

n_0	$n_0 + 2$ prime	factorisation of $n_0^2 + n_0 + 1$; two-prime extensions (p, q)
1	yes	3; (3, 5)
3	yes	13; (5, 17)
15	yes	241; (17, 257)
255	yes	$97 \cdot 673$; (257, 65537) and (353, 929)
65535	yes	$193 \cdot 22253377$; none
83623935	yes	$67 \cdot 180097 \cdot 579535339$; none
4294967295	no	$2^{64} - 2^{32} + 1$, prime; none
6992962672132095	no	$13 \cdot 37 \cdot 109 \cdot 30802180789 \cdot 30280941915943441$; none

The chains stop for the following reasons. For $n_0 = 65535$ the trivial divisor gives $p = 65537$ and $q = 2^{32} + 1 = 641 \cdot 6700417$, which is Euler's factorisation of F_5 , and the divisor 193 gives $p = 65729$, a prime, but $q = 22318913 = 3037 \cdot 7349$. For $n_0 = 4294967295$ the one-prime extension fails because $n_0 + 2 = 2^{32} + 1$ is composite once more. Figure 16 shows the hyperbolas for $n_0 = 255$ and $n_0 = 65535$ with their lattice points.

Remark 7.1. For three new primes, $Q = spq$ with $s < p < q$, write $x = s - 1$, $y = p - 1$, $z = q - 1$. Equation (7.1) becomes

$$xyz = n_0(xy + yz + zx) + n_0(x + y + z) + n_0 + 1,$$

and dividing by n_0xyz shows that $1/x + 1/y + 1/z$ falls short of $1/n_0$ by $(n_0(x + y + z) + n_0 + 1)/(n_0xyz)$, which is less than $4/x^2$ because $x < y < z$. Hence x , the smallest of the three, satisfies $3/x > 1/n_0 - 4/x^2$ and $1/x < 1/n_0$, that is, $n_0 < x < 3n_0 + 2$. For fixed x the equation is bilinear in y and z :

$$((x - n_0)y - n_0(x + 1))((x - n_0)z - n_0(x + 1)) = (x - n_0)(n_0x + n_0 + 1) + n_0^2(x + 1)^2,$$

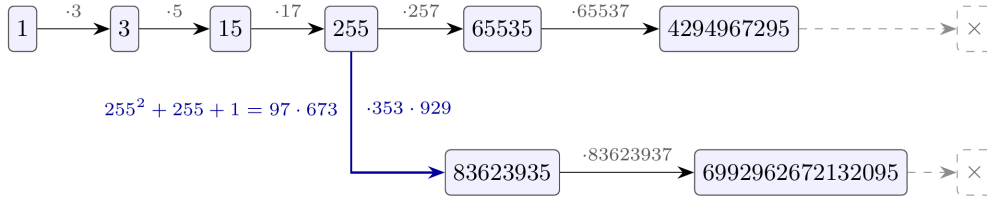


FIGURE 15. The Fermat-type solutions of $n + 1 = 2\varphi(n)$ as the closure of 1 under extensions. Horizontal arrows are one-prime extensions $n_0 \mapsto n_0(n_0 + 2)$, labelled by the prime $n_0 + 2$. The bent arrow is the two-prime extension of 255 by (353, 929), coming from the factorisation $65281 = 97 \cdot 673$. Dashed arrows end the two chains: 4294967295 and 6992962672132095 are composite, and neither $n_0^2 + n_0 + 1$ has a factorisation that yields two primes.

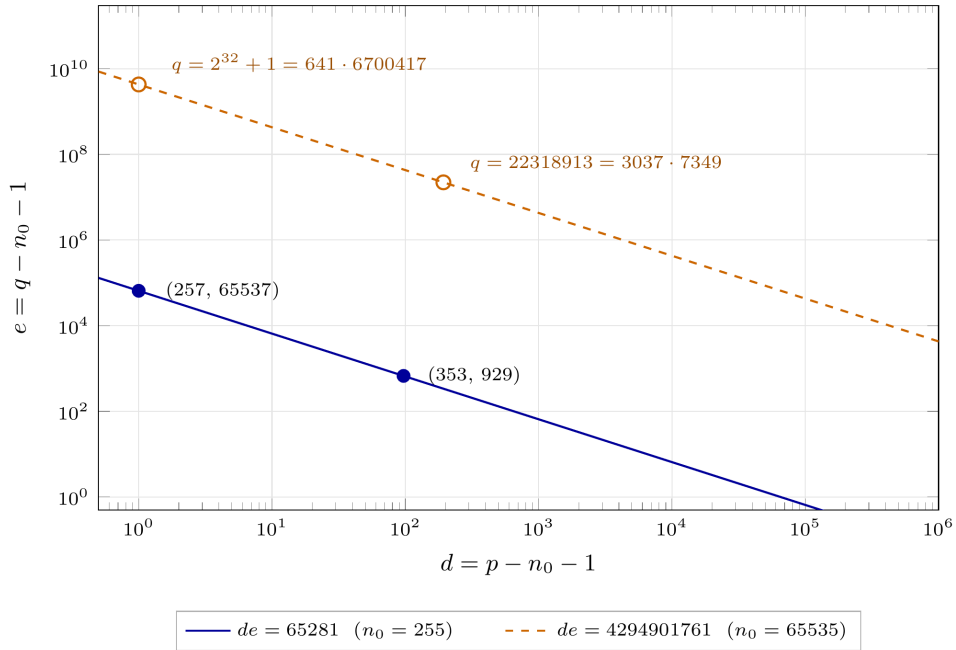


FIGURE 16. The two-prime extensions of n_0 are the lattice points $(d, e) = (p - n_0 - 1, q - n_0 - 1)$ on $de = n_0^2 + n_0 + 1$ with both p and q prime, shown on logarithmic axes. For $n_0 = 255$ both lattice points with $d \leq e$ give primes (filled). For $n_0 = 65535$ both lattice points give a prime p and a composite q (open).

so the pair (y, z) comes from the divisors of an integer of size about $n_0^2 x^2$. For $n_0 = 4294967295$ there are about $3.8 \cdot 10^8$ primes s in the range, and $1.3 \cdot 10^8$ of them pass the congruence prune: $s \equiv 2 \pmod{3}$ as Lemma 2.2 requires, and $s \not\equiv 1$ modulo 5, 17, 257 and 65537. Each of them leaves an integer of about forty digits to factor. This is the prefix 3, 5, 17, 257, 65537 that dominates the case $k = 8$ in Section 7.4.

7.4. Eight prime factors. Let n satisfy (1.2) with $\omega(n) = 8$. Then $n > 3$, and n is odd and squarefree by Lemma 2.1. If $3 \nmid n$, then $\omega(n) \geq 16$ by Theorem 1.4; hence $3 \mid n$. By Lemma 2.2 the quotient M is either 2 or at least 5, and in the second case $\omega(n) \geq 1540$. Hence $n + 1 = 2\varphi(n)$

and $p_1 = 3$, and it suffices to run the search with $\varepsilon = +1$ and $k = 8$ over the prefixes that begin with 3.

The traversal down to depth 5 takes a few seconds. It yields 10459 prefixes $(p_1, \dots, p_5)$ with a non-empty interval for p_6 , and 10458 of them begin with 3. Their intervals have total length $1.3045 \cdot 10^{10}$, and most of it belongs to a few prefixes (Figure 18, left): $8.59 \cdot 10^9$ to 3, 5, 17, 257, 65537, that is, to the solution 4294967295 itself, where $C_5 = 1$; $1.23 \cdot 10^9$ to 3, 5, 17, 257, 65543; and $1.67 \cdot 10^8$ to 3, 5, 17, 353, 929, the solution 83623935, where again $C_5 = 1$. Thirty-four prefixes have an interval longer than 10^7 , and 326 one longer than 10^6 . The intervals contain 228260561 primes s that pass the congruence prune, 131788942 of them for the prefix 3, 5, 17, 257, 65537. Each pair of a prefix and such a prime $s = p_6$ leaves the two-prime problem of Proposition 4.1, in which N has between 15 and 40 digits.

These problems lie far to the left of those of Figure 9. For the prefix 3, 5, 17, 257, 65537 and $p_6 = s$ one has $A' = (2^{32} - 1)s$ and $B' = 2^{31}(s - 1)$, so that the modulus is $c = s - 2^{32}$, and $N = 2A'B' + c$ is close to $2^{64}s^2$. As s runs over $[2^{32} + 1, 3 \cdot 2^{32} - 1]$, the ratio c^3/N stays below $2.1 \cdot 10^{-10}$, and the number $(N/c^3)^{1/2}$ that governs the methods of Section 4 grows from $7 \cdot 10^4$ at the upper end to $4 \cdot 10^{16}$ at the first admissible prime, $s = 4294967357$, where $c = 61$. We therefore combine the sum route with trial division and with a complete factorisation, as follows.

Fix a prefix and an admissible prime s , define A' , B' , c , N as in Section 4, and let $t = cp - 2B'$ be the divisor of Proposition 4.1. Since $p \geq s + 2$, it satisfies $t \geq t_0 = c(s + 2) - 2B'$, and we choose a split point t_d with $t_0 \leq t_d$.

- (i) The divisors with $t < t_d$ are found by running p over the integers with $t_0 \leq cp - 2B' < t_d$ and testing whether t divides $A'p + \varepsilon$, as in Proposition 3.5. Since $c(A'p + \varepsilon) = A't + N$ and $\gcd(t, c) = 1$ by Proposition 4.1, this is the same as testing whether t divides N , and the program does that when $A'p$ is too large for 128-bit arithmetic.
- (ii) For the divisors with $t_d \leq t \leq \sqrt{N}$ we use the sum $\sigma = p + q$ and the product $\pi = pq$. By (3.5),

$$c\pi = 2B'\sigma - 2B' + \varepsilon, \quad (q - p)^2 = \sigma^2 - 4\pi, \quad (7.2)$$

so σ lies in the class of $(2B' - \varepsilon)(2B')^{-1}$ modulo c , and $t + N/t = c\sigma - 4B'$. As $x + N/x$ decreases on $(0, \sqrt{N}]$, the sum runs over its class in the range $2\sqrt{N} + 4B' \leq c\sigma \leq t_d + N/t_d + 4B'$, and each value is tested by asking whether $\sigma^2 - 4\pi$ is a perfect square. This is the route of Section 4.4, written in terms of p and q instead of u and v : the two sums differ by the constant $(2r + 4B')/c$.

- (iii) Both loops are sieved by congruences. Let $p < q$ be integers with $A'pq + 1 = 2B'(p - 1)(q - 1)$. No p_i divides $p - 1$ or $q - 1$, since it would then divide 1. If moreover p and q have no prime factor up to 61, which holds for primes larger than s because every admissible s exceeds 61, then modulo every prime $\ell \leq 61$ they avoid the residue 0, and also the residue 1 when ℓ is one of the p_i ; in particular they are odd and congruent to 2 modulo 3. Call these residues excluded. Since $X^2 - \sigma X + \pi = (X - p)(X - q)$, the roots of this polynomial modulo a prime ℓ are p and q modulo ℓ . The sieve discards only values of σ that are odd, or not congruent to 1 modulo 3, or for which $\sigma^2 - 4\pi$ is not a square modulo one of 2^8 , 3^2 , 5^2 and 7^2 , or for which, for some prime $\ell \leq 43$, the polynomial has no root modulo ℓ or a root at an excluded residue; and in the trial division it discards only values of p at an excluded residue modulo some prime $\ell \leq 61$. Every other value is tested exactly. So the sieve keeps every value that comes from integers p and q without prime factors up to 61, and in particular every value that comes from primes.
- (iv) If the estimated work of (i) and (ii) exceeds four milliseconds, the time we allow for factoring N , then N is factored completely with PARI/GP [30], every prime factor is

proved prime,¹² and all divisors t of N with $t_0 \leq t \leq \sqrt{N}$ in the class of Proposition 4.1 are listed.

Since c is odd and prime to 3, the admissible values of σ form one class modulo $6c$. The work of (i) and (ii) is therefore about $(t_d - t_0)/c$ values of p and $(t_d + N/t_d - 2\sqrt{N})/(6c^2)$ values of σ , and t_d is chosen to minimise it (Figure 17), with the two kinds of steps weighted by their measured costs. With equal weights the minimum lies near $t_d = (N/6c)^{1/2}$ and amounts to about $2(N/6c^3)^{1/2}$ steps, as in Section 4.4. By the argument of that section, (i) and (ii) together find every divisor $t \geq t_0$ of N in the class up to $\sqrt{N}$, and so does (iv); the sieve (iii) loses none that comes from integers p and q without prime factors up to 61. Where σ and π are too large for 128-bit arithmetic, (ii) is carried out in multiprecision arithmetic. Now let n be a solution with $p_1 = 3$ and $k = 8$. By Theorem 3.6 its first five primes form one of the prefixes and p_6 is one of the admissible primes s of its interval, and by Proposition 4.1 its last two primes give a divisor $t \geq t_0$ in the class. Hence n is found. Every completion $s < p < q$ in integers that is found is checked against $A'pq + 1 = 2B'(p-1)(q-1)$ exactly, and p and q are tested for primality.

The computation examined all 228260561 pairs; in 1967265 of them N was factored, with up to 39 digits (Appendix A.8 gives the size of the run). It found 21 completions $s < p < q$ in integers and no solution. Sixteen of the completions lie under the prefix 3, 5, 17, 353, 929 with $s = 83623937$, where $3 \cdot 5 \cdot 17 \cdot 353 \cdot 929 \cdot s$ is the solution $n_0 = 6992962672132095$. They are the pairs $p = n_0 + 1 + d$, $q = n_0 + 1 + e$ of Theorem 1.6(ii), one for each factorisation $n_0^2 + n_0 + 1 = de$ with $d < e$, and by Table 8 none of them consists of two primes. In each of the other five, listed in Table 9, p or q is divisible by one of $p_1, \dots, p_5$. So (1.2) has no solution with eight prime factors, and together with Section 7.1 this proves Theorem 1.3.

TABLE 9. The completions $s < p < q$ in integers of the case $k = 8$ other than the sixteen two-prime extensions of 6992962672132095, with the factorisations of p and q .

$s = p_6$	p	q
prefix 3, 5, 17, 257, 65537		
4294967459	$17 \cdot 43 \cdot 673 \cdot 230521189693$	$5 \cdot 409642859 \cdot 26363459459$
4294967969	27419312914203719	$5 \cdot 19 \cdot 5011 \cdot 7699 \cdot 21393991009$
4596195557	65533391633	$17 \cdot 20923232569978145771052877$
prefix 3, 5, 17, 353, 929		
83625827	$3^2 \cdot 5^3 \cdot 13 \cdot 252862031$	$3 \cdot 5 \cdot 7 \cdot 13 \cdot 113 \cdot 794656817253868193$
83760029	$3 \cdot 5 \cdot 3431155289$	$3 \cdot 87389293 \cdot 45294677621$

8. CHECKS ON THE COMPUTATIONS

A search that finds nothing proves something only if it would have found what it looks for. Every search of this paper was therefore carried out by at least two programs written independently, and every count reported here is the same for all of them. For Theorems 1.1 and 1.4 one of the two programs never factors an integer, so that its negative answers depend on no primality proof. The programs were also run on equations that do have solutions: on

¹²With the option `factor_proven` set, PARI/GP certifies every prime factor it returns. A factor below 2^{64} is certified by its Baillie–PSW test, which no composite number below 2^{64} passes; the PARI/GP documentation records that this was checked against the table of base-2 pseudoprimes below 2^{64} of Feitsma and Galway [12]. A larger factor is certified by the $n-1$ method [5] when enough of $n-1$ is known, and otherwise by the test of Adleman, Pomerance and Rumely [1] in the form of Cohen and Lenstra [9].

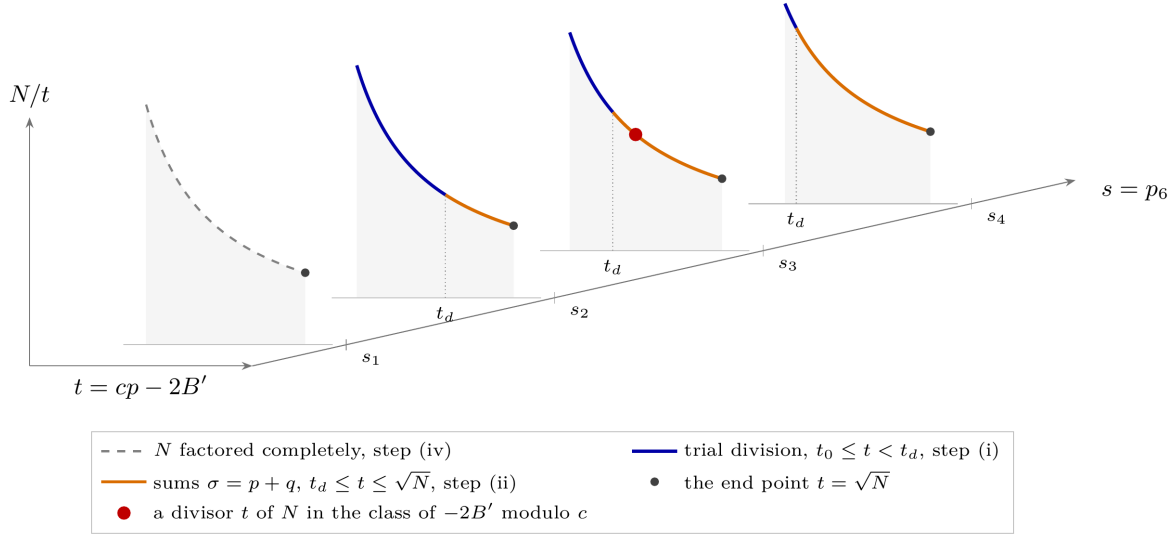


FIGURE 17. Steps (i), (ii) and (iv) for four admissible values $s_1 < s_2 < s_3 < s_4$ of p_6 , not to scale. The divisors $t = cp - 2B'$ of N with $t_0 \leq t \leq \sqrt{N}$ lie on the branch $t \mapsto N/t$. Trial division covers $t < t_d$ and the sums cover $t_d \leq t \leq \sqrt{N}$. The split point t_d is chosen for each s to minimise the work, and where both routes would be slow, as for s_1 , the integer N is factored completely.

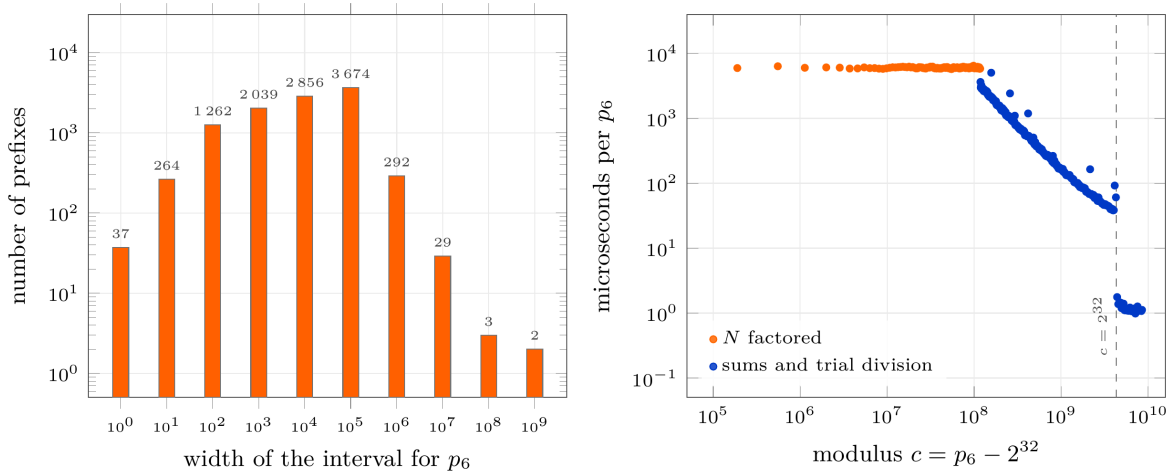


FIGURE 18. The case $k = 8$ of the companion equation. Left: the 10458 prefixes $(3, p_2, \dots, p_5)$ by the width of the interval for p_6 , on logarithmic axes; the two in the last bar end in 65537 and 65543. Right: processor time per admissible p_6 under the prefix 3, 5, 17, 257, 65537, one point for each piece of the run, against the modulus $c = p_6 - 2^{32}$. For c below about 10^8 the integer N is factored, and each p_6 costs about six milliseconds in all; above that the sums and the trial division are faster, and their cost falls as c grows. Beyond $c = 2^{32}$, that is for $p_6 > 2^{33}$, the condition $p > p_6$ leaves only a short range of divisors.

the equation $2^k(n-1) = (2^k + h)\varphi(n)$, whose 56 solutions with $k = 4$ and $k = 5$ in the ranges of Appendix B they all return, and on (1.2), where they return every known solution with at

most six prime factors at the right depth and nothing else. The divisor routines of Section 4 were compared with complete factorisations and tested on planted divisors, and the program of Section 7.4 was tested in the same ways and partly run twice. Appendix A gives the details.

9. THE NEXT CASES

The methods of Section 4 settle the case $k = 15$ in a quarter of an hour on a single processor, and the case of eight primes of the companion equation in 6.9 hours. This section explains why they do not reach the next cases, and which properties of the primes any further progress has to use.

9.1. Lehmer's equation with sixteen primes. By Theorem 1.2(i) a solution with sixteen prime factors would lie below 2^{512} . For $k = 16$ the traversal down to depth 12 is again a matter of seconds. It produces 90092 prefixes with four primes still to choose, and their intervals for p_{13} have total length $1.6 \cdot 10^9$ (Figure 19). They contain $1.2 \cdot 10^8$ primes, of which $5.4 \cdot 10^7$ pass the congruence prune. The widest interval, of length $1.6 \cdot 10^8$, belongs once more to the prefix ending in 25867. Each admissible p_{13} now leaves three primes to choose, and every admissible p_{14} then leaves a two-prime problem of the kind solved in Section 4. Near the lower end of a long interval the shortfall C_{13}/A_{13} after p_{13} is tiny, so the interval for p_{14} is very long. If $T - 1$ is the shortfall after twelve primes, and x_0 is about $1/(T - 1)$, a choice $p_{13} = x_0 + d$ leaves a shortfall of about $(T - 1)^2 d$ and hence an interval of length about $2/((T - 1)^2 d)$ for p_{14} . Summing over the admissible primes p_{13} , and counting the primes p_{14} with density $1/\log x$ and the share of them that passes the prune, gives to first order some $4 \cdot 10^{14}$ pairs of primes (p_{13}, p_{14}) , of which about $9 \cdot 10^{13}$ are admissible, two thirds of these from the single prefix ending in 25867, where $T - 1 = 1.8 \cdot 10^{-8}$. The integers N of the resulting two-prime problems are as large as 10^{86} , and about 10^{77} on average. At the average speed of the sum route for $k = 15$, 24 microseconds per problem, the admissible pairs alone would take $2 \cdot 10^9$ processor seconds. The problems with a small modulus add to this. Close to the lower end of each interval for p_{14} the modulus c is small: about eleven per cent of the problems have $c^3 < N$, where the work grows like $(N/c^3)^{1/2}$, some 10^9 have $c^3/N < 10^{-12}$, and for the smallest c the integer N would have to be factored. With these the total is of the order of 10^{10} processor seconds, some three centuries. The case $k = 16$ is therefore far beyond a single machine. It needs either a computation of that size or a method that treats a whole range of p_{13} at once.

9.2. The companion equation with nine primes. For (1.2), Theorems 1.3 and 1.4 leave open the solutions with $3 \mid n$ and $\omega(n) \geq 9$, and those with $3 \nmid n$ and $\omega(n) \geq 16$. In the first family the Fermat primes make the next case far harder than the last. Under the prefix 3, 5, 17, 257, 65537 every admissible $p_6 = s$ now leaves three primes to choose, and the shortfall after six primes is c/A' with $c = s - 2^{32}$ and $A' = (2^{32} - 1)s$, so that the interval for p_7 runs from about A'/c to $3A'/c$. Near the lower end of the interval for p_6 this is an interval of length about 10^{18} . Counting the admissible primes s exactly up to $s = 2^{32} + 10^7$ and estimating the rest, and counting the primes p_7 with density $1/\log x$ and the share of them that passes the prune, gives some $1.6 \cdot 10^{17}$ two-prime problems from this one prefix, each with an integer N of between 60 and 75 digits. At a microsecond per problem this would take five thousand years of processor time, so the case $k = 9$ with $3 \mid n$ is out of reach of these methods.

9.3. Pseudo-solutions. Apart from the primality tests, the traversal of Section 3 uses only the following properties of the tuple $(p_1, \dots, p_k)$: its entries are odd and increasing, they satisfy

$$x_1 x_2 \cdots x_k + \varepsilon = 2(x_1 - 1)(x_2 - 1) \cdots (x_k - 1), \quad (9.1)$$

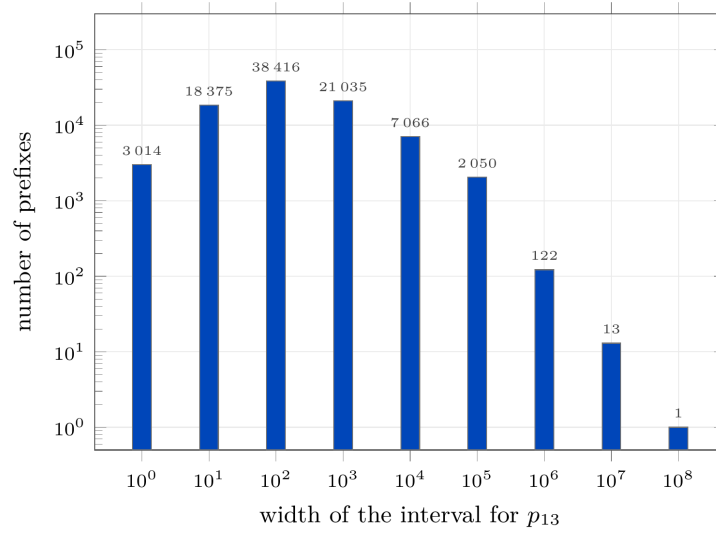


FIGURE 19. The 90092 prefixes of twelve primes for $n - 1 = 2\varphi(n)$ with $k = 16$, by the width of the interval for p_{13} , on logarithmic axes. The single prefix in the last bar ends in 25867.

and $\gcd(x_i, x_j - 1) = 1$ for all i and j . Indeed the proofs of Propositions 3.1, 3.2 and 3.5 use nothing else once $\varphi(n)$ is replaced by $\prod_i (x_i - 1)$. We call a tuple with these properties a *pseudo-solution* if not all its entries are prime.

Proposition 9.1. *Let $n_0 = p_1 \cdots p_m$ be a Fermat-type solution with $m \geq 2$ prime factors $p_1 < \cdots < p_m$. Then $(p_1, \dots, p_m, n_0)$ is a pseudo-solution of (9.1) with $\varepsilon = -1$.*

Proof. The entries are odd and increasing, and n_0 is composite. Since n_0 is squarefree, $\prod_i (p_i - 1) = \varphi(n_0) = (n_0 + 1)/2$, and so

$$p_1 \cdots p_m n_0 - 1 = n_0^2 - 1 = 2 \cdot \frac{n_0 + 1}{2} \cdot (n_0 - 1) = 2 \prod_{i \leq m} (p_i - 1) \cdot (n_0 - 1).$$

Lemma 2.1(ii), applied to n_0 with $\varepsilon = +1$, gives $p_i \nmid p_j - 1$ for all $i, j \leq m$; hence $\gcd(p_i, p_j - 1) = 1$ and, as n_0 is the product of the p_i , $\gcd(n_0, p_j - 1) = 1$. Finally $\gcd(n_0, n_0 - 1) = 1$, and so $\gcd(p_i, n_0 - 1) = 1$ because $p_i \mid n_0$. $\square$

The six Fermat-type solutions in (1.3) with at least two prime factors give pseudo-solutions with $k = 3, 4, 5, 6, 6, 7$ entries, the smallest being $(3, 5, 15)$, for which $3 \cdot 5 \cdot 15 - 1 = 224 = 2 \cdot 2 \cdot 4 \cdot 14$. Run with $\varepsilon = -1$ and $p_{\min} = 3$, the search of Section 3 reaches each of them as a lattice point on the hyperbola (3.5) and rejects it only because its last entry is composite. In the proof of Theorem 1.1 they never arise, because Lemma 2.2 excludes $3 \mid n$, and that lemma rests on primality once more: its proof needs that a prime other than 3 is prime to 3, and the entry 15 is not. Consequently no argument that uses only (9.1), the order of the entries and the conditions $\gcd(x_i, x_j - 1) = 1$ can exclude solutions of $n - 1 = 2\varphi(n)$ with k prime factors for any k with $3 \leq k \leq 7$; the primality of the entries has to enter the argument.

9.4. Pseudo-solutions prime to 3. The conditions $\gcd(x_i, x_j - 1) = 1$ are consequences of (9.1), since a common divisor of $x_1 \cdots x_k$ and $\prod_j (x_j - 1)$ divides ε . What separates the pseudo-solutions of Proposition 9.1 from solutions of Lehmer's equation is the prime 3.

Lemma 9.2. *Let $x_1, \dots, x_k$ be integers with $x_1 \cdots x_k + \varepsilon = M \prod_i (x_i - 1)$, where $\varepsilon = \pm 1$, and suppose that exactly $\nu \geq 1$ of them are divisible by 3. Then $\varepsilon \equiv (-1)^\nu M \pmod{3}$. In particular a solution of (9.1) has ν even if $\varepsilon = -1$, and $\nu = 0$ or ν odd if $\varepsilon = +1$.*

Proof. A common divisor of x_i and $x_j - 1$ divides both $x_1 \cdots x_k$ and $M \prod_i (x_i - 1)$, hence ε , so $\gcd(x_i, x_j - 1) = 1$ for all i and j . Let $3 \mid x_i$. Then no x_j is congruent to 1 modulo 3. Hence $x_j - 1 \equiv -1 \pmod{3}$ if $3 \mid x_j$ and $x_j - 1 \equiv 1 \pmod{3}$ otherwise, and the equation modulo 3 reads $\varepsilon \equiv (-1)^\nu M$. For $M = 2$ this is $\varepsilon \equiv (-1)^{\nu+1}$. $\square$

For $\nu = 1$ this is the congruence of Lemma 2.2. The pseudo-solutions that Proposition 9.1 builds from the solutions in (1.3) have $\nu = 2$, since 3 divides both p_1 and n_0 , whereas the prime factors of n give $\nu \leq 1$. The fact that 3 divides at most one prime factor therefore removes all of them, and by Lemma 9.2 a pseudo-solution of Lehmer's equation with pairwise coprime entries has $\nu = 0$. We call a pseudo-solution with $\nu = 0$ *prime to 3*.

Theorem 9.3. *Let $\varepsilon = \pm 1$.*

- (i) *No pseudo-solution of (9.1) with $k \leq 13$ is prime to 3.*
- (ii) *For $k \leq 15$ no pseudo-solution of (9.1) prime to 3 has $x_1, \dots, x_{k-3}$ prime.*

Proof. The proofs of Propositions 3.1, 3.2 and 3.5 use only (9.1) and the order of the entries, so the traversal of Section 3 applies to odd integers prime to 3, with the congruence prune replaced by $x \not\equiv 0$ modulo the odd primes of B_j and $x \not\equiv 1$ modulo the primes of A_j . For (i) the tree is traversed with integer entries down to depth $k - 3$; for $k = 12$ it has 489 nodes at that depth, below which lie 55423 admissible values of x_{k-2} . For (ii) the traversal with prime entries is run to depth $k - 3$. For $k \leq 12$ in (i), and for every k in (ii), every admissible odd x_{k-2} prime to 3 in the interval is followed by the sum route of Section 4.4, which does not use primality. For $k = 15$ in (ii) these are $1.43 \cdot 10^8$ values of x_{k-2} for each sign, treated in about four minutes per sign; for eight of them per sign the route would be long, and there $2AB + \varepsilon C$ is factored completely, with every prime factor proved prime. No completion occurs.

For $k = 13$ in (i) the tree has 86459 nodes at depth 10, the same for both signs. Their intervals for x_{11} have total length $1.07 \cdot 10^{11}$ and contain $1.64 \cdot 10^{10}$ admissible values; the longest, of length $1.9 \cdot 10^{10}$, belongs to the prefix 5, 7, 13, 17, 19, 23, 25, 37, 119, 147089. These values are treated by the program of Section 7.4, with the conditions on primes replaced by those of the integer prune: modulo 2, modulo 3 and modulo the odd primes of B_{10} the residue 0 is excluded, and modulo the primes of A_{10} the residue 1. Every completion in odd integers prime to 3 avoids these residues, because $\gcd(x_i, x_j - 1) = 1$, so the sieve loses none of them. For 73106 values, near the lower ends of the longest intervals, the integer $2AB + \varepsilon C$, of up to 54 digits, is factored completely instead, with every prime factor proved prime. The program chooses these values from its estimate of the running time, which depends on ε only through the term εC , negligible beside $2AB$; it chooses the same 73106 values for both signs. The run took about 15 hours of processor time per sign, 2.5 of them in PARI/GP, and found no completion in odd integers prime to 3 for either sign. A second program factors each integer $2AB + \varepsilon C$ completely with PARI/GP [30], proves every prime factor prime, and runs through all its divisors in the class of Proposition 4.1. It traverses the integer tree independently, with its own bounds, and repeats (i) for $k \leq 12$; it repeats (ii) for $k \leq 14$, including all 67626 values of x_{11} at $k = 14$, with the same result. The case $k = 13$ of (i) rests on the program of Section 7.4 and its checks in Appendix A.8. Allowed to use the entry 3, the second program finds 1, 1, 2, 8 and 47 solutions of (9.1) in odd integers with $\varepsilon = -1$ for $k = 3, \dots, 7$, all with $\nu \in \{2, 4\}$, and 1, 1, 2, 4 and 18 with $\varepsilon = +1$, all with ν odd; the first program, run on prime prefixes, finds exactly those whose first $k - 3$ entries are prime. $\square$

Theorem 9.3 says that for $k \leq 13$ the search behind Theorem 1.1, which solves (9.1) with $\varepsilon = -1$ and $x_1 \geq 5$, needs no primality beyond the fact that 3 divides at most one prime factor, and that for $k \leq 15$ it needs the primality of the first $k - 3$ factors only.

Tuples of this kind have also been studied by Molnar and Singh [23], who call integers $x_1, \dots, x_r \geq 2$, not necessarily distinct, with $\prod_i (x_i - 1)$ dividing $\prod_i x_i - 1$ a spoof Lehmer factorisation. They give an algorithm that finds all of them with a given number of entries, and list the 31 with odd entries and at most six of them [23, Theorem 7 and Table 1]. Twelve of the 31 have distinct entries, and all twelve have quotient 2. They are exactly the solutions of (9.1) with $\varepsilon = -1$ and $3 \leq k \leq 6$ found by the second program in the proof of Theorem 9.3, so the two computations agree. The six-entry ones include

$$\begin{aligned} (3, 5, 17, 257, 65717, 23794275), & \quad (3, 5, 17, 257, 68975, 1314417), \\ (3, 5, 17, 353, 929, 83623935), & \quad (3, 5, 17, 377, 1217, 2295), \\ (3, 5, 17, 395, 1059, 2303), & \quad (3, 5, 33, 53, 69, 8343), \end{aligned}$$

each of which satisfies $x_1 \cdots x_6 - 1 = 2(x_1 - 1) \cdots (x_6 - 1)$, as direct multiplication shows.

9.5. Congruences. The next proposition shows that Lemma 2.2 is the only obstruction to completing a prefix that a fixed modulus can detect.

Proposition 9.4. *Let $x_1 < \cdots < x_j$ be odd integers such that $A = \prod x_i$ and $B = \prod (x_i - 1)$ are coprime, let $m \geq 2$, and let ℓ^e be a power of a prime ℓ . Unless $\ell = 3$, $3 \mid A$ and $2B \not\equiv \varepsilon \pmod{3}$, there are integers $y_1, \dots, y_m$, none divisible by ℓ and, if $\ell \mid A$, none congruent to 1 modulo ℓ , such that*

$$A y_1 \cdots y_m + \varepsilon \equiv 2B \prod_{i \leq m} (y_i - 1) \pmod{\ell^e}.$$

In the excluded case no such integers exist.

Proof. If $\ell \nmid A$, take $y_1 = \cdots = y_{m-1} = 1$ and y_m with $A y_m \equiv -\varepsilon \pmod{\ell^e}$. If $\ell \mid A$, then $\ell \nmid 2B$. Take $y_3 = \cdots = y_m = 2$ and $y_2 = 1 + u$, where $u \in \{1, 2\}$ and $\ell \nmid 2Bu + \varepsilon$. Such a u exists, since ℓ cannot divide both $2B + \varepsilon$ and $4B + \varepsilon$; for $\ell = 3$ we take $u = 1$, which is allowed because $2B + \varepsilon \equiv 2\varepsilon \pmod{3}$ by hypothesis. Then $y_2, \dots, y_m$ are prime to ℓ and not congruent to 1 modulo ℓ . The congruence is then linear in y_1 with coefficient $K = 2^{m-2}A(1+u) - 2Bu \equiv -2Bu \not\equiv 0 \pmod{\ell}$, so it has a solution y_1 . Reducing modulo ℓ gives $K y_1 \equiv -(2Bu + \varepsilon)$, so $\ell \nmid y_1$, and $K(y_1 - 1) \equiv -\varepsilon$, so $y_1 \not\equiv 1$. In the excluded case every y_i is congruent to 2 modulo 3, so $y_i - 1 \equiv 1$, and the congruence modulo 3 reads $\varepsilon \equiv 2B$. $\square$

By the Chinese remainder theorem the same holds for every modulus. The integers y_i satisfy every condition modulo ℓ that the prune imposes on the remaining primes, namely $y \not\equiv 0$ modulo the primes of B and $y \not\equiv 1$ modulo the primes of A . So no congruence condition removes a node with at least two primes left, except that of Lemma 2.2. With one prime q left the situation changes: the equation $Aq + \varepsilon = 2B(q - 1)$ gives $q = (2B + \varepsilon)/C$ with $C = 2B - A$, so C must divide $2B + \varepsilon$, or equivalently $A + \varepsilon$, and this modulus depends on the prefix.

9.6. What a proof for all k has to use. In both cases the obstruction comes from prefixes whose product $\prod p/(p - 1)$ is very close to 2. Such a prefix forces the next prime to be large and leaves it a long interval. For (1.2) the Fermat primes give the extreme example, with a shortfall of 2^{-31} after five primes, and every prefix near them inherits a shortfall of the same order. By Proposition 9.4 no congruence condition removes such a prefix.

For the companion equation with $3 \mid n$ the tuple $(3, 5, 17, 257, 65537, 2^{32} + 1)$ is a pseudo-solution with pairwise coprime entries and $\nu = 1$. It is excluded only by the factorisation

$2^{32} + 1 = 641 \cdot 6700417$, since $641 - 1$ does not divide $n + 1 = 2^{64}$, so a proof for all k has to use the primality of the large entries.

For Lehmer's equation, and for the companion equation with $3 \nmid n$, the situation is different. A solution of Lehmer's equation with $M = 2$ has $3 \nmid n$ by Lemma 2.2, so its prime factors form a solution of (9.1) prime to 3, and the same holds for a solution of the companion equation with $M = 2$ and $3 \nmid n$. Theorem 9.3 shows that for $k \leq 13$ equation (9.1) has no solution in odd integers prime to 3, prime or not, so for these k the case $M = 2$, $3 \nmid n$ of both equations follows without using that the entries are prime. A first-moment count along the tree, weighting each two-entry problem by the expected number of divisors of $2AB + \varepsilon C$ in its class,¹³ predicts 0.8, 1.7, 5.2 and about 25 pseudo-solutions of Lehmer's equation with the entry 3 allowed for $k = 4, \dots, 7$, against the true 1, 2, 8 and 47. For pseudo-solutions prime to 3 it gives $4.3 \cdot 10^{-6}$ for $k = 12$, almost all of it below the prefix 5, 7, 13, 17, 19, 23, 25, 37, 119, which has the smallest value $C_9 = 2529647$ among the 910 prefixes of length nine in the tree for $k = 13$. Below that prefix it gives $4 \cdot 10^{-6}$, $5 \cdot 10^{-5}$ and $2 \cdot 10^{-4}$ for $k = 12, 13$ and 14. Theorem 9.3 is therefore what the count predicts. For larger k such solutions exist.

Theorem 9.5. *There are pseudo-solutions of (9.1) prime to 3 with $\varepsilon = +1$ for every $k \geq 25$, and with $\varepsilon = -1$ for every $k \geq 26$.*

Proof. Let $(x_1, \dots, x_k)$ be a pseudo-solution prime to 3 with $\varepsilon = +1$, and write $A = x_1 \cdots x_k$ and $B = \prod_i (x_i - 1)$, so that $A + 1 = 2B$. Suppose that $3 \mid B$. Every $x_i - 1$ divides B , so $x_k \leq B + 1 < A < 2B + 1$, and $\gcd(A, 2B) = \gcd(A, A + 1) = 1$.

Append $x = 2B + 1$. Then x is odd and $x \equiv 1 \pmod{3}$; it is prime to every $x_j - 1$, since $\gcd(x, B) = 1$; and $x - 1 = 2B$ is prime to every x_i and to x . Moreover

$$Ax + 1 = 2AB + A + 1 = 2B(A + 1) = 2 \prod_i (x_i - 1) \cdot (x - 1),$$

so the new tuple is a pseudo-solution prime to 3 with $\varepsilon = +1$, and 3 divides $B(x - 1)$.

Append $x = A$ instead. Then x is odd and prime to 3, it is prime to every $x_j - 1$ because $\gcd(A, B) = 1$, and $x - 1 = A - 1$ is prime to every x_i and to x . Moreover $A \cdot A - 1 = (A + 1)(A - 1) = 2B(A - 1)$, so the new tuple is a pseudo-solution prime to 3 with $\varepsilon = -1$.

By induction it suffices to give one pseudo-solution prime to 3 with $\varepsilon = +1$, $k = 25$ and $3 \mid B$. Its first nine entries are the prefix 5, 7, 13, 17, 19, 23, 25, 37, 119 named above, so $3 \mid B$ and the entry 25 is composite. The remaining sixteen entries have between 6 and 131176 digits,¹⁴ and the product has 262353 digits. After 24 entries the defect $2B - A$ equals 29, and 29 divides $2B + 1$; the last entry is $(2B + 1)/29$. The entries are deposited with the data. The equation, the order of the entries, their residues modulo 2 and 3, and all 625 conditions $\gcd(x_i, x_j - 1) = 1$ are checked by two independent programs, one of them written in PARI/GP [30]. $\square$

The tuple was found by a search along the tree of Theorem 9.3. A prefix $x_1, \dots, x_j$ with $3 \mid B$ and defect $c = 2B - A > 0$ is completed by the entry $X = (2B + \varepsilon)/c$ whenever c divides $2B + \varepsilon$ and $X > x_j$, because X then satisfies every other condition: it is odd and prime to 3, $\gcd(X, B)$ divides ε , and $X - 1 = (A + \varepsilon)/c$ is prime to A . Appending x to a prefix gives the defect $cx - 2B$, so the defects of its children form an arithmetic progression with difference c . Starting from the prefixes of the tree for $k = 13$, the search kept at each depth the 10^7 admissible children

¹³An integer t in the class is counted as a divisor of $N = 2AB + \varepsilon C$ with probability $1/t$, so that a problem contributes about $\rho \log(\sqrt{N}/t_0)/C$, where t_0 is the least value of t that the order of the entries allows and the factor $\rho \leq 1$ corrects for the congruence conditions that both entries satisfy modulo small primes. Nodes with many children are sampled rather than enumerated.

¹⁴The tenth and eleventh entries are 147563 and 45571237, the twelfth is 46357278045502805, and from there on each entry has about twice as many digits as the one before (Figure 20).

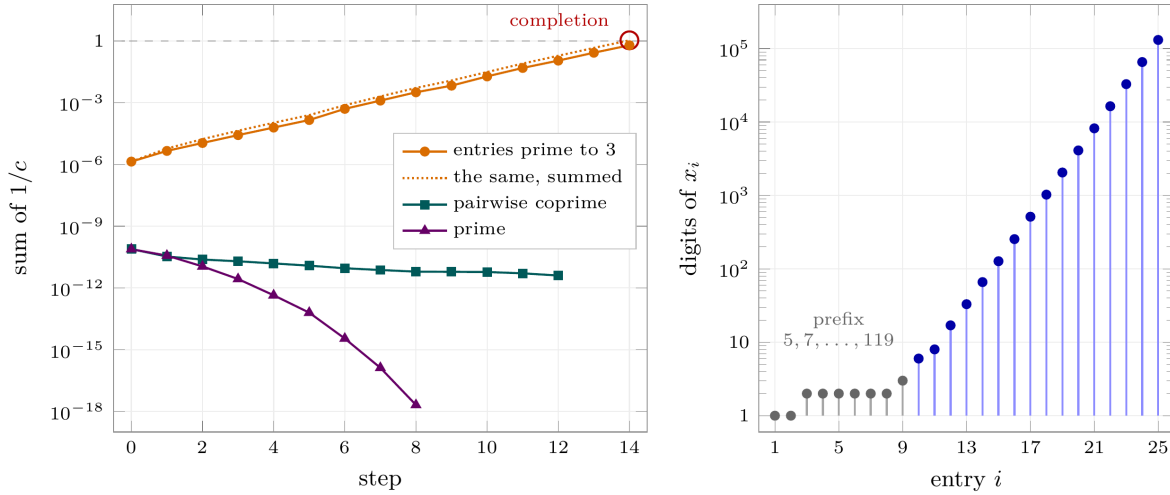


FIGURE 20. Left: the search for pseudo-solutions prime to 3. The sum of $1/c$ over the prefixes kept at each step estimates the number of completions at that step. With 10^7 prefixes kept it grows from $1.4 \cdot 10^{-6}$ to 0.62, its running total passes 1 at step 14, and the tuple of Theorem 9.5 is completed there. With pairwise coprime entries and 10^6 prefixes kept it falls from $7.7 \cdot 10^{-11}$ to $4.0 \cdot 10^{-12}$ in twelve steps, and with prime entries and 5000 prefixes kept it falls from $7.6 \cdot 10^{-11}$ to $2.1 \cdot 10^{-18}$ in eight (Section 9.8). Right: the number of digits of the 25 entries of that tuple, of which the first nine form the prefix 5, 7, 13, 17, 19, 23, 25, 37, 119.

with the smallest defect and tested each of them. The sum of $1/c$ over the prefixes kept at one step, which estimates the number of completions, grew by a factor of about 2.5 per step, and its total over all steps first exceeded 1 at the fourteenth step, where the completion above occurred (Figure 20).

None of these tuples has pairwise coprime entries, and the search does not reach such tuples. Among the 910 prefixes of length 9 of the tree for $k = 13$ with positive defect the smallest defect is 2529647, while among the 48 of them with pairwise coprime entries it is 8544556193. The traversal of the tree for $k = 15$ with pairwise coprime entries visits 63663 nodes, and the 13190 prefixes among them with $0 < cx_j < 2B$, which one more entry can complete, have a sum of $1/c$ of $7.7 \cdot 10^{-11}$, against $1.4 \cdot 10^{-6}$ for the 26821 prefixes from which the search above started. Moreover only about seven per cent of the children of a prefix are admissible and prime to every earlier entry, against fourteen per cent without the second condition. Run from these prefixes with 10^6 children kept at each step, the search finds the sum of $1/c$ falling by a factor of about 0.8 per step, and no completion in twelve steps.

Solutions of (9.1) with $\varepsilon = +1$ give solutions with $\varepsilon = -1$ through the analogue of Theorem 1.6(ii).

Proposition 9.6. *Let $(x_1, \dots, x_k)$ satisfy (9.1) with $\varepsilon = +1$, let $A = x_1 \cdots x_k$, and let $A^2 + A - 1 = de$ with $1 \leq d < e$. Then $(x_1, \dots, x_k, A + 1 + d, A + 1 + e)$ satisfies (9.1) with $\varepsilon = -1$. If $x_1, \dots, x_k$ are pairwise coprime, then so are all $k + 2$ entries if and only if $\gcd(d^2 - 1, A) = \gcd(d - 1, A + 2) = 1$, and this fails whenever 3 or 5 divides A .*

Proof. Write $X = A + 1 + d$ and $Y = A + 1 + e$. Since $2 \prod_i (x_i - 1) = A + 1$, the equation for the longer tuple reads $AXY - 1 = (A + 1)(X - 1)(Y - 1)$, which is equivalent to $(X - A - 1)(Y - A - 1) = A^2 + A - 1$, as in the proof of Theorem 1.6; the entries increase because $x_k \leq A < X < Y$.

Since $de \equiv -1 \pmod{A}$, the integer d is prime to A , so $\gcd(X, A) = \gcd(d+1, A)$ and $\gcd(Y, A) = \gcd(d(e+1), A) = \gcd(d-1, A)$. A prime r dividing X and Y has $d \equiv e \equiv -(A+1)$, hence $(A+1)^2 \equiv A^2 + A - 1$, that is $A \equiv -2$ and $d \equiv 1 \pmod{r}$; conversely, if r divides $A+2$ and $d-1$, then $e \equiv de = A^2 + A - 1 \equiv 1$, and r divides X and Y . If $3 \mid A$, then $A^2 + A - 1 \equiv -1 \pmod{3}$, so $3 \nmid d$ and $3 \mid d^2 - 1$. Finally let $5 \mid A$. Every prime factor r of $A^2 + A - 1$ is odd and different from 5, and $(2A+1)^2 \equiv 5 \pmod{r}$, so 5 is a square modulo r and, by quadratic reciprocity,¹⁵ $r \equiv \pm 1 \pmod{5}$. Hence $d \equiv \pm 1 \pmod{5}$, and 5 divides $\gcd(d^2 - 1, A)$. $\square$

For the Fermat numbers $3, 5, 17, 257, 65537, 2^{32} + 1$, Proposition 9.6 gives pseudo-solutions of Lehmer's equation with $\nu = 2$, and none with pairwise coprime entries, because 3 divides A . Appending $2B + 1 = A + 2$, as in the proof of Theorem 9.5, keeps the entries pairwise coprime, so a pairwise coprime pseudo-solution of the companion equation prime to 3 with $3 \mid B$ gives infinitely many, and Proposition 9.6 turns every divisor d of $A^2 + A - 1$ with $\gcd(d^2 - 1, A) = \gcd(d - 1, A + 2) = 1$ into a pseudo-solution of Lehmer's equation with pairwise coprime entries. By Lemma 9.2 a pseudo-solution of Lehmer's equation with pairwise coprime entries is prime to 3, and none has at most 13 entries, by Proposition 9.10(iii). The smallest k with a pseudo-solution prime to 3 lies between 14 and 25 for $\varepsilon = +1$, and between 14 and 26 for $\varepsilon = -1$.

9.7. Descent, three entries and Korselt's criterion. Equation (9.1) has degree one in each entry. If all entries but x are fixed, and A' and B' are the products of the other entries and of the numbers $x_j - 1$ for them, then the equation reads

$$x(2B' - A') = 2B' + \varepsilon,$$

so the remaining entry is determined by the others, and it exists exactly when $2B' - A'$ is positive and divides $2B' + \varepsilon$. This is the step that completes a prefix in the search above. It also means that the equation has no counterpart of the involution behind descent arguments for equations of degree two in each variable.¹⁶ Fixing $k - 1$ entries leaves at most one solution, so there is no second root to replace an entry with, and a descent of this kind has nothing to act on.

The last two entries are easier to find than the last three for a structural reason. By Proposition 3.5, the equation for the last two entries after a prefix with products A and B is a product of two linear forms set equal to the constant $2AB + \varepsilon C$, and its solutions come from the divisors of that integer. With three entries left this never happens.

Proposition 9.7. *Let A and B be positive integers with $C = 2B - A > 0$, and let*

$$F(x, y, z) = Axyz + \varepsilon - 2B(x-1)(y-1)(z-1),$$

so that x, y, z complete a prefix with products A and B exactly when $F(x, y, z) = 0$.

(i) *With $u = Cx - 2B$, $v = Cy - 2B$ and $w = Cz - 2B$,*

$$uvw - 2AB(u + v + w) = 12AB^2 - 8B^3 + (2B + \varepsilon)C^2 - C^2F(x, y, z).$$

(ii) *There are no linear polynomials L_1, L_2, L_3 in x, y, z with rational coefficients and no rational number κ with $F = L_1L_2L_3 + \kappa$.*

¹⁵For an odd prime $r \neq 5$ the law of quadratic reciprocity gives $\left(\frac{5}{r}\right) = \left(\frac{r}{5}\right)$, since $5 \equiv 1 \pmod{4}$, and the nonzero squares modulo 5 are 1 and 4.

¹⁶For Markov's equation $x^2 + y^2 + z^2 = 3xyz$, fixing y and z leaves a quadratic equation in x with the roots x and $3yz - x$. Replacing the largest entry of a solution by the other root gives a smaller solution, and every solution descends in this way to $(1, 1, 1)$.

Proof. Write $e_1 = x+y+z$ and $e_2 = xy+yz+zx$. Expanding, $F = -Cxyz + 2Be_2 - 2Be_1 + 2B + \varepsilon$, and

$$uvw = C^3xyz - 2BC^2e_2 + 4B^2Ce_1 - 8B^3, \quad u + v + w = Ce_1 - 6B.$$

Since $4B^2C - 2ABC = 2BC^2$, the left side of (i) equals $C^2(Cxyz - 2Be_2 + 2Be_1) - 8B^3 + 12AB^2$, and $Cxyz - 2Be_2 + 2Be_1 = 2B + \varepsilon - F$. This proves (i).

For (ii), suppose that $F = L_1L_2L_3 + \kappa$. The terms of degree three give $\ell_1\ell_2\ell_3 = -Cxyz$ for the homogeneous linear parts ℓ_i of the L_i , and by unique factorisation each ℓ_i is a rational multiple of one of x, y, z , a different one for each i . So $F = \alpha(x+r)(y+s)(z+t) + \kappa$ with rational α, r, s, t . Comparing the coefficients of xyz , of xy, yz, zx , and of x gives $\alpha = -C$, then $\alpha t = \alpha r = \alpha s = 2B$, and finally $\alpha st = -2B$. Hence $s = t = -2B/C$ and $-4B^2/C = -2B$, that is, $C = 2B$ and $A = 0$, which is impossible. $\square$

Part (i) shows what takes the place of the hyperbola (3.5): the product uvw is no longer a fixed integer but moves with $u + v + w$, and by (ii) no choice of linear forms turns it into one. The divisors of a single integer therefore do not give the last three primes. This is why, for $k = 16$, every admissible p_{14} has to be enumerated in Section 9.1, and why a method that treats a whole range of primes at once has to rest on something other than divisors.

Primality enters the search only through its primality tests. Let $(x_1, \dots, x_k)$ satisfy (9.1) and put $A = x_1 \cdots x_k$. Every $x_i - 1$ divides $2 \prod_j (x_j - 1) = A + \varepsilon$, so by Fermat's little theorem a prime entry x_i satisfies $a^{A+\varepsilon} \equiv 1 \pmod{x_i}$ for every integer a prime to x_i . For a composite entry this is a real condition.

Lemma 9.8. *Let $(x_1, \dots, x_k)$ satisfy (9.1), put $A = x_1 \cdots x_k$, and let x be an entry with $a^{A+\varepsilon} \equiv 1 \pmod{x}$ for every integer a prime to x . Then x is squarefree, and $q - 1 \mid A + \varepsilon$ for every prime $q \mid x$. If the entries are pairwise coprime and each of them has this property, then A is squarefree and $q - 1 \mid A + \varepsilon$ for every prime $q \mid A$; for $\varepsilon = -1$ and $k \geq 2$ this says, by Korselt's criterion, that A is a Carmichael number.*

Proof. Since $x \mid A$ and $\gcd(A, A + \varepsilon) = 1$, no prime factor of x divides $A + \varepsilon$. Let q be a prime factor of x . If $q^2 \mid x$, the Chinese remainder theorem gives an integer a prime to x with $a \equiv 1 + q \pmod{q^2}$, and then $1 \equiv a^{A+\varepsilon} \equiv 1 + (A + \varepsilon)q \pmod{q^2}$, so $q \mid A + \varepsilon$, which is impossible. Hence x is squarefree. Taking instead for a an integer prime to x that is a primitive root modulo q gives $q - 1 \mid A + \varepsilon$. For pairwise coprime entries the prime factors of A are those of the entries, each to the first power, which gives the last statement. $\square$

The tuples constructed in the proof of Theorem 9.5 lack the property, since each of them contains the entry 25, which is not squarefree. In the tuple with 25 entries it fails at other entries as well: $147563 = 13 \cdot 11351$ and $45571237 = 17 \cdot 2680661$ have the prime factors 11351 and 2680661, for which $q - 1$ does not divide $A + 1$, and each of the fourteen larger entries, all of them composite, has $2^{A+1} \not\equiv 1 \pmod{x_i}$. Of the composite entries only $119 = 7 \cdot 17$ has the property, as 6 and 16 divide $A + 1$. For the pseudo-solution $(3, 5, 17, 257, 65537, 2^{32} + 1)$ of the companion equation, where $A + 1 = 2^{64}$, the property fails at the last entry, since $2^{32} + 1 = 641 \cdot 6700417$ and $640 \nmid 2^{64}$; this is the factorisation that excludes it in Section 9.6.

Lemma 9.8 therefore singles out a class of tuples between the integer tuples of Section 9.6 and the prime ones. The prime factors of a solution of (1.1) or (1.2) with $M = 2$ and $3 \nmid n$ form a solution of (9.1) in pairwise coprime odd integers prime to 3, each of which satisfies $a^{A+\varepsilon} \equiv 1 \pmod{x_i}$ for every a prime to x_i . No tuple in this section has these properties: those of Propositions 9.1 and 9.6 and of Theorem 9.5, and the spoof factorisations above, all have two entries with a common factor or contain the entry 3. For $k \leq 13$ no such tuple exists, by Theorem 9.3.

9.8. Primality of the largest entry. The tuples of Sections 9.6 and 9.7 weaken the primality of every entry. Here we keep the primality of every entry but the largest and drop it there. The largest entry is the one that the step at the start of Section 9.7 determines from the others, and in the tuple $(3, 5, 17, 257, 65537, 2^{32} + 1)$ of Section 9.6 it is the entry whose primality decides the matter.

In the next lemma and the remarks after it, $p_1 < \dots < p_j$, with $j \geq 1$, are odd primes with $p_i \nmid p_l - 1$ for all i and l , and

$$A = p_1 \cdots p_j, \quad B = (p_1 - 1) \cdots (p_j - 1), \quad C = 2B - A.$$

A prime factor of A that divides B divides some $p_l - 1$, so $\gcd(A, B) = 1$.

Lemma 9.9. *If $C \geq 1$, then $\gcd(C, 2AB) = 1$. Suppose moreover that C divides $2B + \varepsilon$, and put $X = (2B + \varepsilon)/C$.*

- (i) $X - 1 = (A + \varepsilon)/C$ and $\gcd(X, 2B) = \gcd(X - 1, A) = 1$.
- (ii) If $C = 1$, then $X = A$ for $\varepsilon = -1$ and $X = A + 2$ for $\varepsilon = +1$.
- (iii) Let $\varepsilon = -1$. If $3 \mid A$, then $3 \mid X$. If $3 \nmid A$, then $3 \nmid X$ and $3 \nmid C$.

Proof. Since $\gcd(A, B) = 1$, we have $\gcd(C, B) = \gcd(A, B) = 1$ and $\gcd(C, A) = \gcd(2B, A) = 1$, and C is odd because A is.

(i) From $CX = 2B + \varepsilon$ we get $C(X - 1) = 2B + \varepsilon - C = A + \varepsilon$. So X divides $2B + \varepsilon$, which is prime to $2B$, and $X - 1$ divides $A + \varepsilon$, which is prime to A .

(ii) If $C = 1$, then $A = 2B - 1$ and $X = 2B + \varepsilon = A + 1 + \varepsilon$.

(iii) If $3 \mid A$, then 3 is one of the p_i , and every other p_l is congruent to 2 modulo 3 , since $3 \nmid p_l - 1$. Hence $B \equiv 2$, $2B - 1 \equiv 0$ and $C = 2B - A \equiv 1 \pmod{3}$, and $3 \mid X$. If $3 \nmid A$, then either some $p_l \equiv 1 \pmod{3}$ and $3 \mid B$, or every $p_l \equiv 2 \pmod{3}$ and $B \equiv 1 \pmod{3}$. In both cases $3 \nmid 2B - 1 = CX$. $\square$

By the step at the start of Section 9.7, a prime $p > p_j$ makes $n = Ap$ a solution of $n - 1 = 2\varphi(n)$ exactly when $C \geq 1$, C divides $2B - 1$ and $p = X$. Since $2B - 1 = C + (A - 1)$, the condition on the first j primes is the single divisibility $C \mid A - 1$, and the only further condition is that X be prime. This is the congruence of the remark after Proposition 9.4, whose modulus depends on the prefix. Once C divides $A - 1$ the identity $AX - 1 = 2B(X - 1)$ holds exactly, so every condition that Lehmer's equation imposes modulo a power of a prime q is contained in this one divisibility, including $v_q(n - 1) \geq \sum_i v_q(p_i - 1)$, which Korselt's criterion weakens to the largest of these valuations. For $\varepsilon = -1$, part (ii) is Proposition 9.1 read from the prefix, and part (iii) gives the case $M = 2$ of Lemma 2.2.

The next proposition asks nothing of the largest entry except that it be prime to the others. Its first part holds for every k .

Proposition 9.10. *Let $k \geq 2$, let $p_1 < \dots < p_{k-1}$ be odd primes, and let $x > p_{k-1}$ be an integer prime to $p_1 \cdots p_{k-1}$.*

- (i) *If $(p_1, \dots, p_{k-1}, x)$ satisfies (9.1) with $\varepsilon = -1$, then $p_i \nmid p_l - 1$ for all i and l , and, with A, B and C formed from $p_1, \dots, p_{k-1}$, we have $3 \nmid A$, $C \geq 5$, $C \mid A - 1$ and $x = (2B - 1)/C$. This holds with $x = p_k$ for every composite solution $n = p_1 \cdots p_k$ of $n - 1 = 2\varphi(n)$.*
- (ii) *If $k \leq 15$, then $(p_1, \dots, p_{k-1}, x)$ does not satisfy (9.1) with $\varepsilon = -1$.*
- (iii) *For $k \leq 13$, equation (9.1) with $\varepsilon = -1$ has no solution in pairwise coprime integers greater than 1.*

Proof. (i) Put $A = p_1 \cdots p_{k-1}$ and $B = (p_1 - 1) \cdots (p_{k-1} - 1)$. A common divisor of Ax and $B(x - 1)$ divides $Ax - 2B(x - 1) = 1$, so $\gcd(A, B) = 1$ and $p_i \nmid p_l - 1$ for all i and l , and the equation reads $Cx = 2B - 1$. So $C \geq 1$, C divides $2B - 1 = C + (A - 1)$, hence also $A - 1$, and

$x = X$ in the notation of Lemma 9.9. Since x is prime to A , part (iii) of that lemma gives $3 \nmid A$ and $3 \nmid C$, and part (ii) gives $C \neq 1$, since $X = A$ is not prime to $A > 1$. As C is odd, $C \geq 5$. By Lemma 2.1(i) a composite solution of $n - 1 = 2\varphi(n)$ is odd and squarefree, say $n = p_1 \cdots p_k$ with primes $p_1 < \cdots < p_k$ and $k \geq 2$. Then $n - 1 = 2 \prod_i (p_i - 1)$, so $(p_1, \dots, p_k)$ satisfies (9.1) with $\varepsilon = -1$, and $x = p_k$ is prime to the other entries.

(ii) Suppose that the tuple satisfies (9.1) with $\varepsilon = -1$ and $k \leq 15$. By part (i), $3 \nmid A$, and Lemma 9.9(iii) gives $3 \nmid x$. The entries are odd, since x divides $2B - 1$, and increasing, and they satisfy (9.1) with $\varepsilon = -1$ and hence the conditions $\gcd(x_i, x_l - 1) = 1$. If x is prime, then Ax is a composite solution of $\varphi(n) \mid n - 1$ with at most fifteen prime factors, which Theorem 1.1 excludes. Otherwise the tuple is a pseudo-solution prime to 3 whose first $k - 3$ entries are prime, which Theorem 9.3(ii) excludes.

(iii) Let pairwise coprime integers greater than 1 form such a solution. They are odd, since an even entry would make the left side of (9.1) odd and the right side even. By Lemma 9.2 the number of entries divisible by 3 is even, and pairwise coprime entries include at most one multiple of 3, so no entry is divisible by 3. If every entry is prime, Theorem 1.1 excludes the solution; otherwise it is a pseudo-solution prime to 3, which Theorem 9.3(i) excludes. $\square$

For the companion equation the analogue of Proposition 9.10(ii) is false. The tuple $(3, 5, 17, 257, 65537, 2^{32} + 1)$ of Section 9.6 has five prime entries, and its last entry $2^{32} + 1 = 641 \cdot 6700417$ is prime to them. With $A = 2^{32} - 1$ and $B = 2^{31}$ it satisfies $Ax + 1 = 2^{64} = 2B(x - 1)$, which is (9.1) with $\varepsilon = +1$. It comes from $C = 1$, which by Lemma 9.9(ii) gives the entry $A + 2$ for $\varepsilon = +1$. For $\varepsilon = -1$ the same step gives the entry A , which is not prime to the others; it produces the pseudo-solutions of Proposition 9.1 and those of Theorem 9.5 with $\varepsilon = -1$, and the condition $C \geq 5$ of Proposition 9.10(i) excludes it.

Proposition 9.10(ii) rests on the searches, and they give no shortcut beyond $k = 15$. After a prefix of $k - 2$ primes, with A, B and C now formed from these, the identity (3.5), whose proof uses only the equation, turns the last two entries $y < x$ into a factorisation $(Cy - 2B)(Cx - 2B) = 2AB - C$, and Proposition 9.10(ii) asks only that y be prime and x be prime to the other entries. Its extension to $k = 16$ therefore needs the traversal of Section 9.1, with its $9 \cdot 10^{13}$ two-prime problems.

The next lemma shows how the primality of the prefix keeps the defects large.

Lemma 9.11. *Let $x_1 < \cdots < x_j$ be integers greater than 1 with $A = x_1 \cdots x_j$ and $B = (x_1 - 1) \cdots (x_j - 1)$, suppose that $C = 2B - A \geq 2$ is prime to $2B$, and put $y_0 = \lfloor 2B/C \rfloor + 1$. An entry y appended to the prefix gives a positive defect exactly when $y \geq y_0$, and then*

$$Cy - 2B = r + (y - y_0)C, \quad r = Cy_0 - 2B, \quad 1 \leq r \leq C - 1.$$

In particular the defect decreases only for $y = y_0$. Moreover C divides $2B - 1$ exactly when $r = C - 1$, and then $(2B - 1)/C = y_0 - 1$.

Proof. Appending y gives the defect $2B(y - 1) - Ay = Cy - 2B$, which is positive exactly when $y > 2B/C$, that is, when $y \geq y_0$. Since $C \geq 2$ is prime to $2B$, it does not divide $2B$, so $r = C - (2B \bmod C)$ lies between 1 and $C - 1$. Finally $r = C - 1$ exactly when $2B \bmod C = 1$, that is, when $C \mid 2B - 1$, and then $2B - 1 = C \lfloor 2B/C \rfloor = C(y_0 - 1)$. $\square$

Along a path of integers every step may take the entry y_0 , and the defects can fall quickly: in the search behind Theorem 9.5 the smallest defect fell from $1.9 \cdot 10^6$ after the first step to 29 at the fourteenth. Along a path of independent primes, where $\gcd(C, 2B) = 1$ by Lemma 9.9, the defect falls only at a step where $\lfloor 2B/C \rfloor + 1$ happens to be prime. Of the 13190 prefixes from which the search with pairwise coprime entries starts, 12170 consist of primes, with a sum of $1/C$ of $7.6 \cdot 10^{-11}$. A search of the same kind in which every new entry is a probable prime,

started from these 12170 prefixes and keeping at each step the 5000 children with the smallest defect, saw the smallest defect rise from $1.2 \cdot 10^{11}$ to $2.2 \cdot 10^{18}$ in eight steps. The sum of $1/C$, which estimates the number of completions by an integer, fell to $2.1 \cdot 10^{-18}$, with a total of $1.3 \cdot 10^{-10}$ over all steps and no completion (Figure 20). With integer entries, as in the search behind Theorem 9.5, and the same prefixes and number of children, the sum rose instead, to a total of $1.9 \cdot 10^{-8}$ after twelve steps. These counts are heuristic, and no result of this paper depends on them.

For Lehmer's equation with $M = 2$ the tuples of this section compare as follows. The prime factors of a composite solution of $n - 1 = 2\varphi(n)$ form a solution of (9.1) with $\varepsilon = -1$ of each of three kinds: in odd integers prime to 3; in pairwise coprime odd integers that pass the test of Lemma 9.8; and in primes followed by a largest entry prime to them. Solutions of the first kind exist for every $k \geq 26$ (Theorem 9.5). Solutions of the second and third kinds have pairwise coprime entries, so none has $k \leq 13$ (Proposition 9.10(iii)), and none of the third kind has $k \leq 15$ (Proposition 9.10(ii)). Neither of these two kinds contains the other: the second asks of every entry a property weaker than primality, and the third asks every entry but the largest to be prime and nothing of the largest except that it be prime to the others. By Proposition 9.10(i) a solution of the third kind comes from a prefix of independent odd primes with $C \geq 5$, $C < (2B - 1)/p_{k-1}$ and $C \mid 2B - 1$. By Proposition 9.4 no congruence modulo a fixed number other than that of Lemma 2.2 removes a prefix with two or more entries left, so excluding these prefixes for $k \geq 16$ needs the modulus C of each prefix.

9.9. Prime factors of the last defect. Let $n = p_1 \cdots p_k$ be a composite solution of $n - 1 = 2\varphi(n)$, and form A , B and C from $p_1, \dots, p_{k-1}$ as in Section 9.8. We call C the *last defect* of n . By Proposition 9.10(i), $C \geq 5$, $C \mid A - 1$ and $p_k = (2B - 1)/C$, so the first $k - 1$ prime factors determine the last through C (Figure 21). By Corollary 2.6, every composite solution of $\varphi(n) \mid n - 1$ with at most 16000 prime factors satisfies $n - 1 = 2\varphi(n)$ and is prime to 3. The prime factors of C restrict the other prime factors of n in the way that the prime 3 does in Lemma 2.2.

Lemma 9.12. *Let $n = p_1 \cdots p_k$ be a composite solution of $n - 1 = 2\varphi(n)$ with last defect C , and let ℓ be a prime factor of C .*

- (i) $\ell \neq p_i$ and $p_i \not\equiv 1 \pmod{\ell}$ for every $i < k$.
- (ii) $A \equiv 1$ and $2B \equiv 1 \pmod{C}$, so that $\prod_{i < k} (1 - 1/p_i) \equiv \frac{1}{2} \pmod{C}$.
- (iii) $\prod_{i < k} \frac{p_i}{p_i - 1} < 2 < \prod_{i \leq k} \frac{p_i}{p_i - 1}$.

Proof. By Lemma 9.9, $\gcd(C, 2AB) = 1$. (i) The prime ℓ divides neither $A = p_1 \cdots p_{k-1}$ nor $B = (p_1 - 1) \cdots (p_{k-1} - 1)$. (ii) The first congruence is $C \mid A - 1$, and the second is $2B - 1 = p_k C$. As A and B are prime to C , the product $\prod_{i < k} (1 - 1/p_i) = B/A$ is congruent to $\frac{1}{2}$ modulo C . (iii) The product over $i < k$ is $A/B = 2 - C/B < 2$, and the product over $i \leq k$ is $n/\varphi(n) = 2 + 1/\varphi(n)$ by Lemma 2.1(iii). $\square$

Part (i) removes from the first $k - 1$ prime factors the prime ℓ and the primes congruent to 1 modulo ℓ , which by the prime number theorem for arithmetic progressions make up the proportion $1/(\ell - 1)$ of all primes, and part (iii) then needs many of the primes that remain. To measure this, let $\ell \geq 5$ be prime and $K \geq 1$. We call a set S of K primes an ℓ -set if its elements are at least 5, different from ℓ and not congruent to 1 modulo ℓ , if S is independent, and if

$$\prod_{p \in S} \frac{p}{p - 1} < 2 < \prod_{p \in S} \frac{p}{p - 1} \cdot \frac{q_S}{q_S - 1}, \quad (9.2)$$

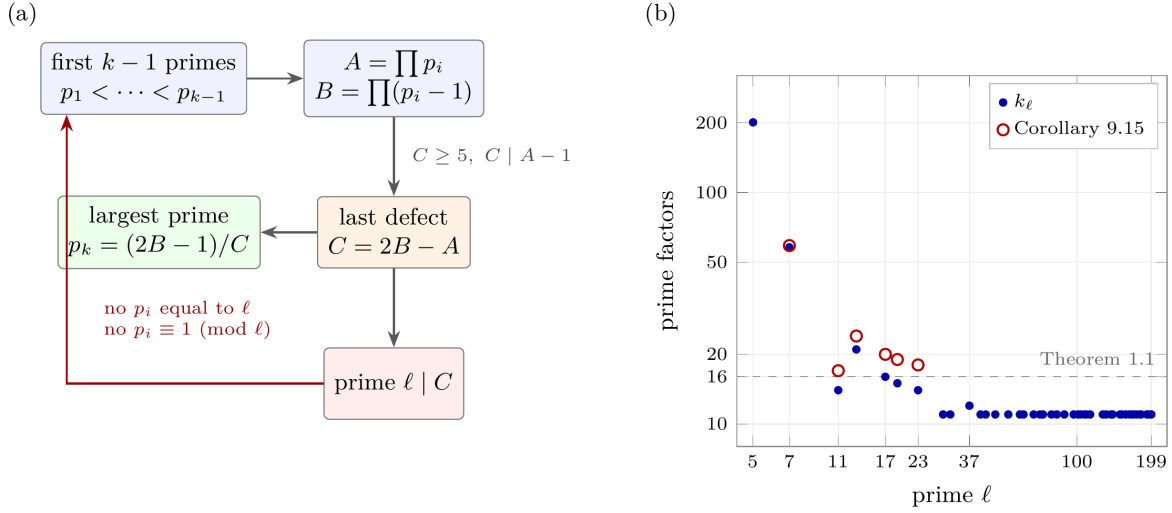


FIGURE 21. (a) The first $k-1$ prime factors of a solution of $n-1 = 2\varphi(n)$ determine A , B and the last defect C , and C determines the largest prime factor. A prime ℓ dividing C excludes ℓ and the class 1 modulo ℓ from the first $k-1$ prime factors (Lemma 9.12). (b) The least number k_ℓ of prime factors that this exclusion allows, for the primes $\ell \leq 199$ (Proposition 9.13), and the larger lower bounds of Corollary 9.15 (open circles). The dashed line is the bound $\omega(n) \geq 16$ of Theorem 1.1.

where q_S is the least prime larger than the elements of S that is not congruent to 1 modulo any of them. Let k_ℓ be the least value of $K+1$ for which an ℓ -set of size K exists. If ℓ divides the last defect of a solution $n = p_1 \cdots p_k$, then $S = \{p_1, \dots, p_{k-1}\}$ is an ℓ -set: its elements are at least 5 by Proposition 9.10(i), it is independent by Lemma 2.1(ii), Lemma 9.12(i) gives the conditions modulo ℓ , and $p_k \geq q_S$, because p_k is a prime larger than p_{k-1} and not congruent to 1 modulo any p_i , so that Lemma 9.12(iii) gives (9.2). Hence $k \geq k_\ell$.

Proposition 9.13. *We have $k_5 = 201$, $k_7 = 58$, $k_{11} = 14$, $k_{13} = 21$, $k_{17} = 16$, $k_{19} = 15$, $k_{23} = 14$ and $k_{37} = 12$, and $k_\ell = 11$ for every other prime ℓ with $29 \leq \ell \leq 199$. A composite solution of $n-1 = 2\varphi(n)$ whose last defect is divisible by ℓ has at least k_ℓ prime factors.*

Proof. The second assertion was shown above. For the first we use the search of Proposition 2.5 with three changes. The allowed primes are those from 5 on that differ from ℓ and are not congruent to 1 modulo ℓ . The bound $B(S)$ of a set S with fewer than K elements contains, besides the factors for S and for the $K - |S|$ smallest candidates for S , the factor $q/(q-1)$ for the least prime q above the last of these candidates that is not congruent to 1 modulo an element of S ; this prime is not restricted modulo ℓ , since Lemma 9.12 says nothing about p_k modulo ℓ . Finally, a set whose product is at least 2 is discarded, since every set that contains it has a larger product. If T is an ℓ -set of size K whose $|S|$ smallest elements form S , then the other elements of T are candidates for S and q_T is at least the prime q just described, so the right-hand side of (9.2) for T is at most $B(S)$, and the argument in the proof of Proposition 2.5 shows that the search creates T . For every ℓ the search finds no ℓ -set of size less than $k_\ell - 1$ and finds one of size $k_\ell - 1$.¹⁷ Sets of size K are decided by (9.2) in exact rational arithmetic, and the bounds

¹⁷For $\ell = 17$ the only ℓ -set of size 15 is $\{5, 7, 13, 19, 23, 37, 59, 67, 73, 83, 89, 97, 107, 109, 163\}$, with $q_S = 173$ and product 1.98918...; its last defect $C = 15724416449794753166951$ is congruent to 6 modulo 17.

TABLE 10. The ℓ -sets of size $k - 1$ for the pairs (ℓ, k) of Proposition 9.14, and the number of them whose last defect C is divisible by ℓ . None of them has $C \mid A - 1$.

ℓ	k	ℓ -sets	$\ell \mid C$	ℓ	k	ℓ -sets	$\ell \mid C$
7	58	70829	11708	19	15	1	0
11	14	17	2	19	16	40	1
11	15	803	93	19	17	1926	113
11	16	582751	61964	19	18	319369	17685
13	21	118	12	23	14	4	0
13	22	11965	974	23	15	95	4
13	23	3254401	273205	23	16	5348	251
17	16	1	0	23	17	1766427	81086
17	17	42	1				
17	18	2836	161				
17	19	935248	59643				

are handled as in Proposition 2.5, so that rounding can add sets to the search but never remove one. Two programs written independently carry out the search (Appendix A.11). $\square$

For $\ell = 11$ and for $17 \leq \ell \leq 199$ the value of k_ℓ is at most the bound 16 of Theorem 1.1. Deciding the equation on every ℓ -set goes further.

Proposition 9.14. *For each pair (ℓ, k) in Table 10, no composite solution of $n - 1 = 2\varphi(n)$ with k prime factors has a last defect divisible by ℓ .*

Proof. For such a solution $\{p_1, \dots, p_{k-1}\}$ is an ℓ -set of size $k - 1$, as shown above, and with A , B and C formed from its elements, ℓ divides C and C divides $A - 1$. Continued past the first ℓ -set, the search of Proposition 9.13 lists every ℓ -set of size $k - 1$. Table 10 gives their number and the number of them with $\ell \mid C$; none of these has $C \mid A - 1$. $\square$

Corollary 9.15. *Let n be a composite solution of $\varphi(n) \mid n - 1$ with $k \leq 16000$ prime factors. Then $n - 1 = 2\varphi(n)$; let C be its last defect.*

- (i) *If $\ell \mid C$, then $k \geq 201$ for $\ell = 5$, $k \geq 59$ for $\ell = 7$, $k \geq 17$ for $\ell = 11$, $k \geq 24$ for $\ell = 13$, $k \geq 20$ for $\ell = 17$, $k \geq 19$ for $\ell = 19$ and $k \geq 18$ for $\ell = 23$.*
- (ii) *If $k = 16$, then every prime factor of C is at least 29; in particular $C \geq 29$.*

Proof. By Corollary 2.6, $n - 1 = 2\varphi(n)$. (i) Proposition 9.13 gives $k \geq k_\ell$, and Proposition 9.14 excludes the values of k from k_ℓ up to one less than the stated bound. (ii) The number C is odd, and $3 \nmid C$ by Lemma 9.9(iii), since $3 \nmid A$. For $k = 16$, part (i) shows that no prime from 5 to 23 divides C . As $C \geq 5$, it has a prime factor, and every prime factor is at least 29. $\square$

For the primes ℓ with $29 \leq \ell \leq 199$ the values of k_ℓ lie below the bound of Theorem 1.1, and Lemma 9.12 bounds the prime factors of C only from below. A solution with sixteen prime factors whose last defect has only prime factors from 29 on is therefore not excluded by this section; excluding it needs the traversal of Section 9.1.

Remark 9.16. Lehmer's conjecture is equivalent to the following assertion, which this paper does not prove: *there are no finite set S of odd primes and no integer $M \geq 2$ such that, with $A = \prod_{p \in S} p$ and $B = \prod_{p \in S} (p - 1)$, the number $D = MB - A$ is positive, divides $A - 1$, and makes $(MB - 1)/D$ a prime larger than every element of S .* Indeed, let n be a composite

solution of $\varphi(n) \mid n - 1$, let q be its largest prime factor, S the set of its other prime factors and $M = (n - 1)/\varphi(n)$. Then $n = Aq$, $\varphi(n) = B(q - 1)$, and $n - 1 = M\varphi(n)$ reads $qD = MB - 1$; so $D \geq 1$, D divides $MB - 1 = D + (A - 1)$ and hence $A - 1$, and $q = (MB - 1)/D$. Conversely, for S and M as in the assertion the number $n = Aq$ with $q = (MB - 1)/D$ is composite and satisfies $n - 1 = M\varphi(n)$, since $qD = MB - 1$ is the same equation. By Theorem 1.1 a set S with these properties has at least 15 elements, at least 16000 if $M \geq 3$, and at least 10^8 if $3 \in S$. For $M = 2$ the number D is the last defect, with the properties of Proposition 9.10(i) and Corollary 9.15. The assertion asks that one number attached to each set S never be a prime above the elements of S . The integer tuples of Theorem 9.5 show that a proof has to use the primality of the elements of S , and Proposition 9.4 shows that no congruence modulo a fixed number other than that of Lemma 2.2 can replace the modulus D , which depends on S . For the companion equation the same construction with $D = 1$ gives the Fermat numbers: the Fermat primes $S = \{3, 5, 17, 257, 65537\}$ give $2^{32} + 1$. For Lehmer's equation $D = 1$ would give $(MB - 1)/D = A$, which is not a prime larger than the elements of S ; so $D > 1$, and the assertion is not a question about Fermat numbers.

APPENDIX A. IMPLEMENTATION AND CHECKS

This appendix describes how the computations were carried out and checked. None of it is used in the proofs beyond the statements of Section 8.

A.1. Two implementations. The first implementation is recursive and works with exact rational arithmetic. At a terminal node it enumerates p_{k-1} over its interval when the interval is short or D is large, and otherwise enumerates the divisors of D . The second was written separately. It traverses the tree iteratively with an explicit stack, works only with the integers A_j and B_j , computes the interval endpoints from a floating-point guess corrected by integer comparisons, and at every terminal node enumerates p_{k-1} over the whole interval. It never factors anything. For Theorems 1.1 and 1.4 the second implementation is the proof and the first a corroboration. For Theorem 1.3 with $k = 7$ the roles are reversed, for the reason given in Section 7.1, and the factorisations used are all certified. In every run the two implementations visited the same number of nodes. For $k = 15$ the route of Section 4.4, which factors nothing, plays the part of the second implementation, and the two lattice implementations of Section 4 corroborate it. The case $k = 8$ of Theorem 1.3 rests on the program of Section 7.4, whose checks are described in Section A.8. The search behind Theorem 1.2 was likewise carried out by two independent programs (Appendix A.9), and so were the searches of Proposition 2.5 and of Section 9.9 (Appendix A.11).

A.2. A second implementation of the lattice method. We implemented the lattice method twice. The first implementation follows Section 4: its boxes are intervals in u , and the candidates in a box come from Lemma 4.3. The second takes its nodes at depth $k - 3$ from the rational bounds of the first program of Appendix A.1, generates the primes s by a different routine, cuts the hyperbola into boxes along v rather than u , and lists the points of the coset in a box from a basis of the lattice $\{(u, v) : r'u + rv \equiv 0 \pmod{c}\}$ that is Lagrange-reduced for the norm in which the box is a square;¹⁸ a point is accepted when it satisfies (4.1) exactly. It switches to factoring at a lower threshold, and it certifies every prime factor with FLINT. The two share no code beyond the arithmetic libraries.

¹⁸A basis e_1, e_2 of a plane lattice is Lagrange-reduced for a positive definite quadratic form Q when $Q(e_1) \leq Q(e_2) \leq Q(e_2 \pm e_1)$. For a box with sides w_u and w_v the program takes $Q(u, v) = (w_v u)^2 + (w_u v)^2$, writes the four corners of the box in the coordinates of the reduced basis, and runs over the integer points of the smallest rectangle that contains them. Since the box is convex, this rectangle contains the coordinates of every lattice point in the box whatever the basis; the reduction only keeps the rectangle small.

A.3. Fifteen primes. For the 33865004 two-prime problems of Section 5.2, the first lattice implementation factored N completely in the 473 cases where the estimated number of boxes exceeded its threshold, and took 19 minutes of processor time in all. The second factored N in 1772 cases and took 45 minutes. The route of Section 4.4, which factors nothing, took 13 minutes (Table 11).

TABLE 11. The case $k = 15$ of $n - 1 = 2\varphi(n)$, by the two lattice implementations of Section 4 and the route of Section 4.4. The three agree on each of the 55443 pieces into which the work was divided.

	first	second	sums
prefixes of twelve primes	54 985	54 985	54 985
admissible choices of p_{13}	33 865 004	33 865 004	33 865 004
boxes	82 882 379	61 384 414	–
lattice points tested	30 664 171	25 527 573	–
trial divisions	–	–	116 427 765
square tests	–	–	82 568 071
complete factorisations of N	473	1 772	0
completions $s < p < q$ in integers	0	0	0
processor time	19 min	45 min	13 min

A.4. Equations with solutions. Both implementations accept a general equation

$$b(n + \varepsilon) = a\varphi(n) \quad (\text{A.1})$$

with positive integers a, b , the bounds of Section 3 holding with 2 replaced by a/b and (3.5) becoming $(Cp - aB)(Cq - aB) = b(aAB + \varepsilon C)$ with $C = aB - bA$. For $\varepsilon = -1$, $b = 2^k$ and $a = 2^k + h$, equation (A.1) says that $\varphi(n)/2^k = \prod_i \frac{p_i - 1}{2}$ divides $n - 1$ with quotient $2^k + h$, and it has solutions. We ran both implementations with $p_{\min} = 3$ for $k = 4$, $1 \leq h \leq 40$, and for $k = 5$, $1 \leq h \leq 85$. Both returned the same 56 solutions, 20 with $k = 4$ and 36 with $k = 5$, listed in Appendix B; each was verified by direct arithmetic. The bounds and prunes exercised here are exactly those on which Theorems 1.1, 1.3 and 1.4 depend.

A.5. A sieve. As a check that shares no code with the search, we computed $\varphi(n)$ for all $n \leq 10^8$ by a sieve. There is no composite $n \leq 10^8$ with $\varphi(n) \mid n - 1$; the $n \leq 10^8$ with $\varphi(n) \mid n + 1$ are 1, 2, 3, 15, 255, 65535, 83623935; and the odd squarefree $n \leq 10^8$ with $\omega(n) \in \{4, 5\}$ satisfying (A.1) for the parameters above are exactly the ten members of the list in Appendix B below 10^8 .

A.6. The known solutions. For $\varepsilon = +1$ and $p_{\min} = 3$ the search is its own test: Table 7 shows that every known Fermat-type solution with at most six prime factors is found, at the right depth, and nothing else.

A.7. The last three primes. The methods of Section 4 were checked in five ways.

Brute force. The routines that list the divisors of N in a residue class were compared with the list read off from a known factorisation. The two lattice routines were run on 4000 and 3674 random integers N of up to 93 digits, with up to seven prime factors of up to thirteen digits, some of them repeated. The moduli c had between 2 and 55 per cent of the bits of N , so that c^3/N ranged from about 10^{-82} to 10^{44} , and the classes were chosen at random or among the classes of actual divisors. Of these instances 2764 and 2586 were handled by boxes, and the rest, where the number of boxes would have exceeded the limit, by the factoring fallback. The sum

route was run on 3100 integers $N \equiv r^2 \pmod{c}$ of up to 46 digits, about half of them products of two planted members of the class, with c^3/N between about 10^{-13} and 10^{28} . All lists agreed.

Planted divisors. We took 1400 pairs of a prefix and a prime s from the search, 400 of them just above the lower ends of the forty longest intervals, where c^3/N is smallest, and replaced N by the product of two members $t \leq t^*$ of the class of about the same size, with t chosen log-uniformly; the values of c^3/N ranged from $10^{-7.7}$ to 10^{15} . The sum route found the planted divisor in all 1400 cases. The lattice routines were run with factoring switched off, a limit of 20000 boxes and a limit of five seconds per instance. They found the planted divisor in 1287 and 1210 cases and stopped at a limit in the others. Nothing they returned was wrong, and whenever two routines finished they returned the same list. The time limit is needed because $r' = r$ makes the coset (4.2) a union of lines $u + v = \text{const}$. A planted divisor puts a long segment of such a line inside the box that contains it, and the lattice routines examine every point of that segment. This affects only the running time, and the runs of the search had no time limit.

Equations with solutions. Used as a complete search from depth $k-3$, both lattice implementations return the 56 solutions of (A.1) for $k=4$ and $k=5$. For (1.2) with $p_{\min}=3$ and $3 \leq k \leq 7$ they return exactly the solutions 255, 65535, 83623935, 4294967295 and 6992962672132095 of Table 7, and nothing with $k=7$. Since every factorisation in these runs is certified, this re-derives the case $3 \leq k \leq 7$ of Theorem 1.3 independently of Section 7.1. The sum route, applied to the first $k-2$ primes of each of these 61 solutions, returns the solution. For $7 \leq k \leq 14$ and both signs of ε all three find no solution, and the number of choices of p_{k-2} they treat equals the number of nodes at depth $k-2$ found in Sections 5 and 7.

The lattice route alone. For each of the 61 known solutions we took $s = p_{k-2}$ from the solution and asked the first implementation, with factoring switched off, for the divisors of N in the class. In all 36 cases with $c > 1$ the divisor $t = cp_{k-1} - aB'$ was among them; when $c = 1$ every integer lies in the class and only factoring applies. The second implementation assumes $\gcd(r, c) = 1$, which holds in the search by Proposition 4.1 but in only one of these cases, and there it found t as well.

Agreement. For each sign of ε the three computations divided the case $k=15$ into the same 55443 tasks, counted the same number of primes p_{13} in every task, and returned no completion in integers in any task.

A.8. The eight-prime search. The computation of Section 7.4 was cut into 10897 pieces, each a range of s under one prefix, and run on four processors. It made $3.4 \cdot 10^{11}$ trial divisions; the sums ranged over $6.7 \cdot 10^{13}$ values of σ , and the sieve (iii) passed $2.1 \cdot 10^8$ of them on to the exact test. The whole run took 6.9 hours of processor time, 3.2 of them in PARI/GP.

The program was tested in four ways before the run, and part of the run was repeated afterwards.

Known solutions. Run with prime entries, $\varepsilon = +1$ and $p_{\min} = 3$ from depth $k-3$ for $3 \leq k \leq 7$, the program returns the solutions 255, 65535, 83623935, 4294967295 and 6992962672132095, and no other completion with prime entries. The runs were repeated with every divisor sent to the trial division (i), with every divisor sent to the sum route (ii), and with the multiprecision arithmetic used for every quantity; all runs return the same completions among those in which p and q have no prime factor up to 61 other than themselves.

Odd integers. With the conditions modulo 3 removed, and with the congruence prune of Theorem 9.3 in place of the conditions on primes, the program solves (9.1) in odd integers. For $3 \leq k \leq 7$ it finds 1, 1, 2, 8 and 47 solutions with $\varepsilon = -1$, and 1, 1, 2, 4 and 18 with $\varepsilon = +1$, the numbers found by the independent program of Theorem 9.3, and again all runs agree.

Complete factorisation. We took 400 random pairs of a prefix of the case $k=8$ and an admissible prime s , and all 2468 admissible primes s in three ranges under the prefix

3, 5, 17, 257, 65537: just above 2^{32} , where c is small and most N are factored; around 2^{33} ; and around $s = 4596195557$, which has a completion in integers. The completions found by the program, together with those found by factoring the 633 integers N it passed on to PARI/GP, coincide with those found by factoring N for every one of the 2868 pairs. There are three of them, and they reappear in Table 9.

Thirteen entries. The case $k = 13$ of Theorem 9.3(i) uses the program with odd entries prime to 3. For each sign we took 600 values of x_{11} , 270 of them at nine places of the longest interval and 330 under random nodes, and compared the completions found by the program with those read off from a complete factorisation of N , which has up to 55 digits. The two agree: for none of these values does N have a divisor in the class that yields integers $x_{11} < p < q$.

Repetition. After the run, 13 pieces chosen at random with a fixed seed among the 441 whose counts the run records separately, and the piece that contains the four completions in which p and q have no prime factor up to 61, were run again with other parameters. The sums were weighted more than three times as heavily against the trial division, so that the split points t_d move, and N was factored whenever the estimated work exceeded 1.5 milliseconds instead of four. The repetition treated 7415547 admissible values of p_6 and factored 196063 integers N . It found the same number of admissible values of p_6 in each of the 13 pieces as the run, and the same four completions.

A.9. The size bound. Two independent programs carried out the search of Theorem 6.15, one of them written in PARI/GP [30]. The first applies rule (iii) of Section 6.3 by solving the quadratic inequality, and the second locates the omitted interval by bisection; they use different data structures for the tree. For $s = 1, \dots, 7$ both report the same trees. Either program completes the search for $s = 7$ in less than half a minute on one processor core.

A.10. Primality proofs. Every prime factor of an integer N factored in the route of Theorem 4.4 is proved prime: below ψ_{13} by the strong probable-prime test to the first thirteen prime bases, which is a proof there by the work of Sorenson and Webster [35], and above it by the primality prover of FLINT [13]. That prover uses the $n - 1$ and $n + 1$ methods of Brillhart, Lehmer and Selfridge [5] when enough of $n - 1$ or $n + 1$ can be factored, and otherwise the Jacobi sum test of Adleman, Pomerance and Rumely [1] in the form given by Cohen and Lenstra [9].

A.11. The last defect. Two programs written independently carry out the searches of Propositions 9.13 and 9.14. The first uses a table of the primes below $3 \cdot 10^7$, bounds each child both with the candidates of its parent and with its own, and evaluates the bounds in double precision with a margin of 10^{-9} in the logarithm; a second run with the margin 10^{-6} gives the same values of k_ℓ . The second, written in PARI/GP [30], generates the candidates one prime at a time without a table, bounds each child only with the candidates of its parent, computes the logarithms to 38 digits with a margin of 10^{-20} , and decides every bound within the margin in exact rational arithmetic. Both decide every set of full size exactly. For every prime $\ell \leq 199$ the two programs give the same value of k_ℓ , and for every pair in Table 10 the same two counts. Each program finds all the values of k_ℓ in less than a minute, and the longest listing in the table, of the 3254401 sets for $(\ell, k) = (13, 23)$, takes each of them about five minutes. No ℓ -set in the table has both $\ell \mid C$ and $C \mid A - 1$, so no primality test enters these results.

APPENDIX B. SOLUTIONS USED IN THE CHECKS

The following 56 integers $n = p_1 \cdots p_k$ satisfy $2^k(n - 1) = (2^k + h)\varphi(n)$ for the indicated h . They are the complete output of both implementations for $k = 4$, $1 \leq h \leq 40$, and $k = 5$, $1 \leq h \leq 85$, with $p_1 \geq 3$, listed in increasing order of n down the columns. Each was verified by direct arithmetic.

TABLE 12. Solutions with $k = 4$, $1 \leq h \leq 40$.

$p_1, \dots, p_4$	h	$p_1, \dots, p_4$	h
3, 5, 47, 89	15	13, 29, 433, 2969	2
5, 19, 29, 163	6	7, 97, 139, 8581	3
7, 17, 229, 251	4	13, 29, 383, 28879	2
3, 29, 263, 521	9	11, 47, 1109, 15497	2
13, 37, 151, 271	2	29, 41, 1291, 15199	1
3, 29, 197, 1559	9	11, 47, 1049, 77477	2
17, 23, 83, 1709	2	7, 61, 859, 183397	3
29, 71, 163, 193	1	31, 37, 2377, 66499	1
41, 43, 97, 491	1	19, 157, 9697, 15511	1
37, 41, 137, 829	1	19, 157, 8353, 20887	1

TABLE 13. Solutions with $k = 5$, $1 \leq h \leq 85$.

$p_1, \dots, p_5$	h	$p_1, \dots, p_5$	h
5, 7, 13, 139, 653	19	5, 47, 317, 40253, 499027	9
11, 13, 17, 109, 379	9	19, 269, 443, 1747, 1852699	2
3, 17, 29, 317, 4643	21	23, 71, 983, 1031, 103437589	2
11, 17, 107, 173, 1373	6	19, 199, 757, 5059, 23698783	2
5, 19, 37, 73, 19739	12	31, 43, 233, 318629, 12310321	2
19, 73, 97, 103, 613	3	19, 313, 389, 1277, 590838019	2
11, 17, 127, 131, 2879	6	11, 47, 1051, 67967, 102302009	4
5, 7, 269, 317, 4283	15	7, 59, 1721, 41813, 958695209	6
5, 53, 233, 857, 5689	9	41, 167, 14411, 322633, 1891541	1
31, 53, 149, 569, 3407	2	31, 37, 2411, 47681, 5732949913	2
13, 17, 271, 1567, 5573	5	31, 37, 2297, 2639701, 1379629439	2
7, 13, 17, 1553, 480499	11	37, 463, 827, 1667389, 4325665973	1
17, 23, 131, 199, 167099	4	37, 353, 1867, 25109743, 844718249	1
11, 83, 163, 239, 210461	4	43, 139, 61729, 212167, 32068788511	1
13, 167, 349, 1747, 8237	3	37, 367, 1553, 21098893, 41563512479	1
19, 181, 1439, 3767, 17683	2	43, 139, 47857, 57253657, 72147837793	1
3, 11, 269, 17837, 3861929	21	37, 353, 1867, 24384917, 20504256743083	1
43, 317, 331, 953, 166357	1	43, 139, 47819, 1145259251, 653813324977	1

DECLARATIONS

Data and code availability. The programs that carry out the searches, the threshold computations and the checks of Appendix A, their recorded output, the factorisations of the fifteen integers of Table 3, the logs of the three-prime searches of Section 5.2, the program and the journal of the eight-prime search of Section 7.4, the programs of Sections 2.1 and 6.3 and their output, the programs that check the numbers of Table 1 and of the proof of Proposition 6.11, the programs and logs of Section 9, the entries of the tuple of Theorem 9.5 with the check of Lemma 9.8 on them, Lean 4 proofs [26], checked with Mathlib [21], of many statements that do not depend on a search (among them Theorem 6.2, Proposition 6.8, the results of Section 9.7 and Lemmas 9.9 and 9.11), and the sources of this paper are available at <https://github.com/creelie/eLhmr> and archived on Zenodo [4]. Earlier versions of the same programs are archived at [doi:10.5281/zenodo.23072268](https://doi.org/10.5281/zenodo.23072268).

Funding. No funding was received for conducting this study.

Declaration of competing interest. The author has no competing interests to declare that are relevant to the content of this article.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work the author used Claude (Anthropic) to write and run the programs, to write the Lean proofs and to draft text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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